

Supplementary Appendix for Efficient Inference for Inverse Reinforcement Learning and Dynamic Discrete Choice Models

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This appendix is organized around the main reader tasks left after the core results. Section [A](#) collects the reward-normalization extensions and the assumptions used by Theorem [6](#). Section [B](#) gives self-contained detailed statements for the example estimators, so each example’s estimator, additional assumptions, and asymptotic conclusion can be read together. Section [C](#) reports additional simulation results. The remaining sections contain proofs, technical lemmas for the soft Bellman equation, and a brief expanded related-work discussion.

A Extended applications under reward normalization

This section collects the appendix-only extensions of the normalized-reward framework and the detailed assumptions referenced in Theorem [6](#).

A.1 Detailed assumptions for normalized-reward functionals

We state here the assumptions used by Theorem [6](#) and by the normalized-reward extensions below.

- (C1*) For all $(r, k) \in \mathcal{H} \otimes \mathcal{K}$ in an L^2 -neighborhood of $(r_{0,\nu,\gamma,g}, k_0)$, there exist Gâteaux derivatives $\partial_r \tilde{m}(\cdot, r, k) : T_{\mathcal{H}} \rightarrow L^2(\lambda)$ and $\partial_k \tilde{m}(\cdot, r, k) : T_{\mathcal{K}}(k) \rightarrow L^2(\lambda)$ such that, for all $h \in T_{\mathcal{H}}$ and $\eta \in T_{\mathcal{K}}(k)$,

$$\left. \frac{d}{dt} E_0[\tilde{m}(A, S, r + th, (1 + t\eta)k)] \right|_{t=0} = E_0[\partial_r \tilde{m}(A, S, r, k)(h) + \partial_k \tilde{m}(A, S, r, k)(\eta)].$$

- (C2*) (*Continuity of derivatives*) The linear functionals $h \mapsto E_0[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)(h)]$ and $\beta \mapsto E_0[\partial_k \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)(\beta)]$ are continuous, with respective domains $(T_{\mathcal{H}}, \|\cdot\|_{\mathcal{H},0})$ and $(T_{\mathcal{K}}, \|\cdot\|_{\mathcal{K},0})$.
- (C3*) (*Lipschitz continuity*) The map $(r, k) \mapsto E_0[\tilde{m}(A, S, r, k)]$ is locally Lipschitz continuous at $(r_{0,\nu,\gamma,g}, k_0)$ with respect to $(\mathcal{H}, \|\cdot\|_{\mathcal{H},0}) \times (\mathcal{K}, \|\cdot\|_{\mathcal{K}})$.

A.2 Softmax values under reward normalization

Write $\bar{r}_0 := r_{0,\nu,\gamma,g}$. Let $\pi_{\bar{r}_0,k_0}^{\star,\gamma'}$, $q_{\bar{r}_0,k_0}^{\star,\gamma'}$, $V_{\bar{r}_0,k_0}^{\star,\gamma'}$, $v_{\bar{r}_0,k_0}^{\star,\gamma'}$, $\Phi_{\bar{r}_0,k_0}^{\star,\gamma'}$, $\rho_{\bar{r}_0,k_0}^{\star,\gamma'}$, $\tilde{\rho}_{\bar{r}_0,k_0}^{\star,\gamma'}$, $d_{\bar{r}_0,k_0}^{\star,\gamma'}$, and $\tilde{d}_{\bar{r}_0,k_0}^{\star,\gamma'}$ denote the analogues of the objects in Theorem [10](#), but with reward $\bar{r}_0$ and target discount γ' .

Theorem 1 (EIF for softmax value under reward normalization). *Suppose the analogue of Condition D2 holds for $\pi_{\bar{r}_0, k_0}^{*, \gamma'}$, and suppose the soft Bellman fixed point is locally Lipschitz in the sense of Condition D3, with r_0 replaced by $\bar{r}_0$. Assume also that Conditions C1*–C3* hold for $\tilde{m}(a, s, r, k) = V_{r, k}^{*, \gamma'}(s)$. Then the normalized softmax value parameter from Example 2 is pathwise differentiable with efficient influence function*

$$\begin{aligned} \chi_0(s, a, s') &= \alpha_0(a, s) + \alpha_0(a, s) \{r_0(a, s) - g(s) + \gamma V_{r_0 - g, k_0}^{\nu, \gamma}(s') - q_{r_0 - g, k_0}^{\nu, \gamma}(a, s)\} \\ &\quad + \frac{\gamma'}{\tau^*} \tilde{d}_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) \{ \Phi_{\bar{r}_0, k_0}^{*, \gamma'}(s') - v_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) \} \\ &\quad + d_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) \{ \bar{r}_0(a, s) + \gamma' V_{\bar{r}_0, k_0}^{*, \gamma'}(s') - q_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) \} + V_{\bar{r}_0, k_0}^{*, \gamma'}(s) - \Psi(P_0), \end{aligned}$$

where

$$\alpha_0(a, s) := d_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) + \tau^{*-1} \tilde{d}_{\bar{r}_0, k_0}^{*, \gamma'}(a, s) - \frac{\nu(a | s)}{\pi_0(a | s)} \left\{ \rho_{\bar{r}_0, k_0}^{*, \gamma'}(s) + \tau^{*-1} \tilde{\rho}_{\bar{r}_0, k_0}^{*, \gamma'}(s) \right\}.$$

A.3 Structural coefficients under reward normalization

The representation in Lemma 7 underlies the normalization argument from Section 4.4. It shows that linear averages of the normalized reward depend only on the anchored centered log-behavior-policy reward.

Under the classical normalization of Rust (1987), $\nu(a | s) = \mathbb{1}\{a = a^*\}$, this gives

$$E_0[w(A, S)r_0^\dagger(A, S)] = E_0[w(A, S)\{r_0(A, S) - r_0(a^*, S)\}].$$

Thus linear functionals of the normalized reward, and the structural quantities derived from them, are identified directly from r_0 , without imposing parametric structure on $r_0^\dagger$.

Theorem 2 (EIF for structural coefficients under reward normalization). *Assume $x : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^p$ is bounded and $G_0 := E_0[x(A, S)x(A, S)^\top]$ is nonsingular. Let $\bar{r}_0 := r_{0, \nu, \gamma, g}$ and*

$$\alpha_0^x(a, s) := x(a, s) - \frac{\nu(a | s)}{\pi_0(a | s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | s) x(\tilde{a}, s).$$

Then the EIF for $b_0 := E_0[x(A, S)\bar{r}_0(A, S)]$ is

$$\begin{aligned} \chi_{b, 0}(s, a, s') &:= \alpha_0^x(a, s) + \alpha_0^x(a, s) \{r_0(a, s) - g(s) + \gamma V_{r_0 - g, k_0}^{\nu, \gamma}(s') - q_{r_0 - g, k_0}^{\nu, \gamma}(a, s)\} \\ &\quad + x(a, s)\bar{r}_0(a, s) - b_0, \end{aligned}$$

and the EIF for $\theta_0 := G_0^{-1}b_0$ is

$$\chi_{\theta, 0}(s, a, s') = G_0^{-1} \left[\chi_{b, 0}(s, a, s') - \{x(a, s)x(a, s)^\top - G_0\} \theta_0 \right].$$

B Detailed statements for example estimators

This section groups the example-specific estimator displays, the additional assumptions deferred from Section 6.2, and the resulting asymptotic conclusions. The main-text theorem labels remain unchanged; we restate the first three results only in detailed, unnumbered form for readability.

B.1 Example 1a: value of a fixed policy

Let $\pi_n := \exp\{r_n\}$, let $q_n^{\pi, \gamma}$ estimate $q_{r_0, k_0}^{\pi, \gamma}$, and define $V_n^{\pi, \gamma}(s) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} | s) q_n^{\pi, \gamma}(\tilde{a}, s)$. Let $\rho_n^{\pi, \gamma}$ estimate the state occupancy ratio $\rho_0^{\pi, \gamma}$. The estimator of $\psi_0 = E_0[V_{r_0, k_0}^{\pi, \gamma}(S)]$ is

$$\begin{aligned} \psi_n := & \frac{1}{n} \sum_{i=1}^n V_n^{\pi, \gamma}(S_i) + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma}(S_i) \frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} \{r_n(A_i, S_i) + \gamma V_n^{\pi, \gamma}(S'_i) - q_n^{\pi, \gamma}(A_i, S_i)\} \\ & + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma}(S_i) \left\{ \frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} - 1 \right\}. \end{aligned}$$

The additional conditions for Theorem 12 are stated below.

- (E1) (*Boundedness*) There exists $M \in (0, \infty)$ such that $\|\rho_n^{\pi, \gamma}\|_{L^\infty(\mu)} + \|q_n^{\pi, \gamma}\|_{L^\infty(\lambda)} + \|\pi^{-1}\|_{L^\infty(\lambda)} < M$ for all n almost surely.
- (E2) (*Sample splitting*) The estimators r_n , $\rho_n^{\pi, \gamma}$, $q_n^{\pi, \gamma}$ are constructed using data independent of $\mathcal{D}_n$.
- (E3) (*Consistency*) $\|\rho_n^{\pi, \gamma} - \rho_0^{\pi, \gamma}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|q_n^{\pi, \gamma} - q_0^{\pi, \gamma}\|_{\mathcal{H}, r_0} = o_p(1)$.

Detailed version of Theorem 12. Assume D1, C5(b), E1, and E1–E3. Then Theorem 8 holds for the estimator above with influence function χ_0 given in Theorem 9.

B.2 Example 1b: value of a counterfactual softmax policy

Let r_n estimate r_0 , let v_n^* estimate the soft Bellman fixed point v_{r_0, k_0}^* , and define

$$\pi_n^*(a | s) \propto 1\{a \in \mathcal{A}^*\} \exp\left\{\frac{1}{\tau^*} \{r_n(a, s) + \gamma v_n^*(a, s)\}\right\}.$$

Let q_n^* estimate q_{r_0, k_0}^* , set

$$V_n^*(s) := \sum_{\tilde{a} \in \mathcal{A}^*} \pi_n^*(\tilde{a} | s) q_n^*(\tilde{a}, s),$$

and let d_n^* and $\tilde{d}_n^*$ estimate the ordinary and total advantage-weighted state-action representers. Define

$$\rho_n^*(s) := \sum_{a \in \mathcal{A}} \pi_n(a | s) d_n^*(a, s), \quad \tilde{\rho}_n^*(s) := \sum_{a \in \mathcal{A}} \pi_n(a | s) \tilde{d}_n^*(a, s).$$

The resulting estimator of $\psi_0 = E_0\{V_{r_0, k_0}^*(S)\}$ is

$$\begin{aligned}\psi_n := & \frac{1}{n} \sum_{i=1}^n V_n^*(S_i) + \frac{1}{n} \sum_{i=1}^n d_n^*(A_i, S_i) \{r_n(A_i, S_i) + \gamma V_n^*(S'_i) - q_n^*(A_i, S_i)\} \\ & + \frac{\gamma}{n\tau^*} \sum_{i=1}^n \tilde{d}_n^*(A_i, S_i) \left[\tau^* \log \left\{ \sum_{a \in \mathcal{A}^*} \exp\left(\frac{1}{\tau^*} \{r_n(a, S'_i) + \gamma v_n^*(a, S'_i)\}\right) \right\} - v_n^*(A_i, S_i) \right] \\ & + \frac{1}{n} \sum_{i=1}^n \left\{ d_n^*(A_i, S_i) + \frac{1}{\tau^*} \tilde{d}_n^*(A_i, S_i) - \rho_n^*(S_i) - \frac{1}{\tau^*} \tilde{\rho}_n^*(S_i) \right\}.\end{aligned}$$

The additional conditions for Theorem 13 are stated below.

(E1*) (*Boundedness*) There exists $M \in (0, \infty)$ such that, for all n almost surely,

$$\|d_n^*\|_{L^\infty(\lambda)} + \|\tilde{d}_n^*\|_{L^\infty(\lambda)} + \|q_n^*\|_{L^\infty(\lambda)} + \|v_n^*\|_{L^\infty(\lambda)} + \|r_n\|_{L^\infty(\lambda)} < M.$$

(E2*) (*Sample splitting*) The nuisance estimators d_n^* , $\tilde{d}_n^*$, q_n^* , v_n^* , and r_n are constructed using data independent of $\mathcal{D}_n$ (or via cross-fitting).

(E3*) (*Consistency*)

$$\|d_n^* - d_{r_0, k_0}^*\|_{\mathcal{H}, r_0} + \|\tilde{d}_n^* - \tilde{d}_{r_0, k_0}^*\|_{\mathcal{H}, r_0} + \|q_n^* - q_{r_0, k_0}^*\|_{\mathcal{H}, r_0} + \|v_n^* - v_{r_0, k_0}^*\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, r_0} = o_p(1).$$

Detailed version of Theorem 13. Assume D2, C5(b), E1*, and E1*–E3*. Then Theorem 8 holds for the estimator above with influence function χ_0 given in Theorem 10.

B.3 Example 2: fixed-policy value under reward normalization

In addition to the example-specific conditions below, the normalized-reward EIF used in Theorem 11 relies on Conditions C1*–C3* from Appendix A.1.

Let $\pi_n := \exp\{r_n\}$, let $q_n^{\nu, \gamma}$ estimate $q_{r_0 - g, k_0}^{\nu, \gamma}$, and define $V_n^{\nu, \gamma}(s) := \sum_{\tilde{a} \in \mathcal{A}} \nu(\tilde{a} | s) q_n^{\nu, \gamma}(\tilde{a}, s)$ and $r_{n, \nu, \gamma, g}(a, s) := g(s) + q_n^{\nu, \gamma}(a, s) - V_n^{\nu, \gamma}(s)$. Let $q_{n, \nu}^{\pi, \gamma'}$ estimate $q_{r_0, k_0, \nu}^{\pi, \gamma'}$, set $V_{n, \nu}^{\pi, \gamma'}(s) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} | s) q_{n, \nu}^{\pi, \gamma'}(\tilde{a}, s)$, and let $\rho_n^{\pi, \gamma'}$ estimate $\rho_0^{\pi, \gamma'}$. The corresponding estimator of $\psi_0 = E_0[V_{r_0, k_0, \nu}^{\pi, \gamma'}(S)]$ is

$$\begin{aligned}\psi_n := & \frac{1}{n} \sum_{i=1}^n V_{n, \nu}^{\pi, \gamma'}(S_i) \\ & + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma'}(S_i) \frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} \{r_{n, \nu, \gamma, g}(A_i, S_i) + \gamma' V_{n, \nu}^{\pi, \gamma'}(S'_i) - q_{n, \nu}^{\pi, \gamma'}(A_i, S_i)\} \\ & + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma'}(S_i) \left(\frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} - \frac{\nu(A_i | S_i)}{\pi_n(A_i | S_i)} \right) \{r_n(A_i, S_i) - g(S_i) + \gamma V_n^{\nu, \gamma}(S'_i) - q_n^{\nu, \gamma}(A_i, S_i)\} \\ & + \frac{1}{n} \sum_{i=1}^n \rho_n^{\pi, \gamma'}(S_i) \left(\frac{\pi(A_i | S_i)}{\pi_n(A_i | S_i)} - \frac{\nu(A_i | S_i)}{\pi_n(A_i | S_i)} \right).\end{aligned}$$

The additional conditions for Theorem 14 are stated below.

(E1**) (*Boundedness*) There exists $M \in (0, \infty)$ such that, for all n almost surely,

$$\|\rho_n^{\pi, \gamma'}\|_{L^\infty(\mu)} + \|q_n^{\pi, \gamma'}\|_{L^\infty(\lambda)} + \|q_n^{\nu, \gamma}\|_{L^\infty(\lambda)} + \|r_n\|_{L^\infty(\lambda)} + \|\pi^{-1}\|_{L^\infty(\lambda)} + \|\nu^{-1}\|_{L^\infty(\lambda)} < M.$$

(E2**) (*Sample splitting*) The nuisance estimators r_n , $\rho_n^{\pi, \gamma'}$, $q_n^{\pi, \gamma'}$, and $q_n^{\nu, \gamma}$ are constructed using data independent of $\mathcal{D}_n$.

(E3**) (*Consistency*) $\|\rho_n^{\pi, \gamma'} - \rho_0^{\pi, \gamma'}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|q_n^{\nu, \gamma} - q_0^{\nu, \gamma}\|_{\mathcal{H}, r_0} + \|q_n^{\pi, \gamma'} - q_0^{\pi, \gamma'}\|_{\mathcal{H}, r_0} = o_p(1)$.

Detailed version of Theorem 14. Assume D1, D4, C5(b), E1**, and E1**–E3**. Then Theorem 8 holds for the estimator above with influence function χ_0 given in Theorem 11.

B.4 Example 3: structural coefficients under reward normalization

To estimate θ_0 in Example 3, define $\bar{r}_n(a, s) := g(s) + q_n^{\nu, \gamma}(a, s) - V_n^{\nu, \gamma}(s)$, $\pi_n := \exp\{r_n\}$, $V_n^{\nu, \gamma}(s) := \sum_{\tilde{a}} \nu(\tilde{a} | s) q_n^{\nu, \gamma}(\tilde{a}, s)$, and

$$\alpha_n^x(a, s) := x(a, s) - \frac{\nu(a | s)}{\pi_n(a | s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_n(\tilde{a} | s) x(\tilde{a}, s).$$

Set

$$b_n := \frac{1}{n} \sum_{i=1}^n \left[x(A_i, S_i) \bar{r}_n(A_i, S_i) + \alpha_n^x(A_i, S_i) + \alpha_n^x(A_i, S_i) \{r_n(A_i, S_i) - g(S_i) + \gamma V_n^{\nu, \gamma}(S_i') - q_n^{\nu, \gamma}(A_i, S_i)\} \right],$$

and $G_n := n^{-1} \sum_{i=1}^n x(A_i, S_i) x(A_i, S_i)^\top$, $\theta_n := G_n^{-1} b_n$. The example-specific rate condition from the main text is

(E1***) (*Nuisance rates*)

$$\|r_n - r_0\|_{\mathcal{H}, r_0} + \|\mathcal{T}_{k_0, \nu, \gamma}(q_n^{\nu, \gamma} - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0} = o_p(n^{-1/2}).$$

The additional conditions for Theorem 3 are stated below.

(E2***) (*Boundedness*) There exists $M \in (0, \infty)$ such that, for all n almost surely,

$$\|q_n^{\nu, \gamma}\|_{L^\infty(\lambda)} + \|r_n\|_{L^\infty(\lambda)} < M.$$

(E3***) (*Sample splitting*) The nuisance estimators r_n and $q_n^{\nu, \gamma}$ are constructed using data independent of $\mathcal{D}_n$.

(E4***) (*Consistency*)

$$\|r_n - r_0\|_{\mathcal{H}, r_0} + \|q_n^{\nu, \gamma} - q_0^{\nu, \gamma}\|_{\mathcal{H}, r_0} = o_p(1).$$

Theorem 3 (Asymptotic linearity for Example 3). Assume C5(b) and E1***–E4***. Then

$$\theta_n - \theta_0 = \frac{1}{n} \sum_{i=1}^n \chi_{\theta, 0}(S_i, A_i, S_i') + o_p(n^{-1/2}), \quad \sqrt{n}(\theta_n - \theta_0) \rightsquigarrow N(0, \Sigma_0),$$

where $\Sigma_0 := \text{Var}_{P_0}\{\chi_{\theta,0}(S, A, S')\}$.

C Simulation implementation and diagnostics

Neural FORE. Actions are represented by four-dimensional one-hot vectors. Each neural FORE fit uses the variational log-partition objective, minibatches of 512, five gradient steps per fitted iteration, gradient-norm clipping at 10, and a log-ratio cap of 10. Four common target-action draws are reused across candidates within a selection event. The pilot library crosses hidden-layer widths (64, 64) and (128, 128), learning rates 3×10^{-4} and 10^{-3} , weight decays 10^{-3} and 10^{-2} , and iteration budgets 10, 30, and 100. Ten pilot seeds yield 160 A-PBV selection events across examples, endpoint sample sizes, folds, and signed components. The pilot freezes one architecture and optimizer configuration by selection frequency across these events; ties are broken by mean within-cell rank, smaller width, and stronger regularization. During the confirmatory run the stopping library is expanded to 30, 100, and 300 iterations after the pilot showed that 100 iterations frequently remained on a decreasing validation path. Neural ratio optimization uses a deterministic maximum of 20,000 rows from each outer training fold. This budget affects only FORE; logging-policy and Bellman fitting use the complete outer training fold and the EIF is evaluated on every untouched outer-fold observation. In a four-seed $n = 100000$ audit, capped FORE differed from the exact-ratio estimate by RMSE 0.0014 and at most 0.0022, retained zero clipping and failures, and reduced per-component ratio-fit time by about a factor of 3.4 relative to full-data FORE. Setting the cap to zero recovers the full-data appendix diagnostic. For Examples 1a–1b, the existing fine-grid Bellman evaluation is held fixed as a high-accuracy nuisance benchmark so that this study isolates estimated-policy and estimated-ratio error; neither its transition kernel nor any grid truth enters FORE or A-PBV. The data-fusion study estimates its transition and Bellman nuisances from the observed transition source instead.

Main and stress designs. The primary logging law is a state-dependent quadratic multinomial logit. The quadratic sieve is correctly specified, action probabilities remain bounded away from zero, and the action omitted from the adaptive target has about 5% marginal probability. The ordinary-ratio 99th percentile is approximately 12.4 and the ESS fraction approximately 0.93 in the numerical design check. The appendix stress design uses the original soft-MDP logging law; its corresponding values are approximately 24.2 and 0.71. We use a 41×41 numerical grid. Relative to a 61×61 reference, this changes the adaptive estimand by 0.0017, the target policy by less than 6×10^{-4} uniformly, and the stationary-weighted ordinary and signed ratios by about 0.2% and 1.1% in relative L^2 error.

Held-out A-PBV score. Let w_i and w_j be two positive normalized ratio candidates and define $\ell_{ij} = \log(w_i/w_j)$. Using held-out empirical transition and initial-source terms, candidate i is compared with candidate j using

$$\hat{S}_{ij} = \mathbb{P}_n[w_i(X)\{\ell_{ij}(X) - \gamma(\pi\ell_{ij})(S')\}] - (1 - \gamma)\mathbb{P}_{0,n}\ell_{ij}(X_0) - \frac{1 - \gamma}{4}\mathbb{P}_n\{\sqrt{w_i(X)} - \sqrt{w_j(X)}\}^2.$$

The selected index is the first element of $\arg \min_i \max_j \hat{S}_{ij}$. Because the action space has size four, $(\pi\ell_{ij})(S')$ and the initial-state analogue are evaluated by exact action summation. Candidate

fitting and validation use a deterministic 60/20/20 split within each outer training fold: the middle block selects the ordinary ratio and the last block selects the signed components. Within each 20% validation block, the same held-out rows estimate the transition and initial-source terms of the empirical A-PBV objective; this improves validation precision without touching the outer evaluation fold. Simulation truth does not enter either selection.

FORE returns the normalized discounted state–action ratio w . The EIF in the main paper uses the unnormalized convention, so the implementation sets $\hat{d} = w/(1 - \gamma)$ and recovers the state ratio by

$$\hat{\rho}(s) = \sum_a \hat{\pi}_0(a | s) \hat{d}(s, a).$$

This permits the numerically stable correction $\hat{d}(S, A) \hat{\varepsilon}(S, A) + \hat{d}(S, A) - \hat{\rho}(S)$ without separately dividing by a fitted behaviour propensity.

Signed advantage-weighted ratio. On the ratio-training data define

$$Z_i = \hat{d}^*(S_i, A_i) \{ \hat{q}^*(S_i, A_i) - \hat{V}^*(S_i) \}, \quad m_{\pm} = \mathbb{P}_n(Z_i)_{\pm}.$$

We use the successor states weighted by $(Z_i)_+$ and $(Z_i)_-$ as two initial source measures, fit their normalized FORE ratios w_+ and w_- separately, and set

$$\hat{\tilde{d}}(x) = \hat{z}(x) + \frac{\gamma}{1 - \gamma} \{ m_+ w_+(x) - m_- w_-(x) \}, \quad \hat{z}(s, a) := \hat{d}^*(s, a) \{ \hat{q}^*(s, a) - \hat{V}^*(s) \}.$$

The Jordan fits estimate only the strictly future component because their initial measures are supported on successor states. For EIF evaluation we add the current term $\hat{z}$, evaluate the total ratio at every action, and set

$$\hat{\tilde{\rho}}(s) := \sum_a \hat{\pi}_0(a | s) \hat{\tilde{d}}(s, a).$$

The current state average is zero at the population truth, but it need not be zero for estimated nuisances and is therefore retained. A component is set to zero when its empirical source mass is below 10^{-10} . Positive and negative components have independent A-PBV selections.

Data fusion. The transition source records (S, A, S') and the known logging propensities but withholds rewards. The auxiliary source records only (S, Y) over the state domain, with $E(Y | S = s) = g(s) = \sum_a \nu(a | s) r_0(s, a)$ for a known uniform randomized normalization policy ν ; actions are absent by design. We fit g once using 10^6 outcome-summary observations with noise standard deviation 0.2. A deterministic 80/20 split selects a small tanh network on quadratic features by held-out outcome mean-squared error; three source-only refits are averaged and intercept-calibrated, then frozen across all transition replications. The selected regression has grid RMSE 0.0024 and induces an estimand change of 0.0030 at target discount 0.8. Within each outer training fold, an action-specific degree-three ridge sieve estimates the two conditional transition means under a diagonal Gaussian law, with regularization selected by held-out next-state mean-squared error and innovation scales estimated on that validation block. The fitted transition is discretized on the fixed evaluation grid, after which the normalization and target-policy Bellman equations are solved

deterministically. These selectors receive neither the oracle transition kernel nor reward or estimand truth. Before the confirmatory run, the oracle-grid change in the target induced by replacing g with $\hat{g}$ must be smaller than 10% of the pilot median D-IRL standard error. This is a labeled readiness diagnostic, not an estimator or hyperparameter selection rule. The reported confidence intervals condition on the frozen outcome source and do not include its sampling uncertainty.

Reproducibility and oracle diagnostics. Each run stores raw estimates, Monte Carlo summaries, candidate-level A-PBV rows, the frozen configuration, random seeds, ratio-tail diagnostics, runtime, and failures. Oracle ratios and fine-grid targets are used only in explicitly labeled diagnostic output after fitting. In particular, a population-grid check verifies that the positive-minus-negative FORE normalization above reconstructs the oracle signed state-action ratio. All estimator comparisons use common data-generating seeds; changing the ratio estimator never changes the sampled dataset. The execution order is a two-seed smoke test, the truth-blind selection pilot, and the locked 300-replication confirmatory study. Example 1a and data fusion use $n \in \{2500, 5000, 10000\}$, while Example 1b uses $n \in \{25000, 50000, 100000\}$.

Confirmatory fit diagnostics. The locked study contains 2,700 successful replications and no ratio failure, nonfinite inferential quantity, or clipping event. For the ordinary ratio, A-PBV never selects 30 iterations. The shares selecting 300 iterations are 22%, 40%, and 64% across the three Example 1a sample sizes; 77–79% in Example 1b; and 31%, 59%, and 79% in the data-fusion study. We report this boundary behavior rather than using it to alter the locked library. The pre-confirmatory exact-ratio audits described above show that the resulting large-sample FORE estimates are already close to the exact-ratio benchmark. Across the adaptive-policy cells, the signed positive and negative components have mean normalized masses 1.000–1.001, effective-sample fractions 0.68–0.73, mean 99th percentiles 14.0–18.7, and no clipping. The strict artifact assembler verifies all predeclared seeds, fold-level stopping choices, finite telemetry, frozen-input hashes, and raw-to-summary agreement. Coverage calibration is computed only after fitting and is never used to select or suppress a reported cell.

D Expanded related work

Dynamic discrete choice models. Existing econometric work on DDC and dynamic games is predominantly parametric, focusing on identification and efficient estimation of finite-dimensional structural reward (utility) parameters. Classical approaches include the nested fixed-point method of Rust (1987), the two-step conditional choice probability approach of Hotz and Miller (1993), and simulation-based estimators (Hotz et al., 1994; Aguirregabiria and Mira, 2010). These methods do not allow for fully nonparametric reward models or target the reward-dependent value functionals—such as policy values and value differences—central to IRL. Our work extends this line of research to settings with nonparametric reward representations and modern machine-learning-based nuisance estimation. Unlike classical DDC estimators, our framework avoids repeated dynamic programming or Monte Carlo simulation inside the estimator, instead reducing inference to standard supervised-learning and offline reinforcement learning components. To our knowledge, there is no existing derivation of efficient influence functions for such functionals in softmax IRL or in DDC models with nonparametric rewards, nor any debiased, machine-learning-based inference framework

for this setting.

While identification and semiparametric efficiency have been studied extensively in DDC models, this work concerns largely parametric utility specifications. For dynamic discrete games, [Bajari et al. \(2015\)](#) derive identification and efficiency results when per-period rewards are parametrized by a finite-dimensional vector, treating equilibrium objects and transition probabilities nonparametrically. Foundational identification results include [Hotz and Miller \(1993\)](#), [Hotz et al. \(1994\)](#), and [Magnac and Thesmar \(2002\)](#). [Blundell and Horowitz \(2007\)](#) develop nonparametric specification tests for dynamic discrete choice models, and [Blevins \(2014\)](#) study nonparametric identification of dynamic decision processes with discrete and continuous choices. Other work addresses latent heterogeneity or shock distributions: [Kasahara and Shimotsu \(2009\)](#) analyze identification with unobserved heterogeneity, [Buchinsky et al. \(2010\)](#) propose a diagnostic for zero semiparametric information in fixed-effects DDC models, and [Buchholz et al. \(2021\)](#) consider models with unspecified shock distributions. Additional semiparametric results include [Arcidiacono and Miller \(2011\)](#) and [Norets and Tang \(2014\)](#).

A complementary line of research studies identification of counterfactuals under minimal structure. [Kalouptsi et al. \(2021\)](#) show that certain counterfactuals may be point- or set-identified even when utilities are not. Other identification results include [Abbring and Daljord \(2020\)](#) on discount-factor identification and [Kalouptsi et al. \(2017\)](#) on nonidentification of counterfactuals in dynamic games. These papers consider broader DDC settings but do not yield the exact identification of value differences or normalized rewards that arises under the softmax (Gumbel-shock) framework we impose. Following [Rust \(1987\)](#), we adopt the widely used Gumbel specification. Extending our analysis to more general shock distributions is an interesting direction for future work.

E Proofs for Section 3

Proof of Lemma 1. The key observation is that when $r_0 = \log \pi_0$, the log-partition term is zero, so $v_{r_0, k_0} \equiv 0$ solves the Bellman constraint. We then compare the induced softmax representation with that of $r_0^\dagger$ to identify the shaping potential. By definition, we have

$$\begin{aligned} \sum_{a \in \mathcal{A}} \exp\{r_0(a, s)\} &= \sum_{a \in \mathcal{A}} \exp\{\log \pi_0(a | s)\} \\ &= \sum_{a \in \mathcal{A}} \pi_0(a | s) \\ &= 1. \end{aligned}$$

Hence, $\log \sum_{a \in \mathcal{A}} \exp\{r_0(a, s)\} = 0$, and

$$0 = \int \log \sum_{a' \in \mathcal{A}} \exp\{r_0(a', s') + \gamma \cdot 0\} k_0(s' | a, s) \mu(ds').$$

It follows that 0 solves the soft Bellman equation with respect to the reward r_0 . Moreover, the induced softmax policy satisfies

$$\frac{\exp\{r_0(a, s)\}}{\sum_{\tilde{a} \in \mathcal{A}} \exp\{r_0(\tilde{a}, s)\}} = \exp\{r_0(a, s)\} = \pi_0(a \mid s),$$

so it coincides with the true behaviour policy and therefore maximizes the population log-likelihood. Thus, r_0 is also a solution to the soft Bellman-constrained log-likelihood in (3).

Furthermore, we have

$$\exp\{r_0(a, s)\} = \pi_0(a \mid s) = \frac{\exp\{r_0^\dagger(a, s) + \gamma v_0^\dagger(a, s)\}}{\sum_{\tilde{a} \in \mathcal{A}} \exp\{r_0^\dagger(\tilde{a}, s) + \gamma v_0^\dagger(\tilde{a}, s)\}}.$$

Taking logarithms of both sides yields

$$r_0(a, s) = r_0^\dagger(a, s) + \gamma v_0^\dagger(a, s) - \log \sum_{\tilde{a} \in \mathcal{A}} \exp\{r_0^\dagger(\tilde{a}, s) + \gamma v_0^\dagger(\tilde{a}, s)\}.$$

The soft value function is defined as

$$V_0^\dagger(s) = \log \left(\sum_{a' \in \mathcal{A}} \exp\{r_0^\dagger(a', s) + \gamma v_0^\dagger(a', s)\} \right).$$

By the soft Bellman equation,

$$v_0^\dagger(a, s) = \int V_0^\dagger(s') k_0(s' \mid a, s) \mu(ds').$$

Substituting this expression, we obtain

$$r_0(a, s) = r_0^\dagger(a, s) + \gamma \int V_0^\dagger(s') k_0(s' \mid a, s) \mu(ds') - V_0^\dagger(s).$$

□

Proof of Theorem 3. For any reward r , let v_{r, k_0} solve the soft Bellman equation and set $q_r(a, s) := r(a, s) + \gamma v_{r, k_0}(a, s)$ and $\Lambda_r(s) := \log \sum_{a \in \mathcal{A}} \exp\{q_r(a, s)\}$. Then the criterion in (3) equals

$$E_0\{\Lambda_r(S) - q_r(A, S)\} = E_0\{-\log \pi_r(A \mid S)\}, \quad \pi_r(a \mid s) := \exp\{q_r(a, s) - \Lambda_r(s)\}.$$

Since $A \mid S \sim \pi_0(\cdot \mid S)$,

$$E_0\{-\log \pi_r(A \mid S)\} = E_0[\text{KL}\{\pi_0(\cdot \mid S), \pi_r(\cdot \mid S)\}] - E_0\left[\sum_a \pi_0(a \mid S) \log \pi_0(a \mid S)\right].$$

Thus r minimizes (3) if and only if $\pi_r(\cdot \mid s) = \pi_0(\cdot \mid s)$ for P_0 -a.e. s .

Let $\bar{r}_0 := r_0 - g$ and $q := q_{r_0 - g, k_0}^{\nu, \gamma} = \mathcal{T}_{k_0, \nu, \gamma}^{-1}(\bar{r}_0)$, so $q = \bar{r}_0 + \gamma \mathcal{P}_{k_0} \nu q$. Define $c^\star := -\nu q$,

$r^\star := g + (I - \nu)q$, and $v^\star := \mathcal{P}_{k_0}c^\star = -\mathcal{P}_{k_0}\nu q$. Since $q - \gamma\mathcal{P}_{k_0}\nu q = \bar{r}_0$,

$$r^\star + \gamma v^\star = g + (I - \nu)q - \gamma\mathcal{P}_{k_0}\nu q = r_0 + c^\star.$$

Therefore

$$\Lambda_{r^\star}(s) = \log \sum_{a \in \mathcal{A}} \exp\{r_0(a, s) + c^\star(s)\} = c^\star(s),$$

because $\sum_a e^{r_0(a, s)} = \sum_a \pi_0(a | s) = 1$. Hence $v^\star = \mathcal{P}_{k_0}c^\star = \mathcal{P}_{k_0}\Lambda_{r^\star}$, so $v^\star$ solves the soft Bellman equation for $r^\star$. Also

$$\pi_{r^\star}(a | s) = \exp\{r_0(a, s) + c^\star(s) - \Lambda_{r^\star}(s)\} = \pi_0(a | s),$$

so $r^\star$ minimizes (3). Finally, $\nu r^\star = \nu g + \nu(I - \nu)q = g$, and therefore $r^\star$ is normalized.

For uniqueness, let $\tilde{r} \in \mathcal{H}$ be any normalized minimizer, and let $\tilde{v}$ solve its soft Bellman equation. Since $\tilde{r}$ induces π_0 , we have

$$\tilde{r}(a, s) + \gamma\tilde{v}(a, s) - \tilde{\Lambda}(s) = r_0(a, s), \quad \tilde{\Lambda}(s) := \log \sum_{a \in \mathcal{A}} \exp\{\tilde{r}(a, s) + \gamma\tilde{v}(a, s)\}.$$

Thus $\tilde{r} = r_0 + \tilde{\Lambda} - \gamma\tilde{v}$. Since $\tilde{v} = \mathcal{P}_{k_0}\tilde{\Lambda}$, the normalization $\nu\tilde{r} = g$ gives

$$\tilde{\Lambda} = -\nu(r_0 - g) + \gamma\nu\mathcal{P}_{k_0}\tilde{\Lambda}.$$

The right-hand side is a γ -contraction in sup norm. Thus this fixed-point equation has a unique bounded solution. Applying ν to $q = \bar{r}_0 + \gamma\mathcal{P}_{k_0}\nu q$ shows that $c^\star = -\nu q$ satisfies the same equation, $c^\star = -\nu(r_0 - g) + \gamma\nu\mathcal{P}_{k_0}c^\star$. Hence $\tilde{\Lambda} = c^\star$, $\tilde{v} = \mathcal{P}_{k_0}\tilde{\Lambda} = -\mathcal{P}_{k_0}\nu q$, and

$$\tilde{r} = r_0 + \tilde{\Lambda} - \gamma\tilde{v} = r_0 - \nu q + \gamma\mathcal{P}_{k_0}\nu q = g + q - \nu q = r^\star.$$

Thus the normalized solution is unique.

Writing $q = q_{r_0-g, k_0}^{\nu, \gamma}$ statewise gives

$$r^\star(a, s) = g(s) + q_{r_0-g, k_0}^{\nu, \gamma}(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \nu(\tilde{a} | s) q_{r_0-g, k_0}^{\nu, \gamma}(\tilde{a}, s).$$

Moreover, if $\gamma > 0$, then

$$v^\star(a, s) = -\mathcal{P}_{k_0}\nu q(a, s) = \gamma^{-1}\{r_0(a, s) - g(s) - q_{r_0-g, k_0}^{\nu, \gamma}(a, s)\}.$$

Hence $r^\star = r_{0, \nu, \gamma, g}$ and $v^\star = v_{0, \nu, \gamma, g}$, as claimed. □

Proof of Theorem 2. By Lemma 1,

$$r_0(a, s) = r_0^\dagger(a, s) + \gamma \int V_0^\dagger(s') k_0(s' | a, s) \mu(ds') - V_0^\dagger(s).$$

By the definition of the Q -function under policy π , we have

$$q_{r_0, k_0}^{\pi, \gamma}(a, s) = r_0(a, s) + \gamma \int \sum_{a' \in \mathcal{A}} \pi(a' | s') q_{r_0, k_0}^{\pi, \gamma}(a', s') k_0(s' | a, s) \mu(ds').$$

Substituting the expression for $r_0(a, s)$ gives

$$\begin{aligned} q_{r_0, k_0}^{\pi, \gamma}(a, s) &= r_0^\dagger(a, s) + \gamma \int V_0^\dagger(s') k_0(s' | a, s) \mu(ds') - V_0^\dagger(s) \\ &\quad + \gamma \int \sum_{a' \in \mathcal{A}} \pi(a' | s') q_{r_0, k_0}^{\pi, \gamma}(a', s') k_0(s' | a, s) \mu(ds'). \end{aligned}$$

Rearranging terms,

$$q_{r_0, k_0}^{\pi, \gamma}(a, s) + V_0^\dagger(s) = r_0^\dagger(a, s) + \gamma \int \sum_{a' \in \mathcal{A}} \pi(a' | s') (q_{r_0, k_0}^{\pi, \gamma}(a', s') + V_0^\dagger(s')) k_0(s' | a, s) \mu(ds').$$

Thus, $q_{r_0, k_0}^{\pi, \gamma}(a, s) + V_0^\dagger(s)$ satisfies the Bellman equation with reward $r_0^\dagger$. By uniqueness of the solution, it follows that

$$q_{r_0, k_0}^{\pi, \gamma}(a, s) + V_0^\dagger(s) = q_{r_0^\dagger, k_0}^{\pi, \gamma}(a, s),$$

or equivalently,

$$q_{r_0, k_0}^{\pi, \gamma}(a, s) = q_{r_0^\dagger, k_0}^{\pi, \gamma}(a, s) - V_0^\dagger(s).$$

Taking expectations with respect to the initial state distribution ρ_0 and the policy π , we obtain

$$V_{r_0, k_0}(\pi; \gamma) = V_{r_0^\dagger, k_0}(\pi; \gamma) - E_0[V_0^\dagger(S)].$$

Finally, for any two policies π_1 and π_2 ,

$$\begin{aligned} V_{r_0, k_0}(\pi_1; \gamma) - V_{r_0, k_0}(\pi_2; \gamma) &= (V_{r_0^\dagger, k_0}(\pi_1; \gamma) - E_0[V_0^\dagger(S)]) - (V_{r_0^\dagger, k_0}(\pi_2; \gamma) - E_0[V_0^\dagger(S)]) \\ &= V_{r_0^\dagger, k_0}(\pi_1; \gamma) - V_{r_0^\dagger, k_0}(\pi_2; \gamma). \end{aligned}$$

This establishes the claim. □

F Proofs for Section 4.3

We first derive general differentiability facts for the parameter map and then assemble them into the EIF and von Mises expansion stated in Section 4.3.

The following lemma shows that r_P is a solution to a moment-matching problem. In the main text, we set $\mathcal{H} := L^\infty(\lambda)$. For generality, the lemma below allows $\mathcal{H} \subseteq L^\infty(\lambda)$ to be a linear subspace that includes the state-only functions, i.e., $L^\infty(\mu)$. For example, $\mathcal{H}$ may correspond to an additive or partially linear model.

We will use that r_P is the solution to the multinomial logit problem

$$\begin{aligned} r_P &:= \operatorname{argmin}_{r \in \mathcal{H}} -E_P[r(A, S)] \\ \text{subject to } &\sum_{a \in \mathcal{A}} \exp\{r(a, s)\} = 1, \quad \mu\text{-a.e. } s \in \mathcal{S}. \end{aligned} \tag{1}$$

Lemma 4 (Softmax moment matching and identification). *Let $\mathcal{H} \subset L^\infty(\lambda)$ be a linear class that contains all state-only functions, i.e., $L^\infty(\mu)$. Let r_P denote the solution to (1) for this choice of $\mathcal{H}$. Then*

$$E_P[h(A, S)] = E_P\left[\sum_{a \in \mathcal{A}} \exp\{r_P(a, S)\} h(a, S)\right], \quad \forall h \in \mathcal{H}.$$

Proof of Lemma 4. The equality-constrained optimization problem defining r_P in (1) is equivalent to the convex inequality-constrained problem

$$\begin{aligned} r_P &:= \operatorname{argmin}_{r \in \mathcal{H}} -E_P[r(A, S)] \\ \text{subject to } &\sum_{a \in \mathcal{A}} \exp\{r(a, s)\} \leq 1, \quad \mu\text{-a.e. } s \in \mathcal{S}. \end{aligned} \tag{2}$$

The solution satisfies the boundary constraint

$$\sum_{a \in \mathcal{A}} \exp\{r_P(a, s)\} = 1, \quad \mu\text{-a.e. } s \in \mathcal{S}.$$

Since $\mathcal{H}$ contains constants, Slater's condition holds (take $r \equiv -M$ with M large), so strong duality and KKT apply. Let $\lambda \in L_+^1(\mu)$ be the Lagrange multiplier for the pointwise constraint, with Lagrangian

$$\mathcal{L}(r, \lambda) = -E_P[r(A, S)] + \int \lambda(s) \left(\sum_a e^{r(a, s)} - 1 \right) \mu(ds).$$

Stationarity at r_P in any direction $h \in \mathcal{H}$ gives

$$-E_P[h(A, S)] + \int \lambda(s) \left(\sum_a e^{r_P(a, s)} h(a, s) \right) \mu(ds) = 0.$$

In particular, since $\mathcal{H}$ is assumed to contain all state-only functions. We have, for state-only directions $h(a, s) \equiv c(s)$ with $c \in L^\infty(\mu)$,

$$\int \left\{ -\rho_P(s) + \lambda(s) \sum_a e^{r_P(a, s)} \right\} c(s) \mu(ds) = 0,$$

hence

$$\rho_P(s) = \lambda(s) \sum_{a \in \mathcal{A}} e^{r_P(a, s)} \quad \mu\text{-a.e.}$$

Complementary slackness is pointwise: $\lambda(s) (\sum_a e^{r_P(a, s)} - 1) = 0$ μ -a.e. Since $\rho_P > 0$ μ -a.e., the previous display implies $\lambda(s) > 0$ μ -a.e., so the constraint binds μ -a.e., i.e., $\sum_a e^{r_P(a, s)} = 1$ and

consequently $\lambda(s) = \rho_P(s)$ μ -a.e.

Returning to the stationarity identity for general $h \in \mathcal{H}$,

$$E_P[h(A, S)] = \int \lambda(s) \sum_a e^{r_P(a, s)} h(a, s) \mu(ds) = \int \rho_P(s) \sum_a e^{r_P(a, s)} h(a, s) \mu(ds) = E_P\left[\sum_a e^{r_P(a, S)} h(a, S)\right],$$

which is the claim. $\square$

For each $r \in \mathcal{H}$, define the integral operator $\mathcal{L}_r : L^\infty(\lambda) \rightarrow L^2(\lambda)$ by

$$\mathcal{L}_r(h)(a, s) := \left\{ \sum_{\tilde{a} \in \mathcal{A}} \exp\{r(\tilde{a}, s)\} h(\tilde{a}, s) \right\} - h(a, s).$$

The first-order optimality conditions for r_P , given in (2), can then be written as

$$E_P[\mathcal{L}_{r_P}(h)(A, S)] = 0, \quad \forall h \in \mathcal{H}.$$

The Gâteaux derivative of $\mathcal{L}_r(h)$ in the direction $h' \in L^\infty(\lambda)$ is

$$\dot{\mathcal{L}}_r(h, h')(a, s) := \left. \frac{d}{dt} \mathcal{L}_{r+th'}(h)(a, s) \right|_{t=0} = \sum_{\tilde{a} \in \mathcal{A}} \exp\{r(\tilde{a}, s)\} h'(\tilde{a}, s) h(\tilde{a}, s).$$

Lemma 5 (First-order expansion of $\mathcal{L}_r$). *Fix $r, h, h' \in L^\infty(\lambda)$, and set $r' := r + h'$. Suppose r, r' , and h are essentially bounded by M . Then*

$$\left| E_P[\{\mathcal{L}_{r'}(h) - \mathcal{L}_r(h) - \dot{\mathcal{L}}_r(h, h')\}(A, S)] \right| \lesssim E_P[\dot{\mathcal{L}}_r(h', h')(A, S)],$$

where the implicit constant depends only on M .

Proof of Lemma 5. By definition,

$$\mathcal{L}_r(h)(a, s) = \sum_{\tilde{a}} e^{r(\tilde{a}, s)} h(\tilde{a}, s) - h(a, s), \quad \dot{\mathcal{L}}_r(h, h')(a, s) = \sum_{\tilde{a}} e^{r(\tilde{a}, s)} h'(\tilde{a}, s) h(\tilde{a}, s).$$

Hence,

$$\begin{aligned} \mathcal{L}_{r'}(h)(a, s) - \mathcal{L}_r(h)(a, s) - \dot{\mathcal{L}}_r(h, h')(a, s) &= \sum_{\tilde{a}} \left(e^{r(\tilde{a}, s) + h'(\tilde{a}, s)} - e^{r(\tilde{a}, s)} - e^{r(\tilde{a}, s)} h'(\tilde{a}, s) \right) h(\tilde{a}, s) \\ &= \sum_{\tilde{a}} e^{r(\tilde{a}, s)} (e^{h'(\tilde{a}, s)} - 1 - h'(\tilde{a}, s)) h(\tilde{a}, s). \end{aligned}$$

Using the inequality $|e^u - 1 - u| \leq \frac{1}{2} e^{|u|} u^2$ for all $u \in \mathbb{R}$, we obtain

$$|\mathcal{L}_{r'}(h)(a, s) - \mathcal{L}_r(h)(a, s) - \dot{\mathcal{L}}_r(h, h')(a, s)| \leq \frac{1}{2} \sum_{\tilde{a}} e^{r(\tilde{a}, s)} e^{|h'(\tilde{a}, s)|} |h(\tilde{a}, s)| |h'(\tilde{a}, s)|^2.$$

If r , r' , and h are essentially bounded by M , then

$$\begin{aligned} |\mathcal{L}_{r'}(h)(a, s) - \mathcal{L}_r(h)(a, s) - \dot{\mathcal{L}}_r(h, h')(a, s)| &\leq \sum_{\tilde{a}} e^{r(\tilde{a}, s)} e^{2M} M |h'(\tilde{a}, s)|^2 \\ &\lesssim \dot{\mathcal{L}}_r(h', h')(a, s), \end{aligned}$$

where the implicit constant depends only on M . Taking expectation under P , we conclude that

$$\left| E_P \left[\{ \mathcal{L}_{r'}(h) - \mathcal{L}_r(h) - \dot{\mathcal{L}}_r(h, h') \} (A, S) \right] \right| \lesssim E_P [\dot{\mathcal{L}}_r(h', h')(A, S)].$$

□

Lemma 6. Assume $(r, k) \mapsto E_0[m(A, S, r, k)]$ is Fréchet differentiable at (r_0, k_0) in the additive kernel argument; that is,

$$E_0 \left[m(A, S, r_0 + h, k_0 + g) - m(A, S, r_0, k_0) - \partial_r m(A, S, r_0, k_0)(h) - \tilde{\partial}_k m(A, S, r_0, k_0)(g) \right] = o(\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}})$$

as $(h, g) \rightarrow (0, 0)$ in the product norm $\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}$. Then Conditions C1-C3 hold.

Proof of Lemma 6. Gâteaux differentiability in Condition C1 follows by evaluating the Fréchet expansion along multiplicative kernel submodels, that is, with additive kernel direction $g = k_0 \beta$ for $\beta \in T_{\mathcal{K}}$. Since multiplication by k_0 is continuous from $(T_{\mathcal{K}}, \|\cdot\|_{\mathcal{K},0})$ to $(\tilde{T}_{\mathcal{K}}, \|\cdot\|_{\mathcal{K}})$, the boundedness of the derivative functionals in Condition C2 also follows from Fréchet differentiability. Let $F(r, k) := E_0[m(A, S, r, k)]$. By the triangle inequality and boundedness of the derivatives,

$$\begin{aligned} |F(r_0 + h, k_0 + g) - F(r_0, k_0)| &= \left| E_0 \left[\partial_r m(A, S, r_0, k_0)(h) + \tilde{\partial}_k m(A, S, r_0, k_0)(g) \right] \right| + o(\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}) \\ &= O(\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}) + o(\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}). \end{aligned}$$

Thus, by the definition of the $o(\cdot)$ term, for any $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}} < \delta$,

$$|F(r_0 + h, k_0 + g) - F(r_0, k_0)| \leq (C_r + C_k + \varepsilon) (\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}),$$

which establishes local Lipschitz continuity at (r_0, k_0) . □

Proof of Theorem 4. By Lemma 4, the solution r_P to (1) satisfies the first-order optimality conditions:

$$E_P[h(A, S)] = E_P \left[\sum_{a \in \mathcal{A}} \exp\{r_P(a, S)\} h(a, S) \right], \quad \forall h \in \mathcal{H}.$$

Equivalently,

$$E_P[\mathcal{L}_{r_P}(h)(A, S)] = 0, \quad \forall h \in \mathcal{H}.$$

For $\delta \in \mathbb{R}$, let $(P_t : t \in (-\delta, \delta))$ be a regular (quadratic mean differentiable) submodel through

P_0 , satisfying $P_t = P_0$ at $t = 0$, with score function $\varphi \in L_0^2(P_0)$. Then, for all $t \in (-\delta, \delta)$,

$$E_{P_t}[\mathcal{L}_{r_{P_t}}(h)(A, S)] = 0, \quad \forall h \in \mathcal{H}.$$

Differentiating both sides at $t = 0$ gives

$$\left. \frac{d}{dt} E_{P_t}[\mathcal{L}_{r_{P_t}}(h)(A, S)] \right|_{t=0} = 0, \quad \forall h \in \mathcal{H}.$$

Computing the derivative and applying the chain rule yields

$$\left. \frac{d}{dt} E_{P_t}[\mathcal{L}_{r_0}(h)(A, S)] \right|_{t=0} + \left. \frac{d}{dt} E_{P_0}[\mathcal{L}_{r_{P_t}}(h)(A, S)] \right|_{t=0} = 0, \quad \forall h \in \mathcal{H}. \quad (3)$$

By the smoothness of the map $r \mapsto \mathcal{L}_r$ (Lemma 5) and the implicit function theorem, the path $t \mapsto r_{P_t}$ is smooth with derivative $\dot{r}_{P_0}(\varphi) \in \mathcal{H}$. Hence,

$$\left. \frac{d}{dt} E_{P_0}[\mathcal{L}_{r_{P_t}}(h)(A, S)] \right|_{t=0} = E_{P_0}[\dot{\mathcal{L}}_{r_0}(h, \dot{r}_{P_0}(\varphi))(A, S)],$$

where, by definition,

$$E_{P_0}[\dot{\mathcal{L}}_{r_0}(h, \dot{r}_{P_0}(\varphi))(A, S)] = \langle h, \dot{r}_{P_0}(\varphi) \rangle_{\mathcal{H}, 0}.$$

Moreover,

$$\left. \frac{d}{dt} E_{P_t}[\mathcal{L}_{r_0}(h)(A, S)] \right|_{t=0} = E_0[\varphi(S', A, S) \mathcal{L}_{r_0}(h)(A, S)].$$

Thus, rearranging terms,

$$\langle h, \dot{r}_{P_0}(\varphi) \rangle_{\mathcal{H}, 0} = -E_0[\varphi(S', A, S) \mathcal{L}_{r_0}(h)(A, S)], \quad \forall h \in \mathcal{H}. \quad (4)$$

With the above identity in hand, we now turn to establishing pathwise differentiability of the functional. Adding and subtracting, we obtain

$$\begin{aligned} E_0[m(A, S, r_{P_t}, k_{P_t})] &= E_0[m(A, S, r_0 + t\dot{r}_{P_0}(\varphi), (1 + t\dot{k}_{P_0}(\varphi))k_0)] \\ &\quad + \left\{ E_0[m(A, S, r_{P_t}, k_{P_t}) - m(A, S, r_0 + t\dot{r}_{P_0}(\varphi), (1 + t\dot{k}_{P_0}(\varphi))k_0)] \right\}. \end{aligned}$$

By C3, it follows that

$$\begin{aligned} &\left| E_0[m(A, S, r_{P_t}, k_{P_t}) - m(A, S, r_0 + t\dot{r}_{P_0}(\varphi), (1 + t\dot{k}_{P_0}(\varphi))k_0)] \right| \\ &\lesssim \|r_{P_t} - r_0 - t\dot{r}_{P_0}(\varphi)\|_{\mathcal{H}, 0} + \left\| \frac{k_{P_t} - k_0}{k_0} - t\dot{k}_{P_0}(\varphi) \right\|_{L^2(P_0)} \\ &= o(t), \end{aligned}$$

where we used that $\|r_{P_t} - r_0 - t\dot{r}_{P_0}(\varphi)\|_{\mathcal{H}, 0} = o(t)$ and $\left\| \frac{k_{P_t} - k_0}{k_0} - t\dot{k}_{P_0}(\varphi) \right\|_{L^2(P_0)} = o(t)$ by differen-

tiability of $t \mapsto r_{P_t}$ and $t \mapsto k_{P_t}$. We conclude that

$$E_0[m(A, S, r_{P_t}, k_{P_t})] = E_0[m(A, S, r_0 + t\dot{r}_{P_0}(\varphi), (1 + t\dot{k}_{P_0}(\varphi))k_0)] + o(t),$$

and hence,

$$\left. \frac{d}{dt} E_0[m(A, S, r_{P_t}, k_{P_t})] \right|_{t=0} = \left. \frac{d}{dt} E_0[m(A, S, r_0 + t\dot{r}_{P_0}(\varphi), (1 + t\dot{k}_{P_0}(\varphi))k_0)] \right|_{t=0},$$

where this derivative exists by [C1](#). Furthermore, by [C1](#),

$$\left. \frac{d}{dt} E_0[m(A, S, r_{P_t}, k_{P_t})] \right|_{t=0} = E_0[\partial_r m(A, S, r_0, k_0)(\dot{r}_{P_0}(\varphi))] + E_0[\partial_k m(A, S, r_0, k_0)(\dot{k}_{P_0}(\varphi))].$$

By boundedness of the partial derivative functionals in [C1](#), the Riesz representers α_0 and β_0 defined above Theorem 4 exist. Therefore,

$$E_0[\partial_r m(A, S, r_0, k_0)(\dot{r}_{P_0}(\varphi))] = \langle \alpha_0, \dot{r}_{P_0}(\varphi) \rangle_{\mathcal{H},0}, \quad E_0[\partial_k m(A, S, r_0, k_0)(\dot{k}_{P_0}(\varphi))] = \langle \beta_0, \dot{k}_{P_0}(\varphi) \rangle_{\mathcal{K},0}.$$

Note that $\dot{k}_{P_0}(\varphi)$ is the score along the submodel k_{P_t} through the conditional density k_0 . Hence, it coincides with the orthogonal projection of the score φ onto the tangent space associated with the likelihood component k_0 , which consists of square-integrable functions of (S', A, S) with mean zero under $k_0(\cdot \mid A, S)\mu(ds')$ conditional on (A, S) . Since β_0 lies in this space, because $\beta_0 \in T_{\mathcal{K}}$, we have

$$\begin{aligned} \langle \beta_0, \dot{k}_{P_0}(\varphi) \rangle_{\mathcal{K},0} &= \langle \beta_0, \dot{k}_{P_0}(\varphi) \rangle_{L^2(P_0)} \\ &= \langle \beta_0, \varphi \rangle_{L^2(P_0)}. \end{aligned}$$

Furthermore, applying (4) with $h = \dot{r}_{P_0}(\varphi)$, we obtain

$$E_0[\partial_r m(A, S, r_0, k_0)(\dot{r}_{P_0}(\varphi))] = -E_0[\varphi(S', A, S) \mathcal{L}_{r_0}(\alpha_0)(A, S)] = \langle -\mathcal{L}_{r_0}(\alpha_0), \varphi \rangle_{L^2(P_0)}.$$

Hence,

$$\left. \frac{d}{dt} E_0[m(A, S, r_{P_t}, k_{P_t})] \right|_{t=0} = \left\langle -\mathcal{L}_{r_0}(\alpha_0) + \beta_0, \varphi \right\rangle_{L^2(P_0)}.$$

Moreover, by the chain rule,

$$\begin{aligned} \left. \frac{d}{dt} \Psi(P_t) \right|_{t=0} &= \left. \frac{d}{dt} E_{P_t}[m(A, S, r_{P_t}, k_{P_t})] \right|_{t=0} \\ &= E_0[\varphi(S', A, S) m(A, S, r_0, k_0)] + \left. \frac{d}{dt} E_0[m(A, S, r_{P_t}, k_{P_t})] \right|_{t=0} \\ &= \left\langle m(\cdot, \cdot, r_0, k_0) - \Psi(P_0), \varphi \right\rangle_{L^2(P_0)} + \left\langle -\mathcal{L}_{r_0}(\alpha_0) + \beta_0, \varphi \right\rangle_{L^2(P_0)} \\ &= \left\langle m(\cdot, \cdot, r_0, k_0) - \Psi(P_0) - \mathcal{L}_{r_0}(\alpha_0) + \beta_0, \varphi \right\rangle_{L^2(P_0)}. \end{aligned}$$

The function

$$m(\cdot, \cdot, r_0, k_0) - \Psi(P_0) - \mathcal{L}_{r_0}(\alpha_0) + \beta_0$$

has mean zero and therefore belongs to the nonparametric tangent space. Hence, it is the EIF under the nonparametric model. By definition,

$$-\mathcal{L}_{r_0}(\alpha_0) = \alpha_0 - \sum_{\tilde{a} \in \mathcal{A}} \exp\{r_0(\tilde{a}, \cdot)\} \alpha_0(\tilde{a}, \cdot).$$

It follows that

$$\chi_0 = m(\cdot, \cdot, r_0, k_0) - \Psi(P_0) - \mathcal{L}_{r_0}(\alpha_0) + \beta_0$$

is the EIF. □

Proof of Theorem 5. By C1 and the definition of the linearization remainder $\text{Rem}_0(r_0, k_0; r, k)$, we have

$$E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] = E_0[\partial_r m(A, S, r_0, k_0)(r - r_0)] + E_0\left[\partial_k m(A, S, r_0, k_0)\left(\frac{k - k_0}{k_0}\right)\right] + \text{Rem}_0(r_0, k_0; r, k)$$

By the definition of the Riesz representers α_0 and β_0 , which exist by C1, we have

$$\begin{aligned} E_0[\partial_r m(A, S, r_0, k_0)(r - r_0)] &= \langle \alpha_0, r - r_0 \rangle_{\mathcal{H}, 0} \\ E_0\left[\partial_k m(A, S, r_0, k_0)\left(\frac{k - k_0}{k_0}\right)\right] &= \left\langle \beta_0, \frac{k - k_0}{k_0} \right\rangle_{\mathcal{K}, 0}. \end{aligned}$$

Moreover, using this fact, by Lemma 7, which we state and prove below,

$$E_0\left[\partial_k m(A, S, r_0, k_0)\left(\frac{k - k_0}{k_0}\right)\right] = -P_0\{\beta\} + O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \left\|\frac{k - k_0}{k_0}\right\|_{L^2(P_0)}\right).$$

Hence,

$$\begin{aligned} E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] &= \langle \alpha_0, r - r_0 \rangle_{\mathcal{H}, 0} - P_0\{\beta\} \\ &\quad + O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \left\|\frac{k - k_0}{k_0}\right\|_{L^2(P_0)}\right) \\ &\quad + \text{Rem}_0(r_0, k_0; r, k). \end{aligned}$$

We now turn to the term $\langle \alpha_0, r - r_0 \rangle_{\mathcal{H}, 0}$. Adding and subtracting,

$$\langle \alpha_0, r - r_0 \rangle_{\mathcal{H}, 0} = \langle \alpha, r - r_0 \rangle_{\mathcal{H}, 0} + \langle \alpha_0 - \alpha, r - r_0 \rangle_{\mathcal{H}, 0}.$$

By definition, we have

$$\langle \alpha, r - r_0 \rangle_{\mathcal{H}, 0} = E_0[\dot{\mathcal{L}}_{r_0}(\alpha, r - r_0)].$$

Thus, applying the second-order expansion in Lemma 5,

$$\begin{aligned} E_0[\dot{\mathcal{L}}_{r_0}(\alpha, r - r_0)] &= E_0[\mathcal{L}_r(\alpha)] - E_0[\mathcal{L}_{r_0}(\alpha)] + O\left(E_0[\dot{\mathcal{L}}_{r_0}(r - r_0, r - r_0)(A, S)]\right) \\ &= E_0[\mathcal{L}_r(\alpha)] - E_0[\mathcal{L}_{r_0}(\alpha)] + O(\|r - r_0\|_{\mathcal{H},0}^2), \end{aligned}$$

where the implicit constant depends only on M . By Lemma 4, the optimality conditions for r_0 imply that $E_0[\mathcal{L}_{r_0}(\alpha)] = 0$, since $\alpha \in T_{\mathcal{H}}(r_0)$. Thus,

$$\begin{aligned} E_0[\dot{\mathcal{L}}_{r_0}(\alpha, r - r_0)] &= E_0[\mathcal{L}_r(\alpha)] + O(\|r - r_0\|_{\mathcal{H},0}^2) \\ &= P_0\mathcal{L}_r(\alpha) + O(\|r - r_0\|_{\mathcal{H},0}^2) \end{aligned}$$

Hence,

$$\langle \alpha_0, r - r_0 \rangle_{\mathcal{H},0} = P_0\mathcal{L}_r(\alpha) + \langle \alpha_0 - \alpha, r - r_0 \rangle_{\mathcal{H},0} + O(\|r - r_0\|_{\mathcal{H},0}^2).$$

Putting everything together, we obtain

$$\begin{aligned} E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] &= P_0\{\mathcal{L}_r(\alpha) - \beta\} + \langle \alpha_0 - \alpha, r - r_0 \rangle_{\mathcal{H},0} \\ &\quad + O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \cdot \left\|\frac{k - k_0}{k_0}\right\|_{L^2(P_0)}\right) + \text{Rem}_0(r_0, k_0; r, k). \end{aligned}$$

Rearranging terms and using the definition $\varphi_{r,k}(\cdot; \alpha, \beta) = -\mathcal{L}_r(\alpha) + \beta$, we obtain

$$\begin{aligned} E_0[m(A, S, r, k)] - E_0[m(A, S, r_0, k_0)] &+ \int \varphi_{r,k}(s', a, s; \alpha, \beta) P_0(ds, da, ds') \\ &= -\langle r - r_0, \alpha - \alpha_0 \rangle_{\mathcal{H},0} + O(\|r - r_0\|_{\mathcal{H},0}^2) + O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \cdot \left\|\frac{k - k_0}{k_0}\right\|_{L^2(P_0)}\right) + \text{Rem}_0(r_0, k_0; r, k). \end{aligned}$$

where the implicit constants depend only on M . This completes the proof. $\square$

Lemma 7 (von Mises expansion for kernel partial derivative). *For all $k \in \mathcal{K}$ and $\beta \in T_{\mathcal{K}}(k)$, it holds that*

$$E_0\left[\partial_k m(A, S, r_0, k_0)\left(\frac{k - k_0}{k_0}\right)\right] = -E_0[\beta(S', A, S)] + O\left(\|\beta_0 - \beta\|_{L^2(P_0)} \left\|\frac{k - k_0}{k_0}\right\|_{L^2(P_0)}\right).$$

Proof. By the first-order optimality conditions defining the Riesz representer β_0 in (10), for all $k \in \mathcal{K}$,

$$E_0\left[\partial_k m(A, S, r_0, k_0)\left(\frac{k - k_0}{k_0}\right)\right] = \left\langle \frac{k - k_0}{k_0}, \beta_0 \right\rangle_{L^2(P_0)}.$$

Hence,

$$\begin{aligned}
E_0 \left[\partial_k m(A, S, r_0, k_0) \left(\frac{k-k_0}{k_0} \right) \right] &= \left\langle \frac{k-k_0}{k_0}, \beta_0 \right\rangle_{L^2(P_0)} \\
&= E_0 \left[\int \beta_0(s', A, S) (k(s' | A, S) - k_0(s' | A, S)) \mu(ds') \right] \\
&= E_0 \left[\int \beta_0(s', A, S) k(s' | A, S) \mu(ds') \right],
\end{aligned}$$

where we used that $\int \beta_0(s', A, S) k_0(s' | A, S) \mu(ds') = 0$ P_0 -almost surely (since $\beta_0 \in T_{\mathcal{K}}(k_0)$).

Adding and subtracting β ,

$$\begin{aligned}
E_0 \left[\partial_k m(A, S, r_0, k_0) \left(\frac{k-k_0}{k_0} \right) \right] &= E_0 \left[\int \left\{ \beta_0 - \beta \right\} (s', A, S) k(s' | A, S) \mu(ds') \right] \\
&\quad + E_0 \left[\int \beta(s', A, S) k(s' | A, S) \mu(ds') \right] \\
&= E_0 \left[\int \left\{ \beta_0 - \beta \right\} (s', A, S) k(s' | A, S) \mu(ds') \right],
\end{aligned}$$

where we used that

$$\int \beta(s', A, S) k(s' | A, S) \mu(ds') = 0$$

P_0 -almost surely, since $\beta \in T_{\mathcal{K}}(k)$.

Decomposing $k = (k - k_0) + k_0$, we obtain

$$\begin{aligned}
E_0 \left[\partial_k m(A, S, r_0, k_0) \left(\frac{k-k_0}{k_0} \right) \right] &= E_0 \left[\int \left\{ \beta_0 - \beta \right\} (s', A, S) (k - k_0)(s' | A, S) \mu(ds') \right] \\
&\quad - E_0 [\beta(S', A, S)].
\end{aligned}$$

Here we used that

$$\int \beta_0(s', A, S) k_0(s' | A, S) \mu(ds') = 0$$

P_0 -almost surely, since $\beta_0 \in T_{\mathcal{K}}(k_0)$, and

$$E_0 \left[\int \beta(s', A, S) k_0(s' | A, S) \mu(ds') \right] = E_0 [\beta(S', A, S)].$$

Rearranging,

$$E_0 \left[\partial_k m(A, S, r_0, k_0) \left(\frac{k-k_0}{k_0} \right) \right] = -E_0 [\beta(S', A, S)] + E_0 \left[\int \left\{ \beta_0 - \beta \right\} (s', A, S) (k - k_0)(s' | A, S) \mu(ds') \right].$$

Finally, the remainder is bounded by Cauchy–Schwarz:

$$\begin{aligned} & E_0 \left[\int \left\{ \beta_0 - \beta \right\} (s', A, S) (k - k_0) (s' | A, S) \mu(ds') \right] \\ &= \left\langle \beta_0 - \beta, \frac{k - k_0}{k_0} \right\rangle_{L^2(P_0)} \leq \| \beta_0 - \beta \|_{L^2(P_0)} \left\| \frac{k - k_0}{k_0} \right\|_{L^2(P_0)}. \end{aligned}$$

This completes the proof. $\square$

F.1 Proofs for Section 4.4

Proof of Lemma 7. Without loss of generality, assume that w is nonnegative. Otherwise, we can decompose it into its positive and negative parts and apply the argument to each separately. Furthermore, we may assume without loss of generality that w has unit expectation, $E_0[w(A, S)] = 1$, since the result of the theorem is invariant to rescaling. Denote the w -weighted expectation of $f(A, S)$ by $E_w[f(A, S)] := E_0[w(A, S) f(A, S)]$.

By Theorem 3, $r_0^\dagger = g + (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1}(r_0 - g)$. Hence,

$$\begin{aligned} E_0[w(A, S) r_0^\dagger(A, S)] &= E_w[r_0^\dagger(A, S)] \\ &= E_w[g(S)] + E_w[(I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1}(r_0 - g)(A, S)]. \end{aligned}$$

Applying the final identity in Lemma 9 with $\pi := \nu$ and $P(da, ds) = w(a, s) \rho_0(s) \pi_0(a | s) \mu(ds)$ (so that $E_P = E_w$), we obtain

$$E_w[(I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1}(r_0 - g)(A, S)] = E_w[r_0(A, S) - g(S) - \sum_{a \in \mathcal{A}} \nu(a | S) \{r_0(a, S) - g(S)\}].$$

Therefore,

$$E_0[w(A, S) r_0^\dagger(A, S)] = E_0[w(A, S) \left\{ g(S) + r_0(A, S) - \sum_{a \in \mathcal{A}} \nu(a | S) r_0(a, S) \right\}].$$

$\square$

Proof of Theorem 6. We apply Theorem 4 to the map

$$m(a, s, r, k) := \tilde{m}(a, s, r_{r, k, \nu, \gamma, g}, k).$$

Conditions C1–C3 hold because C1 is satisfied with $m := \tilde{m}$, while C2* and C3* also hold. Moreover, the map

$$(r, k) \mapsto r_{r, k, \nu, \gamma, g} = g + (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1}(r - g)$$

is Fréchet differentiable and Lipschitz continuous by Lemma 10 and the proof of Lemma 11. Hence, the EIF is given by χ_0 in Theorem 4. We now turn to deriving α_0 and β_0 .

We have

$$E_0[m(A, S, r, k)] = E_0[\tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k)], \quad (5)$$

which rewrites m in terms of $\tilde{m}$ and the transformed reward.

We begin with computing the reward partial derivative. By the chain rule,

$$\begin{aligned} E_0[\partial_r m(A, S, r, k)[\alpha]] &= E_0\left[\partial_r \tilde{m}(A, S, g + (I - \nu)\mathcal{T}_{k,\nu,\gamma}^{-1}(r - g), k)[(I - \nu)\mathcal{T}_{k,\nu,\gamma}^{-1}(\alpha)]\right] \\ &= E_0\left[\partial_r \tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k)[(I - \nu)\mathcal{T}_{k,\nu,\gamma}^{-1}(\alpha)]\right]. \end{aligned}$$

Evaluating at $r = r_0$ and $k = k_0$, we obtain

$$\begin{aligned} E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] &= E_0\left[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[(I - \nu)\mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha)]\right] \\ &= \langle \tilde{\alpha}_0, (I - \nu)\mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha) \rangle_{\mathcal{H},0}, \end{aligned} \quad (6)$$

where we used the Riesz representation: for any $\tilde{\alpha} \in L^2(\rho_0 \otimes \pi_0)$,

$$E_0[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0)[\tilde{\alpha}]] = \langle \tilde{\alpha}_0, \tilde{\alpha} \rangle_{\mathcal{H},0},$$

and the fact that $\langle \cdot, \cdot \rangle_{\mathcal{H},0} = \langle \cdot, \cdot \rangle_{L^2(\rho_0 \otimes \pi_0)}$ when $\mathcal{H} = L^\infty(\lambda)$.

By the proof of Lemma 7, the inner product in (6) simplifies further:

$$\begin{aligned} \langle \tilde{\alpha}_0, (I - \nu)\mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha) \rangle_{L^2(\rho_0 \otimes \pi_0)} &= E_0\left[\tilde{\alpha}_0(A, S) (I - \nu)\mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha)(A, S)\right] \\ &= E_0[\tilde{\alpha}_0(A, S) (I - \nu)(\alpha)(A, S)] \end{aligned} \quad (7)$$

$$= E_0[\tilde{\alpha}_0(A, S) \alpha(A, S)] - E_0\left[\tilde{\alpha}_0(A, S) \sum_{a \in \mathcal{A}} \nu(a | S) \alpha(a, S)\right]. \quad (8)$$

Moreover,

$$\begin{aligned} E_0\left[\tilde{\alpha}_0(A, S) \sum_{a \in \mathcal{A}} \nu(a | S) \alpha(a, S)\right] &= E_0\left[\left\{\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \tilde{\alpha}_0(\tilde{a}, S)\right\} \left\{\sum_{a \in \mathcal{A}} \nu(a | S) \alpha(a, S)\right\}\right] \\ &= E_0\left[\left\{\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \tilde{\alpha}_0(\tilde{a}, S)\right\} \frac{\nu(A | S)}{\pi_0(A | S)} \alpha(A, S)\right]. \end{aligned}$$

Hence,

$$\begin{aligned} \langle \tilde{\alpha}_0, (I - \nu)\mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha) \rangle_{L^2(\rho_0 \otimes \pi_0)} &= E_0[\tilde{\alpha}_0(A, S) \alpha(A, S)] - E_0\left[\left\{\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \tilde{\alpha}_0(\tilde{a}, S)\right\} \frac{\nu(A | S)}{\pi_0(A | S)} \alpha(A, S)\right] \\ &= E_0[\alpha_0(A, S) \alpha(A, S)], \end{aligned} \quad (9)$$

where

$$\alpha_0(a, s) := \tilde{\alpha}_0(a, s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \tilde{\alpha}_0(\tilde{a}, s).$$

Combining (5)–(8), we conclude that

$$E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] = E_0[\alpha_0(A, S) \alpha(A, S)]. \quad (10)$$

We now turn to the kernel partial derivative. By the chain rule,

$$\begin{aligned} E_0[\partial_k m(A, S, r, k)[\beta]] &= E_0 \left[\partial_r \tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k) \left[(I - \nu) \tilde{\partial}_k \left\{ \mathcal{T}_{k,\nu,\gamma}^{-1} \right\} (r - g)[k\beta] \right] \right. \\ &\quad \left. + E_0[\partial_k \tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k) [\beta]] \right]. \end{aligned} \quad (11)$$

By the resolvent derivative identity

$$\tilde{\partial}_k \left(\mathcal{T}_{k,\nu,\gamma}^{-1} \right) [k\beta] = \gamma \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_{k\beta} \nu) \mathcal{T}_{k,\nu,\gamma}^{-1},$$

we obtain

$$\begin{aligned} E_0[\partial_k m(A, S, r, k)[\beta]] &= E_0 \left[\partial_r \tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k) \left[(I - \nu) \gamma \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_{k\beta} \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right] \right. \\ &\quad \left. + E_0[\partial_k \tilde{m}(A, S, r_{r,k,\nu,\gamma,g}, k) [\beta]] \right], \end{aligned} \quad (12)$$

where $r_{r,k,\nu,\gamma,g} := g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g)$. Hence, evaluating at $r = r_0$ and $k = k_0$,

$$\begin{aligned} E_0[\partial_k m(A, S, r_0, k_0)[\beta]] &= E_0 \left[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0) \left[(I - \nu) \gamma \mathcal{T}_{k_0,\nu,\gamma}^{-1} (\mathcal{P}_{k_0\beta} \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} (r_0 - g) \right] \right. \\ &\quad \left. + E_0[\partial_k \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0) [\beta]] \right]. \end{aligned} \quad (13)$$

By the Riesz representation property of $\tilde{\beta}_0$, the second term on the right-hand side of (13) satisfies:

$$E_0[\partial_k \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0) [\beta]] = \langle \tilde{\beta}_0, \beta \rangle_{\mathcal{K},0}. \quad (14)$$

By the Riesz representation property of $\tilde{\alpha}_0$, the first term on the right-hand side of (13) satisfies:

$$E_0 \left[\partial_r \tilde{m}(A, S, r_{0,\nu,\gamma,g}, k_0) \left[(I - \nu) \gamma \mathcal{T}_{k_0,\nu,\gamma}^{-1} (\mathcal{P}_{k_0\beta} \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} (r_0 - g) \right] \right] = \langle \tilde{\alpha}_0, (I - \nu) \gamma \mathcal{T}_{k_0,\nu,\gamma}^{-1} (\mathcal{P}_{k_0\beta} \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} (r_0 - g) \rangle_{L^2} \quad (15)$$

By the proof of Lemma 7, the inner product in (15) simplifies as

$$\langle \tilde{\alpha}_0, (I - \nu) \gamma \mathcal{T}_{k_0,\nu,\gamma}^{-1} (\mathcal{P}_{k_0\beta} \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} (r_0 - g) \rangle_{L^2(\rho_0 \otimes \pi_0)} = \gamma \langle \alpha_0, (\mathcal{P}_{k_0\beta} \nu) q_{r_0-g,k_0}^{\nu,\gamma} \rangle_{L^2(\rho_0 \otimes \pi_0)}, \quad (16)$$

where we used that $q_{r_0-g,k_0}^{\nu,\gamma} = \mathcal{T}_{k_0,\nu,\gamma}^{-1} (r_0 - g)$.

Writing out the action of $\mathcal{P}_{k_0\beta}\nu$, equation (16) becomes

$$\gamma \langle \alpha_0, (\mathcal{P}_{k_0\beta}\nu) q_{r_0-g,k_0}^{\nu,\gamma} \rangle_{L^2(\rho_0 \otimes \pi_0)} = \gamma E_0 \left[\alpha_0(A, S) \left\{ V_{r_0-g,k_0}^{\nu,\gamma}(S') - (\mathcal{P}_{k_0} V_{r_0-g,k_0}^{\nu,\gamma})(A, S) \right\} \beta(S' | A, S) \right]. \quad (17)$$

By Bellman's equation, the bracket in (17) equals

$$r_0(A, S) - g(S) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(S') - q_{r_0-g,k_0}^{\nu,\gamma}(A, S). \quad (18)$$

Hence,

$$\gamma \langle \alpha_0, (\mathcal{P}_{k_0\beta}\nu) q_{r_0-g,k_0}^{\nu,\gamma} \rangle_{L^2(\rho_0 \otimes \pi_0)} = E_0 [\beta_{0,1}(S' | A, S) \beta(S' | A, S)], \quad (19)$$

where

$$\beta_{0,1}(s' | a, s) := \alpha_0(a, s) \left\{ r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s) \right\}.$$

Combining (11)–(19), we conclude that

$$E_0[\partial_k m(A, S, r_0, k_0)[\beta]] = \langle \beta_0, \beta \rangle_{\mathcal{K},0},$$

where

$$\beta_0 = \tilde{\beta}_0 + \beta_{0,1}.$$

□

G Proofs for Section 5

Proof of Theorem 8. Write $\psi_n = P_n \varphi_n$, where we define the estimated uncentered influence function by

$$\varphi_n(s', a, s) := m(a, s, r_n, k_n) + \varphi_{r_n, k_n}(s', a, s; \alpha_n, \beta_n),$$

with $\varphi_{r,k}$ as in Section 4.3. Let

$$\varphi_0(s', a, s) := m(a, s, r_0, k_0) + \varphi_{r_0, k_0}(s', a, s; \alpha_0, \beta_0)$$

so that $P_0 \varphi_0 = \psi_0$ and $P_0 \chi_0 = 0$, where χ_0 is the EIF in Theorem 4. Then

$$\begin{aligned} \psi_n - \psi_0 &= P_n \varphi_n - P_0 \varphi_0 \\ &= P_n(\varphi_0 - P_0 \varphi_0) + P_n(\varphi_n - \varphi_0) \\ &= P_n \chi_0 + (P_n - P_0)(\varphi_n - \varphi_0) + \{P_0 \varphi_n - \psi_0\}. \end{aligned}$$

By Conditions C4–(a), $\|\varphi_n - \varphi_0\|_{L^2(P_0)} = o_P(1)$. By sample splitting (Condition C4) and Markov's inequality,

$$(P_n - P_0)(\varphi_n - \varphi_0) = O_P\left(n^{-1/2} \|\varphi_n - \varphi_0\|_{L^2(P_0)}\right) = o_P(n^{-1/2}).$$

Thus

$$\psi_n - \psi_0 = P_n \chi_0 + o_P(n^{-1/2}) + \{P_0 \varphi_n - \psi_0\}.$$

It remains to control the remainder term $P_0 \varphi_n - \psi_0$. By definition,

$$P_0 \varphi_n - \psi_0 = E_0[m(A, S, r_n, k_n)] - E_0[m(A, S, r_0, k_0)] + E_0[\varphi_{r_n, k_n}(\cdot; \alpha_n, \beta_n)].$$

Applying Theorem 5 with $(r, k, \alpha, \beta) = (r_n, k_n, \alpha_n, \beta_n)$ yields

$$\begin{aligned} \{P_0 \varphi_n - \psi_0\} &= -\langle r_n - r_0, \alpha_n - \alpha_0 \rangle_{\mathcal{H}, r_0} + O(\|r_n - r_0\|_{\mathcal{H}}^2) \\ &\quad + O\left(\left\|\beta_0 - \beta_n\right\|_{L^2(P_0)} \left\|\frac{k_n - k_0}{k_0}\right\|_{L^2(P_0)}\right) + \text{Rem}_0(r_0, k_0; r_n, k_n). \end{aligned}$$

Each term on the right-hand side is $o_P(n^{-1/2})$ by the rate conditions (b)–(c). Therefore,

$$\{P_0 \varphi_n - \psi_0\} = o_P(n^{-1/2}),$$

and we conclude

$$\psi_n - \psi_0 = P_n \chi_0 + o_P(n^{-1/2}).$$

Finally, by the central limit theorem,

$$\sqrt{n}(\psi_n - \psi_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \chi_0(S_i, A_i, S'_i) + o_P(1) \rightsquigarrow N(0, \text{Var}_{P_0}(\chi_0(S, A, S'))),$$

establishing asymptotic linearity and efficiency. $\square$

H Proofs for Section 6

H.1 Technical lemmas

We begin with operator and occupancy-ratio identities that are reused across the example-specific EIF derivations and asymptotic expansions.

Lemma 8 (Invertibility of the Bellman operator under stationarity overlap). *Assume $\gamma < 1$ and D1. Then $\mathcal{T}_{k_0, \pi} : L^2(\lambda) \rightarrow L^2(\lambda)$ admits a bounded inverse, both as an operator on $L^2(\lambda)$ and on $L^2(\pi_0 \otimes \rho_0)$.*

Proof. Recall that $p_{k_0}^{\pi, \gamma}$ is the stationary state μ -density for the counterfactual MDP with policy π and kernel k_0 . We claim that $\mathcal{T}_{k_0, \pi}$ is a γ -contraction in $L^2(\pi \otimes p_{k_0}^{\pi, \gamma})$. Let $\mathcal{P}_{k_0, \pi}$ denote the Markov (transition) operator under k_0 and π , acting on state-only functions by

$$\mathcal{P}_{k_0, \pi} f(s) := \int f(s') k_0(ds' | s, a) \pi(da | s),$$

so that $\mathcal{T}_{k_0, \pi} := I - \gamma \mathcal{P}_{k_0, \pi}$.

For any $f \in L^2(p_{k_0}^{\pi, \gamma})$, Jensen's inequality gives

$$\|\mathcal{P}_{k_0, \pi} f\|_{L^2(p_{k_0}^{\pi, \gamma})}^2 = E_{S \sim p_{k_0}^{\pi, \gamma}} \left[(E[f(S') \mid S])^2 \right] \leq E_{S \sim p_{k_0}^{\pi, \gamma}} [E(f(S')^2 \mid S)] = E_{S' \sim p_{k_0}^{\pi, \gamma}} [f(S')^2] = \|f\|_{L^2(p_{k_0}^{\pi, \gamma})}^2,$$

where all expectations are taken under the stationary distribution $p_{k_0}^{\pi, \gamma}$, and the third equality follows from stationarity of the chain. Hence $\|\mathcal{P}_{k_0, \pi}\|_{L^2(p_{k_0}^{\pi, \gamma}) \rightarrow L^2(p_{k_0}^{\pi, \gamma})} \leq 1$, and therefore

$$\|\gamma \mathcal{P}_{k_0, \pi}\|_{L^2(p_{k_0}^{\pi, \gamma}) \rightarrow L^2(p_{k_0}^{\pi, \gamma})} \leq \gamma < 1.$$

Thus $\mathcal{T}_{k_0, \pi}$ is a perturbation of the identity by a γ -contraction with $\gamma < 1$. By Banach's inversion theorem, $\mathcal{T}_{k_0, \pi}$ is therefore invertible on $L^2(\pi \otimes p_{k_0}^{\pi, \gamma})$.

By [D1](#), the spaces $L^2(\pi \otimes p_{k_0}^{\pi, \gamma})$ and $L^2(\pi_0 \otimes \rho_0)$ are equivalent, with norms that are uniformly comparable. Hence $\mathcal{T}_{k_0, \pi}$ is also invertible on $L^2(\pi_0 \otimes \rho_0)$. Similarly, by the assumptions in [Section 2.3](#), both π_0 and ρ_0 are uniformly bounded above and below. It follows that $L^2(\pi_0 \otimes \rho_0)$ and $L^2(\lambda)$ are equivalent in norm and as spaces. Therefore $\mathcal{T}_{k_0, \pi}$ is also invertible on $L^2(\lambda)$. $\square$

In the following lemma, let P be a distribution over (A, S) , and factor it as $P(\{a\}, ds) = \pi_P(a \mid s) \rho_P(s) \mu(ds)$, where π_P is the induced policy and ρ_P is the marginal density of S . Let $\gamma \in [0, 1)$ and let $k \in \mathcal{K}$ be a transition kernel. We define the (unnormalized) discounted state-action occupancy density ratio as

$$d_{P, k}^{\pi}(a, s) := \rho_{P, k}^{\pi}(s) \frac{\pi(a \mid s)}{\pi_P(a \mid s)}, \quad \rho_{P, k}^{\pi}(s) := \sum_{t=0}^{\infty} \gamma^t \frac{d\mathbb{P}_{k, \pi}}{d\rho_P}(S_t = s), \quad (20)$$

where $\rho_{P, k}^{\pi}$ denotes the discounted state occupancy distribution under policy π and dynamics k .

Lemma 9 (Adjoint identities). *Let $\gamma \in [0, 1)$. Then, for any $f \in L^2(\lambda)$,*

$$E_P[(\Pi \mathcal{T}_{k, \pi, \gamma}^{-1} f)(S)] = E_P[d_{P, k}^{\pi}(A, S) f(A, S)],$$

$$E_P[(\mathcal{T}_{k, \pi, \gamma}^{-1} f)(A, S)] = E_P\left[\left\{1 - \frac{\pi(A|S)}{\pi_P(A|S)} + d_{P, k}^{\pi}(A, S)\right\} f(A, S)\right],$$

and consequently,

$$\begin{aligned} E_P[(I - \Pi) \mathcal{T}_{k, \pi, \gamma}^{-1} f(A, S)] &= E_P\left[\left\{1 - \frac{\pi(A|S)}{\pi_P(A|S)}\right\} f(A, S)\right] \\ &= E_P\left[f(A, S) - \sum_{a \in \mathcal{A}} \pi(a \mid S) f(a, S)\right]. \end{aligned}$$

Proof. Consider the inner product $\langle g, h \rangle_P := E_P[gh]$. Then

$$E_P[\Pi \mathcal{T}_{k, \pi, \gamma}^{-1} f] = \langle 1, \Pi \mathcal{T}_{k, \pi, \gamma}^{-1} f \rangle_P = \langle \Pi^* 1, \mathcal{T}_{k, \pi, \gamma}^{-1} f \rangle_P = \langle (\mathcal{T}_{k, \pi, \gamma}^{-1})^* (\Pi^* 1), f \rangle_P.$$

Under E_P we have $\Pi^*1(a, s) = \pi(a \mid s)/\pi_P(a \mid s)$. Expanding $(\mathcal{T}_{k,\pi,\gamma}^{-1})^*$ as a resolvent series gives

$$(\mathcal{T}_{k,\pi,\gamma}^{-1})^*(\Pi^*1)(a, s) = \sum_{t \geq 0} \gamma^t \frac{p_{k,\pi}(S_t = s)}{\rho_P(s)} \frac{\pi(a \mid s)}{\pi_P(a \mid s)} = d_{P,k}^\pi(a, s),$$

which proves the first identity.

For the second identity, note that

$$E_P[\mathcal{T}_{k,\pi,\gamma}^{-1}f] = \sum_{t \geq 0} \gamma^t \mathbb{E}[f(A_t, S_t)],$$

where the expectation is taken with $S_0 \sim \rho_P$, $A_0 \sim \pi_P(\cdot \mid S_0)$, the dynamics follow k , and for $t \geq 1$ the actions follow π . Split the sum into the $t = 0$ term and $\sum_{t \geq 1}$. The $t = 0$ contribution is $E_P[f(A, S)]$.

For $t \geq 1$, a change of measure back to E_P yields

$$\sum_{t \geq 1} \gamma^t E_P \left[\frac{p_{k,\pi}(S_t = S)}{\rho_P(S)} \frac{\pi(A \mid S)}{\pi_P(A \mid S)} f(A, S) \right] = E_P \left[(\rho_{P,k}^\pi(S) - 1) \frac{\pi(A \mid S)}{\pi_P(A \mid S)} f(A, S) \right].$$

Adding the $t = 0$ term, we obtain

$$E_P[\mathcal{T}_{k,\pi,\gamma}^{-1}f] = E_P \left[\left\{ 1 - \frac{\pi(A \mid S)}{\pi_P(A \mid S)} + \rho_{P,k}^\pi(S) \frac{\pi(A \mid S)}{\pi_P(A \mid S)} \right\} f(A, S) \right].$$

Since $d_{P,k}^\pi(A, S) = \rho_{P,k}^\pi(S) \frac{\pi(A \mid S)}{\pi_P(A \mid S)}$, this gives the desired expression. Subtracting the first identity from the second establishes the final identity. $\square$

Lemma 10 (Smoothness of Bellman operator). *Suppose $\mathcal{T}_{k,\pi,\gamma} : L^2(\lambda) \rightarrow L^2(\lambda)$ is invertible with bounded inverse $\mathcal{T}_{k,\pi,\gamma}^{-1}$. Then the map $(r, k) \mapsto \mathcal{T}_{k,\pi,\gamma}^{-1}r$ is Fréchet differentiable in L^2 , with partial derivatives, for $g \in \tilde{T}_K$ and $h \in L^2(\lambda)$,*

$$\tilde{\partial}_k(\mathcal{T}_{k,\pi,\gamma}^{-1})[g] = \gamma \mathcal{T}_{k,\pi,\gamma}^{-1}(\mathcal{P}_g \Pi) \mathcal{T}_{k,\pi,\gamma}^{-1}, \quad \partial_r(\mathcal{T}_{k,\pi,\gamma}^{-1}r)[h] = \mathcal{T}_{k,\pi,\gamma}^{-1}h.$$

Moreover,

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1}(r+h) - \mathcal{T}_{k,\pi,\gamma}^{-1}(r) - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g](r) - \partial_r(\mathcal{T}_{k,\pi,\gamma}^{-1}r)[h]\|_{L^2(\lambda)} \lesssim \|g\|_K^2 \|r\|_{L^2(\lambda)} + \|g\|_K \|h\|_{L^2(\lambda)}.$$

Proof. We have

$$\tilde{\partial}_k(\mathcal{T}_{k,\pi,\gamma}^{-1})[g] = -\mathcal{T}_{k,\pi,\gamma}^{-1}(\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}[g]) \mathcal{T}_{k,\pi,\gamma}^{-1},$$

where $\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}[g] = -\gamma \mathcal{P}_g \Pi$. Adding and subtracting terms gives

$$\begin{aligned} & \mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1})[g] \\ &= \mathcal{T}_{k,\pi,\gamma}^{-1} (\mathcal{T}_{k+g,\pi,\gamma} - \mathcal{T}_{k,\pi,\gamma}) \mathcal{T}_{k,\pi,\gamma}^{-1} (\mathcal{T}_{k+g,\pi,\gamma} - \mathcal{T}_{k,\pi,\gamma}) \mathcal{T}_{k+g,\pi,\gamma}^{-1} \\ &+ \mathcal{T}_{k,\pi,\gamma}^{-1} (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}[g] - (\mathcal{T}_{k+g,\pi,\gamma} - \mathcal{T}_{k,\pi,\gamma})) \mathcal{T}_{k,\pi,\gamma}^{-1}. \end{aligned}$$

$$\mathcal{T}_{k+g,\pi,\gamma} - \mathcal{T}_{k,\pi,\gamma} = -\gamma \mathcal{P}_g \Pi.$$

$$\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}[g] - (\mathcal{T}_{k+g,\pi,\gamma} - \mathcal{T}_{k,\pi,\gamma}) = 0,$$

since $k \mapsto \mathcal{T}_{k,\pi,\gamma} = I - \gamma \mathcal{P}_k \Pi$ is affine in k . Thus, the final term vanishes, and we obtain

$$\begin{aligned} & \mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1})[g] \\ &= \gamma^2 \mathcal{T}_{k,\pi,\gamma}^{-1} (\mathcal{P}_g \Pi) \mathcal{T}_{k,\pi,\gamma}^{-1} (\mathcal{P}_g \Pi) \mathcal{T}_{k+g,\pi,\gamma}^{-1}. \end{aligned}$$

Taking operator norms and using submultiplicativity,

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1})[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq \|\mathcal{T}_{k,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \|\mathcal{T}_{k+g,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{P}_g \Pi\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2.$$

We claim that $\|\mathcal{P}_g \Pi\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|g\|_{\mathcal{K}}$. Assuming this bound, we find that

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1})[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|\mathcal{T}_{k,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \|\mathcal{T}_{k+g,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|g\|_{\mathcal{K}}^2.$$

Finally, $\|\mathcal{T}_{k,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < \infty$ by definition, and by continuity of $k \mapsto \mathcal{T}_{k,\pi,\gamma}$ in operator norm we have that $\|\mathcal{T}_{k+g,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < \infty$ for all g in an $L^2(\lambda)$ ball around zero. Thus,

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1})[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = O(\|g\|_{\mathcal{K}}^2),$$

as claimed.

Finally, note that since $r \mapsto \mathcal{T}_{k,\pi,\gamma}^{-1} r$ is linear, we have $\partial_r (\mathcal{T}_{k,\pi,\gamma}^{-1} r)[h] = \mathcal{T}_{k,\pi,\gamma}^{-1} h$. Adding and subtracting,

$$\mathcal{T}_{k+g,\pi,\gamma}^{-1}(r+h) - \mathcal{T}_{k,\pi,\gamma}^{-1}(r) - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g](r) - \partial_r (\mathcal{T}_{k,\pi,\gamma}^{-1} r)[h] = \underbrace{(\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g])r}_{\text{2nd-order in } g} + \underbrace{(\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1})h}_{\text{1st-order in } g}.$$

Hence, by the triangle inequality and operator-norm submultiplicativity,

$$\begin{aligned} \|\mathcal{T}_{k+g,\pi,\gamma}^{-1}(r+h) - \mathcal{T}_{k,\pi,\gamma}^{-1}(r) - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g](r) - \mathcal{T}_{k,\pi,\gamma}^{-1} h\|_{L^2(\lambda)} &\leq \|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|r\|_{L^2(\lambda)} \\ &\quad + \|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|h\|_{L^2(\lambda)}. \end{aligned}$$

From above, we have

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1} - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|g\|_{\mathcal{K}}^2,$$

and, by the triangle inequality,

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} + \|g\|_{\mathcal{K}}^2 \lesssim \|g\|_{\mathcal{K}} + \|g\|_{\mathcal{K}}^2.$$

Hence,

$$\|\mathcal{T}_{k+g,\pi,\gamma}^{-1}(r+h) - \mathcal{T}_{k,\pi,\gamma}^{-1}(r) - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma}^{-1}[g](r) - \partial_r (\mathcal{T}_{k,\pi,\gamma}^{-1} r)[h]\|_{L^2(\lambda)} \lesssim \|g\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}} \|h\|_{L^2(\lambda)}.$$

It remains to prove the claim that $\|\mathcal{P}_g \Pi\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|g\|_{\mathcal{K}}$. For $(a, s) \in \mathcal{A} \times \mathcal{S}$, write

$$(\mathcal{P}_g \Pi f)(a, s) = \int_{\mathcal{S}} g(s' \mid a, s) h(s') \mu(ds'), \quad h(s') := \sum_{a' \in \mathcal{A}} \pi(a' \mid s') f(a', s').$$

By Cauchy–Schwarz in $L^2(\mu)$,

$$|(\mathcal{P}_g \Pi f)(a, s)| \leq \left(\int g(s' \mid a, s)^2 \mu(ds') \right)^{1/2} \left(\int h(s')^2 \mu(ds') \right)^{1/2}.$$

Next, by Cauchy–Schwarz over $a' \in \mathcal{A}$,

$$h(s')^2 \leq \left(\sum_{a'} \pi(a' \mid s')^2 \right) \left(\sum_{a'} f(a', s')^2 \right).$$

Integrating in s' and letting $c_\pi := \sup_{s'} \sum_{a'} \pi(a' \mid s')^2 \leq 1$, we get

$$\int h(s')^2 \mu(ds') \leq c_\pi \int \sum_{a'} f(a', s')^2 \mu(ds') = c_\pi \|f\|_{L^2(\#_{\mathcal{A}} \otimes \mu)}^2.$$

Therefore,

$$|(\mathcal{P}_g \Pi f)(a, s)|^2 \leq c_\pi \left(\int g(s' \mid a, s)^2 \mu(ds') \right) \|f\|_{L^2(\#_{\mathcal{A}} \otimes \mu)}^2.$$

Integrating over (a, s) with respect to $\lambda(da, ds)$ yields

$$\|\mathcal{P}_g \Pi f\|_{L^2(\lambda)}^2 \leq c_\pi \left(\iint g(s' \mid a, s)^2 \mu(ds') \lambda(da, ds) \right) \|f\|_{L^2(\#_{\mathcal{A}} \otimes \mu)}^2 = c_\pi \|g\|_{L^2(\lambda \otimes \mu)}^2 \|f\|_{L^2(\#_{\mathcal{A}} \otimes \mu)}^2.$$

Taking square roots and using that $\|f\|_{L^2(\#_{\mathcal{A}} \otimes \mu)}^2 \lesssim |\mathcal{A}| \cdot \|f\|_{L^2(\lambda)}^2$ gives

$$\|\mathcal{P}_g \Pi f\|_{L^2(\lambda)} \lesssim \|g\|_{L^2(\lambda \otimes \mu)} \|f\|_{L^2(\lambda)},$$

where $\lesssim$ only depends on $|\mathcal{A}|$. Thus, $\|\mathcal{P}_g \Pi\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|g\|_{\mathcal{K}}$. □

H.1.1 Lemmas for Example 2

For $(r, k) \in L^\infty(\lambda) \otimes \mathcal{K}$, define the two-resolvent value operator:

$$\mathcal{V}_k^g(r) := \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} \left\{ g + (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1} (r - g) \right\}.$$

Note that $\mathcal{V}_k^g(r)$ coincides with the value function of policy π under the ν -normalized reward $r_{r, k, \nu, \gamma, g} = g + (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1} (r - g)$.

Lemma 11 (Differentiability of the two-resolvent value operator). *Let π, ν be policies. Suppose $\mathcal{T}_{k, \pi, \gamma'} : L^2(\lambda) \rightarrow L^2(\lambda)$ and $\mathcal{T}_{k, \nu, \gamma} : L^2(\lambda) \rightarrow L^2(\lambda)$ are invertible with bounded inverses. Then the*

map $(r, k) \mapsto \mathcal{V}_k^g(r)$ defined by

$$\mathcal{V}_k^g(r) := \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} \left\{ g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right\}$$

is Fréchet differentiable in L^2 , with derivatives

$$\begin{aligned} \partial_r \mathcal{V}_k^g(r)[h] &= \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (h), \\ \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] &= \gamma' \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (\mathcal{P}_\eta \Pi) \mathcal{T}_{k,\pi,\gamma'}^{-1} \left\{ g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right\} \\ &\quad + \gamma \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_\eta \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g). \end{aligned}$$

Moreover, the second-order remainder satisfies

$$\|\mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] - \partial_r \mathcal{V}_k^g(r)[h]\|_{L^2(\lambda)} \lesssim \|\eta\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)} + \|\eta\|_{\mathcal{K}} \|h\|_{L^2(\lambda)},$$

where the implicit constant depends on operator norms of Π , $(I - \nu)$, and the inverses $\mathcal{T}_{k,\pi,\gamma'}^{-1}$ and $\mathcal{T}_{k,\nu,\gamma}^{-1}$.

Proof. Since g is fixed, it follows immediately that, for $h \in L^2(\lambda)$,

$$\partial_r \mathcal{V}_k^g(r)[h] = \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (h).$$

We now turn to the additive kernel derivative. By Lemma 10, we have, for $\eta \in \tilde{T}_{\mathcal{K}}$,

$$\tilde{\partial}_k (\mathcal{T}_{k,\nu,\gamma}^{-1})[\eta] = \gamma \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_\eta \nu) \mathcal{T}_{k,\nu,\gamma}^{-1}, \quad \tilde{\partial}_k \mathcal{P}_{k,\nu}[\eta] = \mathcal{P}_\eta \nu,$$

and similarly,

$$\tilde{\partial}_k (\mathcal{T}_{k,\pi,\gamma'}^{-1})[\eta] = \gamma' \mathcal{T}_{k,\pi,\gamma'}^{-1} (\mathcal{P}_\eta \Pi) \mathcal{T}_{k,\pi,\gamma'}^{-1}, \quad \tilde{\partial}_k (\mathcal{P}_k \Pi)[\eta] = \mathcal{P}_\eta \Pi.$$

By the chain rule and the above identities ,

$$\begin{aligned} \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] &= \Pi \left(\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta] \right) \left\{ g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right\} + \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \left(\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta] \right) (r - g) \\ &= \Pi \left[\gamma' \mathcal{T}_{k,\pi,\gamma'}^{-1} (\mathcal{P}_\eta \Pi) \mathcal{T}_{k,\pi,\gamma'}^{-1} \left\{ g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right\} + \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \gamma \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_\eta \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g) \right]. \end{aligned}$$

Equivalently, using $r_{r,k,\nu,\gamma,g} = g + (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g)$ and $q_{r-g,k}^{\nu,\gamma} = \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g)$,

$$\begin{aligned} \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] &= \gamma' \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (\mathcal{P}_\eta \Pi) \mathcal{T}_{k,\pi,\gamma'}^{-1} (r_{r,k,\nu,\gamma,g}) \\ &\quad + \gamma \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (\mathcal{P}_\eta \nu) q_{r-g,k}^{\nu,\gamma}. \end{aligned}$$

We now turn to the second-order remainder. Adding and subtracting, we have

$$\begin{aligned}\mathcal{V}_{k+\eta}^g(r) - \mathcal{V}_k^g(r) &= \Pi \left\{ \mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} \right\} (r - g) \\ &= \Pi \left\{ (\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1}) (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} + \mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}) \right\} (r - g).\end{aligned}$$

Subtracting the linear term

$$\tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] = \Pi \left\{ (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta]) (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} + \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta]) \right\} (r - g),$$

and adding and subtracting $\Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}) (r - g)$, we obtain

$$\begin{aligned}\mathcal{V}_{k+\eta}^g(r) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] &= \Pi \left\{ (\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} - \tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta]) (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} \right\} (r - g) \\ &\quad + \Pi \left\{ \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1} - \tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta]) \right\} (r - g) \\ &\quad + \Pi \left\{ (\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1}) (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}) \right\} (r - g).\end{aligned}$$

Term 1. By Lemma 10,

$$\|\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta])\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = O(\|\eta\|_{\mathcal{K}}^2).$$

Hence,

$$\begin{aligned}&\left\| \Pi \left\{ (\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta]) (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} \right\} (r - g) \right\|_{L^2(\lambda)} \\ &\leq \|\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1}[\eta])\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|(I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (r - g)\|_{L^2(\lambda)} \\ &= O(\|\eta\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)}),\end{aligned}$$

where the big-oh notation absorbs the constant $\|\mathcal{T}_{k,\nu,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$, and we use that $\|(I - \nu)\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq 1 + \|\nu\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < \infty$, since λ induces a uniform policy on $\mathcal{A}$.

Term 2. By Lemma 10,

$$\|\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta])\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = O(\|\eta\|_{\mathcal{K}}^2).$$

Hence,

$$\begin{aligned}&\left\| \Pi \left\{ \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta])) \right\} (r - g) \right\|_{L^2(\lambda)} \\ &\leq \|\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1}[\eta])\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (r - g)\|_{L^2(\lambda)} \\ &= O(\|\eta\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)}),\end{aligned}$$

where the big-oh notation absorbs the constant $\|\mathcal{T}_{k,\pi,\gamma'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$.

Term 3. By Lemma 10,

$$\|\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1})[\eta]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} = O(\|\eta\|_{\mathcal{K}}^2).$$

Moreover, by the triangle inequality,

$$\|\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|\eta\|_{\mathcal{K}},$$

since $\|(\tilde{\partial}_k \mathcal{T}_{k,\pi,\gamma'}^{-1})[\eta]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq C \|\eta\|_{\mathcal{K}}$ for some constant C . By a symmetric argument, we also obtain

$$\|\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|\eta\|_{\mathcal{K}}.$$

Hence, taking operator norms,

$$\begin{aligned} & \left\| \Pi \left\{ \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1} - (\tilde{\partial}_k \mathcal{T}_{k,\nu,\gamma}^{-1})[\eta]) \right. \right. \\ & \quad \left. \left. + (\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1}) (I - \nu) (\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}) \right\} (r - g) \right\|_{L^2(\lambda)} \\ & \lesssim \|\mathcal{T}_{k+\eta,\pi,\gamma'}^{-1} - \mathcal{T}_{k,\pi,\gamma'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{k+\eta,\nu,\gamma}^{-1} - \mathcal{T}_{k,\nu,\gamma}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|r\|_{L^2(\lambda)} \\ & \lesssim \|\eta\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)}. \end{aligned}$$

Finally, note that

$$\begin{aligned} & \mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] - \partial_r \mathcal{V}_k^g(r)[h] \\ & = \left\{ \mathcal{V}_{k+\eta}^g(r) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] \right\} + \left\{ \mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h] \right\}. \end{aligned}$$

We claim that

$$\|\mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h]\|_{L^2(\lambda)} \lesssim \|\eta\|_{\mathcal{K}} \|h\|_{L^2(\lambda)},$$

in which case

$$\begin{aligned} & \|\mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta] - \partial_r \mathcal{V}_k^g(r)[h]\|_{L^2(\lambda)} \\ & \leq \|\mathcal{V}_{k+\eta}^g(r) - \mathcal{V}_k^g(r) - \tilde{\partial}_k \mathcal{V}_k^g(r)[\eta]\|_{L^2(\lambda)} + \|\mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h]\|_{L^2(\lambda)} \\ & = O(\|\eta\|_{\mathcal{K}}^2 \|r\|_{L^2(\lambda)}) + O(\|\eta\|_{\mathcal{K}} \|h\|_{L^2(\lambda)}), \end{aligned}$$

and the result follows.

Proof of the claim. To bound the difference $\mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h]$, recall that

$$\partial_r \mathcal{V}_k^g(r)[h] = \Pi \mathcal{T}_{k,\pi,\gamma'}^{-1} (I - \nu) \mathcal{T}_{k,\nu,\gamma}^{-1} (h).$$

Hence,

$$\begin{aligned} \mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h] &= \Pi \left(\mathcal{T}_{k+\eta, \pi, \gamma'}^{-1} - \mathcal{T}_{k, \pi, \gamma'}^{-1} \right) (I - \nu) \mathcal{T}_{k+\eta, \nu, \gamma}^{-1}(h) \\ &\quad + \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} (I - \nu) \left(\mathcal{T}_{k+\eta, \nu, \gamma}^{-1} - \mathcal{T}_{k, \nu, \gamma}^{-1} \right) (h). \end{aligned}$$

Taking $L^2(\lambda)$ norms gives

$$\begin{aligned} \left\| \mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h] \right\|_{L^2(\lambda)} \\ \leq \left\| \mathcal{T}_{k+\eta, \pi, \gamma'}^{-1} - \mathcal{T}_{k, \pi, \gamma'}^{-1} \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \left\| (I - \nu) \mathcal{T}_{k+\eta, \nu, \gamma}^{-1}(h) \right\|_{L^2(\lambda)} \\ + \left\| \mathcal{T}_{k, \pi, \gamma'}^{-1} \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \left\| (I - \nu) \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \left\| \mathcal{T}_{k+\eta, \nu, \gamma}^{-1} - \mathcal{T}_{k, \nu, \gamma}^{-1} \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|h\|_{L^2(\lambda)}. \end{aligned}$$

By the Lipschitz continuity of the resolvent maps $k \mapsto \mathcal{T}_{k, \pi, \gamma'}^{-1}$ and $k \mapsto \mathcal{T}_{k, \nu, \gamma}^{-1}$, we have

$$\begin{aligned} \left\| \mathcal{T}_{k+\eta, \pi, \gamma'}^{-1} - \mathcal{T}_{k, \pi, \gamma'}^{-1} \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\lesssim \|\eta\|_{\mathcal{K}}, \\ \left\| \mathcal{T}_{k+\eta, \nu, \gamma}^{-1} - \mathcal{T}_{k, \nu, \gamma}^{-1} \right\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\lesssim \|\eta\|_{\mathcal{K}}. \end{aligned}$$

Substituting these bounds and absorbing the operator norms of $(I - \nu)$, Π , and $\mathcal{T}_{k, \pi, \gamma'}^{-1}$ into the implied constant, we conclude that

$$\left\| \mathcal{V}_{k+\eta}^g(r+h) - \mathcal{V}_{k+\eta}^g(r) - \partial_r \mathcal{V}_k^g(r)[h] \right\|_{L^2(\lambda)} \lesssim \|\eta\|_{\mathcal{K}} \|h\|_{L^2(\lambda)}.$$

□

H.1.2 Lemmas for Example 1b

In the following lemma, we recall the softmax policy $\pi_{r,k}^*$ and the solution $v_{r,k}^*$ of the soft Bellman equation from Example 1b. For each (r, k) , define the operator $(\Pi_{r,k}^* f)(s) := \sum_{\tilde{a} \in \mathcal{A}^*} \pi_{r,k}^*(\tilde{a} | s) f(\tilde{a}, s)$, and the log-partition function $\Phi_{r,k}^*(s) := \tau^* \log \left(\sum_{\tilde{a} \in \mathcal{A}^*} \exp \left((r(\tilde{a}, s) + \gamma v_{r,k}^*(\tilde{a}, s)) / \tau^* \right) \right)$. Define the conditional covariance under $\pi_{r,k}^*$ by

$$\text{Cov}_{\pi_{r,k}^*}(f, g)(s) := (\Pi_{r,k}^*[fg])(s) - (\Pi_{r,k}^* f)(s) (\Pi_{r,k}^* g)(s), \quad f, g \in L^\infty(\lambda).$$

Lemma 12 (Derivatives for soft Bellman operators). *The map $(r, k) \mapsto v_{r,k}^*$ is sup-norm Fréchet differentiable as a map from $L^\infty(\lambda) \otimes L^\infty(\mu \otimes \lambda)$ to $L^\infty(\lambda)$, with*

$$\partial_r v_{r,k}^*[h] = (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_k \Pi_{r,k}^* h, \quad \tilde{\partial}_k v_{r,k}^*[w] = (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_w \Phi_{r,k}^*.$$

Moreover, for any test function f ,

$$\partial_r \Pi_{r,k}^*[h] f = \frac{1}{\tau^*} \text{Cov}_{\pi_{r,k}^*}(f, (\mathcal{T}_{r,k}^*)^{-1} h), \quad \tilde{\partial}_k \Pi_{r,k}^*[w] f = \frac{1}{\tau^*} \text{Cov}_{\pi_{r,k}^*}(f, \gamma \tilde{\partial}_k v_{r,k}^*[w]).$$

Consequently,

$$\partial_r \mathcal{T}_{r,k}^* [h] f = -\frac{\gamma}{\tau^*} \mathcal{P}_k \text{Cov}_{\pi_{r,k}^*} (f, (\mathcal{T}_{r,k}^*)^{-1} h),$$

$$\tilde{\partial}_k \mathcal{T}_{r,k}^* [w] f = -\gamma \mathcal{P}_w \Pi_{r,k}^* f - \frac{\gamma}{\tau^*} \mathcal{P}_k \text{Cov}_{\pi_{r,k}^*} (f, \gamma \tilde{\partial}_k v_{r,k}^* [w]),$$

and

$$\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] [h] = \left(I + \gamma \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_k \right) (\partial_r \Pi_{r,k}^* [h]) (\mathcal{T}_{r,k}^*)^{-1}.$$

$$\tilde{\partial}_k [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] [w] = (I - \gamma \Pi_{r,k}^* \mathcal{P}_k)^{-1} \left[(\tilde{\partial}_k \Pi_{r,k}^* [w]) + \gamma \Pi_{r,k}^* \mathcal{P}_w \Pi_{r,k}^* \right] (\mathcal{T}_{r,k}^*)^{-1}.$$

Proof. This is the direct specialization of Appendix I.2 to the counterfactual system in Example 1b: replace the ambient action set $\mathcal{A}$ by $\mathcal{A}^*$ and the fixed temperature τ by τ^* in Lemmas 21, 22, and 28. This yields the displayed formulas for $\partial_r v_{r,k}^*$, $\tilde{\partial}_k v_{r,k}^*$, $\partial_r \Pi_{r,k}^*$, and $\tilde{\partial}_k \Pi_{r,k}^*$.

Since $\mathcal{T}_{r,k}^* = I - \gamma \mathcal{P}_k \Pi_{r,k}^*$, applying the chain rule yields

$$\partial_r \mathcal{T}_{r,k}^* [h] f = -\gamma \mathcal{P}_k (\partial_r \Pi_{r,k}^* [h]) f = -\frac{\gamma}{\tau^*} \mathcal{P}_k \text{Cov}_{\pi_{r,k}^*} (f, (\mathcal{T}_{r,k}^*)^{-1} h),$$

$$\begin{aligned} \tilde{\partial}_k \mathcal{T}_{r,k}^* [w] f &= -\gamma \mathcal{P}_w \Pi_{r,k}^* f - \gamma \mathcal{P}_k (\tilde{\partial}_k \Pi_{r,k}^* [w]) f \\ &= -\gamma \mathcal{P}_w \Pi_{r,k}^* f - \frac{\gamma}{\tau^*} \mathcal{P}_k \text{Cov}_{\pi_{r,k}^*} (f, \gamma \tilde{\partial}_k v_{r,k}^* [w]). \end{aligned}$$

This establishes the derivative identities for $\mathcal{T}_{r,k}^*$.

Let $G(r, k) := \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}$. Using the product rule and the inverse-derivative identity

$$\tilde{\partial}_k (\mathcal{T}_{r,k}^*)^{-1} [w] = -(\mathcal{T}_{r,k}^*)^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k}^* [w]) (\mathcal{T}_{r,k}^*)^{-1},$$

we get

$$\tilde{\partial}_k G[w] = (\tilde{\partial}_k \Pi_{r,k}^* [w]) (\mathcal{T}_{r,k}^*)^{-1} + \Pi_{r,k}^* \tilde{\partial}_k (\mathcal{T}_{r,k}^*)^{-1} [w].$$

Since $\mathcal{T}_{r,k}^* = I - \gamma \mathcal{P}_k \Pi_{r,k}^*$,

$$\tilde{\partial}_k \mathcal{T}_{r,k}^* [w] = -\gamma (\mathcal{P}_w \Pi_{r,k}^* + \mathcal{P}_k \tilde{\partial}_k \Pi_{r,k}^* [w]).$$

Substituting gives

$$\tilde{\partial}_k G[w] = (\tilde{\partial}_k \Pi_{r,k}^* [w]) (\mathcal{T}_{r,k}^*)^{-1} + \gamma \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} (\mathcal{P}_w \Pi_{r,k}^* + \mathcal{P}_k \tilde{\partial}_k \Pi_{r,k}^* [w]) (\mathcal{T}_{r,k}^*)^{-1}.$$

Regrouping,

$$\tilde{\partial}_k G[w] = \left(I + \gamma \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_k \right) (\tilde{\partial}_k \Pi_{r,k}^* [w]) (\mathcal{T}_{r,k}^*)^{-1} + \gamma \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_w \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}.$$

Finally, by the resolvent identity $(I - AB)^{-1} = I + A(I - BA)^{-1}B$ applied with $A = \gamma \Pi_{r,k}^*$, $B = \mathcal{P}_k$,

$$I + \gamma \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_k = (I - \gamma \Pi_{r,k}^* \mathcal{P}_k)^{-1}, \quad \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} = (I - \gamma \Pi_{r,k}^* \mathcal{P}_k)^{-1} \Pi_{r,k}^*,$$

which yields the stated formula. □

Lemma 13 (Linearization remainder for Example 1b). *Assume D2. Then there exists an L^2 -ball around $(r_0, k_0) \in \mathcal{H} \otimes \mathcal{K}$ such that $(\mathcal{T}_{r,k}^*)^{-1} : L^2(\lambda) \rightarrow L^2(\lambda)$ exists. Let $m(a, s, r, k) = \Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}(r)(s)$. Then*

$$\text{Rem}_0(r_0, k_0; r, k) \lesssim \|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2 + \|v_{r_0,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2,$$

with the implicit constant depending on γ , $\|r_0\|_{L^\infty(\lambda)}$, and $\|(\mathcal{T}_{r_0,k_0}^*)^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$.

Proof. For this proof, suppress the star and write $v_{r,k}$, $\Pi_{r,k}$, and $\mathcal{T}_{r,k}$ for the soft Bellman objects associated with the restricted action set $\mathcal{A}^*$ and fixed temperature τ^* from Example 1b. Then $m(a, s, r, k) = \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)(s)$.

By D2 and Lemma 8 applied with $\pi = \pi_{r_0,k_0}^*$, the inverse $\mathcal{T}_{r_0,k_0}^{-1} : L^2(\lambda) \rightarrow L^2(\lambda)$ exists. Moreover, the proofs of Lemmas 29–31 apply verbatim to this restricted-action, temperature- τ^* system, since Appendix I.2 was established for an arbitrary fixed temperature $\tau > 0$. In particular, there exists an L^2 -ball around $(r_0, k_0) \in \mathcal{H} \otimes \mathcal{K}$ such that $\mathcal{T}_{r,k}^{-1} : L^2(\lambda) \rightarrow L^2(\lambda)$ exists. By the assumed uniform boundedness of the state density ρ_0 , we have

$$\begin{aligned} & \left| E_0 \left[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0) - \partial_r [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [r - r_0] - \tilde{\partial}_k [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [k - k_0] \right] \right| \\ & \lesssim \left\| \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0) - \partial_r [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [r - r_0] - \tilde{\partial}_k [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [k - k_0] \right\|_{L^1(\lambda)}. \end{aligned}$$

By Lemma 31, we have

$$\begin{aligned} & \left\| \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0) - \partial_r [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [r - r_0] - \tilde{\partial}_k [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [k - k_0] \right\|_{L^1(\lambda)} \\ & = O \left(\|r - r_0\|_{L^2(\lambda)}^2 + \|v_{r,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2 + \|k - k_0\|_{\mathcal{K}}^2 \right). \end{aligned}$$

Furthermore, by Lemma 23,

$$\|v_{r,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2 \lesssim \|v_{r,k} - v_{r_0,k}\|_{L^2(\lambda)}^2 + \|v_{r_0,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2 \lesssim \|r - r_0\|_{L^2(\lambda)}^2 + \|v_{r_0,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2.$$

Hence,

$$\begin{aligned} & \left\| \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0) - \partial_r [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [r - r_0] - \tilde{\partial}_k [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [k - k_0] \right\|_{L^1(\lambda)} \\ & = O \left(\|r - r_0\|_{L^2(\lambda)}^2 + \|k - k_0\|_{\mathcal{K}}^2 + \|v_{r_0,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2 \right). \end{aligned}$$

We conclude that

$$\begin{aligned} & \left| E_0 \left[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0) - \partial_r [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [r - r_0] \right. \right. \\ & \quad \left. \left. - \tilde{\partial}_k [\Pi_{r_0,k_0} \mathcal{T}_{r_0,k_0}^{-1}(r_0)] [k - k_0] \right] \right| \\ & = O \left(\|r - r_0\|_{L^2(\lambda)}^2 + \|k - k_0\|_{\mathcal{K}}^2 + \|v_{r_0,k} - v_{r_0,k_0}\|_{L^2(\lambda)}^2 \right). \end{aligned}$$

and, hence,

$$\text{Rem}_0(r_0, k_0; r, k) \lesssim \|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2 + \|v_{r_0, k} - v_{r_0, k_0}\|_{L^2(\lambda)}^2.$$

□

H.2 Proofs of EIFs in examples

Proof of Theorem 9. Write $\lambda_0 := \pi_0 \otimes \rho_0$ and $\mathcal{T}_{k, \pi} := \mathcal{T}_{k, \pi, \gamma} = I - \gamma \mathcal{P}_k \Pi$. By D1 and Lemma 8, the linear operator $\mathcal{T}_{k_0, \pi} : L^2(\lambda) \rightarrow L^2(\lambda)$ is bounded and invertible, with Neumann-series inverse

$$\mathcal{T}_{k_0, \pi}^{-1} = \sum_{t=0}^{\infty} (\gamma \mathcal{P}_{k_0} \Pi)^t.$$

Hence,

$$m(a, s, r, k) = \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} \mid s) (\mathcal{T}_{k, \pi})^{-1}(r)(\tilde{a}, s) = \Pi[(\mathcal{T}_{k, \pi})^{-1}(r)](s)$$

is well defined for all $k \in \mathcal{K}$ and $r \in L^2(\lambda)$ in a ball around (r_0, k_0) .

By the chain rule, for each $h \in T_{\mathcal{H}}$ and $\beta \in T_{\mathcal{K}}$, we have

$$\begin{aligned} \partial_r m(A, S, r_0, k_0)[h] &= \Pi(\mathcal{T}_{k_0, \pi})^{-1}(h)(s), \\ \partial_k m(A, S, r_0, k_0)[\beta] &= \Pi(\tilde{\partial}_k \mathcal{T}_{k, \pi}^{-1}[k_0 \beta] \mid_{k=k_0})(r_0)(s) \\ &= \gamma \Pi \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)(s), \end{aligned}$$

where we used the additive derivative formula with kernel direction $g = k_0 \beta$, so that $\tilde{\partial}_k(\mathcal{T}_{k, \pi}^{-1})[k_0 \beta] = -\mathcal{T}_{k, \pi}^{-1}(\tilde{\partial}_k \mathcal{T}_{k, \pi}[k_0 \beta])\mathcal{T}_{k, \pi}^{-1}$ and $\tilde{\partial}_k \mathcal{T}_{k, \pi}[k_0 \beta] = -\gamma \mathcal{P}_{k_0 \beta} \Pi$.

Moreover, since Π , $\mathcal{P}_{k_0 \beta}$, and $\mathcal{T}_{k_0, \pi}^{-1}$ are all bounded linear operators on $L^2(\lambda)$, it follows that $h \mapsto E_0[\partial_r m(A, S, r_0, k_0)[h]]$ is a bounded linear functional on $(\mathcal{H}, \|\cdot\|_{\mathcal{H}})$. In addition, $\text{ess sup}_{a, s} \left| \frac{\pi_0(a|s)}{\exp\{r_0(a, s)\}} \right| < \infty$ since $r_0 \in L^\infty(\lambda)$, and hence we have $\|\cdot\|_{\mathcal{H}} \lesssim \|\cdot\|_{\mathcal{H}, 0}$, so it is also a bounded linear functional on $(T_{\mathcal{H}}, \|\cdot\|_{\mathcal{H}, 0})$. Furthermore, the map $\beta \mapsto \mathcal{P}_{k_0 \beta}$ is bounded in operator norm from $(T_{\mathcal{K}}, \|\cdot\|_{\mathcal{K}, 0})$ to the space of bounded linear operators on $L^2(\lambda)$. Hence, $\beta \mapsto E_0[\partial_k m(A, S, r_0, k_0)[\beta]]$ is a bounded linear functional, since, by properties of operator norms,

$$\begin{aligned} |E_0[\partial_k m(A, S, r_0, k_0)[\beta]]| &\leq \|\Pi \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)\|_{L^2(\rho_0 \otimes \pi_0)} \\ &\lesssim \|\Pi \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)\|_{L^2(\lambda)} \\ &\lesssim \|\mathcal{P}_{k_0 \beta}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)\|_{L^2(\lambda)} \\ &\lesssim \|\beta\|_{\mathcal{K}, 0}, \end{aligned}$$

where we used that the λ and λ_0 measures are equivalent by assumption. Hence, C1 holds under the stated conditions. Moreover, C3 follows from Lemma 10 applied with the triangle inequality and boundedness of the derivative.

We now determine the precise forms of the Riesz representers α_0 and β_0 appearing in Theorem

4. We begin with α_0 . For each $\alpha \in T_{\mathcal{H}}$, we have

$$\begin{aligned} E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] &= E_0[\Pi(\mathcal{T}_{k_0, \pi})^{-1}(\alpha)(S)] \\ &= E_0[d_0^{\pi, \gamma}(A, S) \alpha(A, S)], \end{aligned}$$

where the second equality follows from Lemma 9, and $d_0^{\pi, \gamma}$ is the discounted state-action occupancy ratio defined previously. It follows that

$$\alpha_0(a, s) = d_0^{\pi, \gamma}(a, s) = \rho_0^{\pi, \gamma}(s) \frac{\pi(a | s)}{\pi_0(a | s)}.$$

Next, we determine β_0 . Note

$$\begin{aligned} E_0[\partial_k m(A, S, r_0, k_0)[\beta]] &= \gamma E_0[\Pi \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)(S)] \\ &= \gamma \langle 1, \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0) \rangle_{\rho_0 \otimes \pi} \\ &= \gamma \langle \frac{\pi}{\pi_0}, \mathcal{T}_{k_0, \pi}^{-1} \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0) \rangle_{\lambda_0} \\ &= \gamma \langle (\mathcal{T}_{k_0, \pi}^{-1})^*(\frac{\pi}{\pi_0}), \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0) \rangle_{\lambda_0} \\ &= \gamma \langle d_0^{\pi, \gamma}, \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0) \rangle_{\lambda_0}, \end{aligned}$$

where $d_0^{\pi, \gamma} = (\mathcal{T}_{k_0, \pi}^{-1})^*(\frac{\pi}{\pi_0})$ by the adjoint Poisson equation. Recall that $V_{r_0, k_0}^{\pi, \gamma} = \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0)$ is the value function. Hence,

$$\begin{aligned} E_0[\partial_k m(A, S, r_0, k_0)[\beta]] &= \gamma \langle d_0^{\pi, \gamma}, \mathcal{P}_{k_0 \beta} \Pi \mathcal{T}_{k_0, \pi}^{-1}(r_0) \rangle_{\lambda_0} \\ &= \gamma \langle d_0^{\pi, \gamma}, \mathcal{P}_{k_0 \beta} V_{r_0, k_0}^{\pi, \gamma} \rangle_{\lambda_0} \\ &= E_0[\gamma d_0^{\pi, \gamma}(A, S) V_{r_0, k_0}^{\pi, \gamma}(S') \beta(S' | A, S)] \\ &= E_0[\gamma d_0^{\pi, \gamma}(A, S) \{V_{r_0, k_0}^{\pi, \gamma}(S') - \mathcal{P}_{k_0} V_{r_0, k_0}^{\pi, \gamma}(A, S)\} \beta(S' | A, S)]. \end{aligned}$$

It follows that

$$\beta_0(s' | a, s) = \gamma d_0^{\pi, \gamma}(a, s) \{V_{r_0, k_0}^{\pi, \gamma}(s') - \mathcal{P}_{k_0} V_{r_0, k_0}^{\pi, \gamma}(a, s)\}.$$

By Bellman's equation,

$$(\mathcal{P}_{k_0} V_{r_0, k_0}^{\pi, \gamma})(a, s) = \frac{q_{r_0, k_0}^{\pi, \gamma}(a, s) - r_0(a, s)}{\gamma}.$$

Substituting into the integrand,

$$\beta_0(s' | a, s) = d_0^{\pi, \gamma}(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^{\pi, \gamma}(s') - q_{r_0, k_0}^{\pi, \gamma}(a, s) \right\}.$$

Putting everything together, we conclude that the EIF of Ψ is

$$\begin{aligned}\chi_0(s, a, s') &= \alpha_0(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \alpha_0(\tilde{a}, s) + \beta_0(s' \mid a, s) + V_{r_0, k_0}^{\pi, \gamma}(s) - \Psi(P_0) \\ &= \alpha_0(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) \alpha_0(\tilde{a}, s) + d_0^{\pi, \gamma}(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^{\pi, \gamma}(s') - q_{r_0, k_0}^{\pi, \gamma}(a, s) \right\} \\ &\quad + V_{r_0, k_0}^{\pi, \gamma}(s) - \Psi(P_0).\end{aligned}$$

In the nonparametric case, where $\mathcal{H} = L^\infty(\lambda)$, we have that $\alpha_0 = d_0^{\pi, \gamma}$ and, hence,

$$\begin{aligned}\chi_0(s, a, s') &= d_0^{\pi, \gamma}(a, s) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) d_0^{\pi, \gamma}(\tilde{a}, s) \\ &\quad + d_0^{\pi, \gamma}(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^{\pi, \gamma}(s') - q_{r_0, k_0}^{\pi, \gamma}(a, s) \right\} + V_{r_0, k_0}^{\pi, \gamma}(s) - \Psi(P_0).\end{aligned}$$

The bound for the linearization remainder follows from Lemma 10. In particular, by boundedness of ρ_0 , we obtain

$$\begin{aligned}|\text{Rem}_0(r_0, k_0; r, k)| &\leq \|\mathcal{T}_{k, \pi}^{-1}(r) - \mathcal{T}_{k_0, \pi}^{-1}(r_0) - \tilde{\partial}_k \mathcal{T}_{k_0, \pi}^{-1}[k - k_0](r_0) - \partial_r(\mathcal{T}_{k_0, \pi}^{-1} r_0)[r - r_0]\|_{L^1(\pi_0 \otimes \rho_0)} \\ &\lesssim \|\mathcal{T}_{k, \pi}^{-1}(r) - \mathcal{T}_{k_0, \pi}^{-1}(r_0) - \tilde{\partial}_k \mathcal{T}_{k_0, \pi}^{-1}[k - k_0](r_0) - \partial_r(\mathcal{T}_{k_0, \pi}^{-1} r_0)[r - r_0]\|_{L^2(\lambda)} \\ &\lesssim \|k - k_0\|_{\mathcal{K}}^2 + \|k - k_0\|_{\mathcal{K}} \|r - r_0\|_{L^2(\lambda)}.\end{aligned}$$

□

Proof of Theorem 11. We define

$$m(a, s, r, k) := \sum_{\tilde{a} \in \mathcal{A}} \pi(\tilde{a} \mid s) \mathcal{T}_{k, \pi, \gamma'}^{-1}(r_{r, k, \nu, \gamma, g})(\tilde{a}, s), \quad r_{r, k, \nu, \gamma, g} = g + (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1}(r - g).$$

As shown in Appendix H.1.1, we can equivalently write

$$m(a, s, r, k) = \mathcal{V}_k^g(r)(s), \quad \mathcal{V}_k^g(r) := \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} \left\{ g + (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1}(r - g) \right\}.$$

By Conditions D1 and D4, together with Lemma 8 applied with $\pi = \pi$ and $\pi = \nu$, the operators $\mathcal{T}_{k_0, \pi, \gamma'} : L^2(\lambda) \rightarrow L^2(\lambda)$ and $\mathcal{T}_{k_0, \nu, \gamma} : L^2(\lambda) \rightarrow L^2(\lambda)$ are bounded and invertible. Thus $\mathcal{V}_k^g(r)$ is well defined in an L^2 -neighborhood of (r_0, k_0) . The Fréchet differentiability of m is established formally in Lemma 11. Specifically, applied with additive direction $k\beta$, that lemma yields the score derivative

$$\begin{aligned}\partial_r \mathcal{V}_k^g(r)[\alpha] &= \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1}(\alpha), \\ \partial_k \mathcal{V}_k^g(r)[\beta] &= \gamma' \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} (\mathcal{P}_{k\beta} \Pi) \mathcal{T}_{k, \pi, \gamma'}^{-1}(r_{r, k, \nu, \gamma, g}) \\ &\quad + \gamma \Pi \mathcal{T}_{k, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k, \nu, \gamma}^{-1} (\mathcal{P}_{k\beta} \nu) \mathcal{T}_{k, \nu, \gamma}^{-1}(r - g).\end{aligned}$$

Hence, for each $\alpha \in T_{\mathcal{H}}$ and $\beta \in T_{\mathcal{K}}$,

$$E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] = E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\alpha)(S) \right], \quad (21)$$

$$\begin{aligned} E_0[\partial_k m(A, S, r_0, k_0)[\beta]] &= \gamma' E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (\mathcal{P}_{k_0 \beta} \Pi) \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (r_0, \nu, \gamma, g)(S) \right] \\ &\quad + \gamma E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (r_0, \nu, \gamma, g)(S) \right] \end{aligned} \quad (22)$$

The linear functionals $\alpha \mapsto E_0[\partial_r m(A, S, r_0, k_0)[\alpha]]$ and $\beta \mapsto E_0[\partial_k m(A, S, r_0, k_0)[\beta]]$ are compositions of bounded operators and are therefore bounded. Hence, Condition C1 holds under the stated assumptions. Condition C3 then follows from the Fréchet differentiability of the map in Lemma 11, together with the triangle inequality and the boundedness of the derivatives.

We now turn to determining the specific form of the EIF. We begin by deriving the Riesz representer β_0 for the kernel partial derivative $E_0[\partial_k m(A, S, r_0, k_0)[\beta]]$. An argument identical to the proof of Theorem 9 shows that the first term on the right-hand side of (22) is

$$\begin{aligned} \gamma' E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (\mathcal{P}_{k_0 \beta} \Pi) \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (r_0, \nu, \gamma, g)(S) \right] &= \gamma' E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (\mathcal{P}_{k_0 \beta} \Pi) q_{r_0, k_0, \nu}^{\pi, \gamma'}(A, S) \right] \\ &= E_0[\beta_{0,1}(S', A, S) \beta(S', A, S)], \end{aligned} \quad (23)$$

where

$$\beta_{0,1}(s', a, s) := d_{0, k_0}^{\pi, \gamma'}(a, s) \{r_0, \nu, \gamma, g(a, s) + \gamma' V_{r_0, k_0, \nu}^{\pi, \gamma'}(s') - q_{r_0, k_0, \nu}^{\pi, \gamma'}(a, s)\}.$$

We now turn to the second term in (22):

$$\gamma E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (r_0 - g)(S) \right] = \gamma E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(S) \right], \quad (24)$$

where we used the identity $V_{r_0 - g, k_0}^{\nu, \gamma} = \nu \mathcal{T}_{k_0, \nu, \gamma}^{-1} (r_0 - g)$. By Lemma 9, the right-hand side of (24) satisfies

$$\gamma E_0 \left[\Pi \mathcal{T}_{k_0, \pi, \gamma'}^{-1} (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(S) \right] = \gamma E_0 \left[d_{0, k_0}^{\pi, \gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(A, S) \right].$$

Moreover, by a change of measure and another application of Lemma 9,

$$\begin{aligned} E_0 \left[d_{0, k_0}^{\pi, \gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(A, S) \right] &= \frac{1}{1 - \gamma'} E_{d_{0, k_0}^{\pi, \gamma'}} \left[(I - \nu) \mathcal{T}_{k_0, \nu, \gamma}^{-1} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(A, S) \right] \\ &= \frac{1}{1 - \gamma'} E_{d_{0, k_0}^{\pi, \gamma'}} \left[\left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(A, S) \right] \\ &= E_0 \left[d_{0, k_0}^{\pi, \gamma'}(A, S) \left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} (\mathcal{P}_{k_0 \beta} V_{r_0 - g, k_0}^{\nu, \gamma})(A, S) \right], \end{aligned} \quad (25)$$

where $E_{d_{0, k_0}^{\pi, \gamma'}}$ denotes expectation under the normalized discounted state-action occupancy measure, so the $(1 - \gamma')$ factors cancel.

Writing out the action of $\mathcal{P}_{k_0\beta}$, we have

$$\begin{aligned}
E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} (\mathcal{P}_{k_0\beta} V_{r_0-g,k_0}^{\nu,\gamma})(A, S) \right] \\
= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} V_{r_0-g,k_0}^{\nu,\gamma}(S') \beta(S' | A, S) \right] \\
= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} \{ V_{r_0-g,k_0}^{\nu,\gamma}(S') - \mathcal{P}_{k_0} V_{r_0-g,k_0}^{\nu,\gamma}(A, S) \} \beta(S' | A, S) \right].
\end{aligned} \tag{26}$$

Hence, combining (24)–(26),

$$\begin{aligned}
\gamma E_0 \left[\Pi \mathcal{T}_{k_0,\pi,\gamma'}^{-1}(I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(\mathcal{P}_{k_0\beta}\nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(r_0 - g) \right] &= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \left\{ 1 - \frac{\nu(A|S)}{\pi(A|S)} \right\} \right. \\
&\quad \left. \times \gamma \{ V_{r_0-g,k_0}^{\nu,\gamma}(S') - \mathcal{P}_{k_0} V_{r_0-g,k_0}^{\nu,\gamma}(A, S) \} \beta(S' | A, S) \right].
\end{aligned}$$

As in the proof of Theorem 9, Bellman's equation shows that

$$\gamma \{ V_{r_0-g,k_0}^{\nu,\gamma}(s') - \mathcal{P}_{k_0} V_{r_0-g,k_0}^{\nu,\gamma}(a, s) \} = r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s).$$

Thus,

$$\gamma E_0 \left[\Pi \mathcal{T}_{k_0,\pi,\gamma'}^{-1}(I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(\mathcal{P}_{k_0\beta}\nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(r_0 - g) \right] = E_0 [\beta_{0,2}(S' | A, S) \beta(S' | A, S)], \tag{27}$$

where

$$\beta_{0,2}(s' | a, s) := d_{0,k_0}^{\pi,\gamma'}(a, s) \left\{ 1 - \frac{\nu(a|s)}{\pi(a|s)} \right\} \{ r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s) \}.$$

Combining (23) and (27), we conclude that

$$E_0 [\partial_k m(A, S, r_0, k_0) [\beta]] = E_0 [\beta_0(S' | A, S) \beta(S' | A, S)],$$

where $\beta_0 := \beta_{0,1} + \beta_{0,2}$ and

$$\begin{aligned}
\beta_0(s' | a, s) &= d_{0,k_0}^{\pi,\gamma'}(a, s) \{ r_{0,\nu,\gamma,g}(a, s) + \gamma' V_{r_0,k_0,\nu}^{\pi,\gamma'}(s') - q_{r_0,k_0,\nu}^{\pi,\gamma'}(a, s) \} \\
&\quad + d_{0,k_0}^{\pi,\gamma'}(a, s) \left\{ 1 - \frac{\nu(a|s)}{\pi(a|s)} \right\} \{ r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s) \}.
\end{aligned} \tag{28}$$

Next, we turn to deriving the Riesz representer α_0 for the reward partial derivative $E_0[\partial_r m(A, S, r_0, k_0)[\alpha]]$. Denote $\alpha_{k_0,\nu} := (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha)$. Applying the identities in Lemma 9 to (21), we obtain

$$\begin{aligned}
E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] &= E_0 \left[\Pi \mathcal{T}_{k_0,\pi,\gamma'}^{-1}(\alpha_{k_0,\nu})(S) \right] \\
&= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \alpha_{k_0,\nu}(A, S) \right] \\
&= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha)(A, S) \right].
\end{aligned} \tag{29}$$

By the same argument used to derive (25)—namely, a change of measure together with Lemma 9—

we further obtain

$$\begin{aligned} E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1}(\alpha)(A, S) \right] &= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) (I - \nu) \alpha(A, S) \right] \\ &= E_0 \left[d_{0,k_0}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu(A|S)}{\pi(A|S)} \right) \alpha(A, S) \right]. \end{aligned}$$

Thus,

$$E_0[\partial_r m(A, S, r_0, k_0)[\alpha]] = E_0[\alpha_0(A, S)\alpha(A, S)], \quad (30)$$

where we define

$$\begin{aligned} \alpha_0(A, S) &:= d_{0,k_0}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu(A|S)}{\pi(A|S)} \right) \\ &= \rho_0^{\pi,\gamma'}(S) \left(\frac{\pi(A|S)}{\pi_0(A|S)} - \frac{\nu(A|S)}{\pi_0(A|S)} \right). \end{aligned}$$

We have

$$\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \frac{\pi(\tilde{a} | S)}{\pi_0(\tilde{a} | S)} = 1 \quad \text{and} \quad \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \frac{\nu(\tilde{a} | S)}{\pi_0(\tilde{a} | S)} = 1.$$

Hence,

$$\begin{aligned} \alpha_0(A, S) - \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | S) \alpha_0(\tilde{a}, S) &= \rho_0^{\pi,\gamma'}(S) \left[\frac{\pi(A | S)}{\pi_0(A | S)} - \frac{\nu(A | S)}{\pi_0(A | S)} - (1 - 1) \right] \\ &= \rho_0^{\pi,\gamma'}(S) \left(\frac{\pi(A | S)}{\pi_0(A | S)} - \frac{\nu(A | S)}{\pi_0(A | S)} \right) \\ &= \alpha_0(A, S). \end{aligned} \quad (31)$$

Thus, by substituting (31) and (28) into the expression for the EIF χ_0 given in Theorem 4, we obtain

$$\begin{aligned} \chi_0(s, a, s') &= \rho_0^{\pi,\gamma'}(s) \left(\frac{\pi(a | s)}{\pi_0(a | s)} - \frac{\nu(a | s)}{\pi_0(a | s)} \right) \\ &\quad + d_{0,k_0}^{\pi,\gamma'}(a, s) \left(1 - \frac{\nu(a | s)}{\pi(a | s)} \right) \{ r_0(a, s) - g(s) + \gamma V_{r_0-g,k_0}^{\nu,\gamma}(s') - q_{r_0-g,k_0}^{\nu,\gamma}(a, s) \} \\ &\quad + d_{0,k_0}^{\pi,\gamma'}(a, s) \{ r_{0,\nu,\gamma,g}(a, s) + \gamma' V_{r_0,k_0,\nu}^{\pi,\gamma'}(s') - q_{r_0,k_0,\nu}^{\pi,\gamma'}(a, s) \} \\ &\quad + m(a, s, r_0, k_0) - \Psi(P_0). \end{aligned} \quad (32)$$

The result then follows from Theorem 3, noting that

$$r_{0,\nu,\gamma,g}(a, s) = g(s) + q_{r_0-g,k_0}^{\nu,\gamma}(a, s) - V_{r_0-g,k_0}^{\nu,\gamma}(s).$$

□

Proof of Theorem 10. By Lemma 12, the map $(r, k) \mapsto E_0[m(A, S, r, k)]$ is Fréchet differentiable in the supremum norm and, hence, Gâteaux differentiable over $r \in \mathcal{H}$ and $k \in \mathcal{K}$. The derivatives

are provided in Lemma 12 and are stated later in this proof. All Bellman operators and policies below are the starred counterfactual objects from Example 1b, and we keep the stars throughout. By D2 and Lemma 8, applied with $\pi = \pi_{r_0, k_0}^*$, the operator $\mathcal{T}_{r_0, k_0}^* : L^2(\lambda) \rightarrow L^2(\lambda)$ admits a bounded inverse, both on $L^2(\lambda)$ and on $L^2(\pi_0 \otimes \rho_0)$. As a consequence, the partial derivatives in the sup-norm space,

$$h \mapsto E_0[\partial_r m(A, S, r, k)(h)] \quad \text{and} \quad \beta \mapsto E_0[\partial_k m(A, S, r, k)(\beta)],$$

admit continuous extensions on the respective domains $(T_{\mathcal{H}}, \|\cdot\|_{\mathcal{H},0})$ and $(T_{\mathcal{K}}, \|\cdot\|_{\mathcal{K},0})$. Furthermore, by Lemma 13, for all (r, k) in an L^2 -ball around (r_0, k_0) , the inverse $(\mathcal{T}_{r, k}^*)^{-1}$ exists, and the linearization remainder based on these derivatives satisfies

$$\text{Rem}_0(r_0, k_0; r, k) = O\left(\|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2 + \|v_{r_0, k} - v_{r_0, k_0}\|_{L^2(\lambda)}^2\right),$$

with an implicit constant depending only on $\max\{\|r\|_{L^\infty(\lambda)}, \|r_0\|_{L^\infty(\lambda)}\}$ and the constants in our assumptions.

Moreover, by D3, we obtain

$$\text{Rem}_0(r_0, k_0; r, k) = O(\|k - k_0\|_{\mathcal{K}}^2 + \|r - r_0\|_{\mathcal{H}}^2).$$

Thus, the map $(r, k) \mapsto E_0[m(A, S, r, k)]$ is Fréchet differentiable, satisfying

$$E_0\left[m(A, S, r + h, k + g) - m(A, S, r, k) - \partial_r m(A, S, r, k)(h) - \tilde{\partial}_k m(A, S, r, k)(g)\right] = o(\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}})$$

as $(h, g) \rightarrow (0, 0)$ in the product norm $\|h\|_{\mathcal{H},0} + \|g\|_{\mathcal{K}}$. Fréchet differentiability implies C1, and by the triangle inequality, the boundedness of the derivatives also implies C3. With these conditions verified, we now turn to determining the explicit form of the EIF.

Identities. We will use the following identities throughout. Arguing as in the proof of Lemma 9, for any state-only function $g \in L^2(\rho_0)$,

$$\begin{aligned} E_0\left[(I - \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0})^{-1} g(S)\right] &= E_0\left[\left(I + \gamma \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0}\right) g(S)\right] \\ &= \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{k_0, \pi_{r_0, k_0}^*} [g(S_t)] = E_0[g(S) \rho_{r_0, k_0}^*(S)]. \end{aligned} \tag{33}$$

The first equality in (33) uses the resolvent identity, the second uses the Neumann-series expansion of the inverse, and the last applies the Radon–Nikodym derivative to express discounted trajectory expectations as a reweighted expectation under the initial state distribution. Define

$$z_{r_0, k_0}^* := d_{r_0, k_0}^* (q_{r_0, k_0}^* - V_{r_0, k_0}^*).$$

By the adjoint definition in (13), for every state–action function h ,

$$E_0[z_{r_0, k_0}^*(A, S) \{(\mathcal{T}_{r_0, k_0}^*)^{-1} h\}(A, S)] = E_0[\tilde{d}_{r_0, k_0}^*(A, S) h(A, S)]. \tag{34}$$

Determining the α_0 component (partial derivative in r).

$$\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} (r)] \Big|_{(r_0, k_0)} [h] = \left(\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] \Big|_{(r_0, k_0)} [h] \right) r_0 + \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} (h).$$

Arguing exactly as in the proof of Theorem 9, it holds that

$$E_0 [\Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} (h)(S)] = E_0 [h(A, S) \alpha_{0,1}(A, S)], \quad (35)$$

where

$$\alpha_{0,1}(a, s) = d_{r_0, k_0}^* (a, s) = \rho_{r_0, k_0}^* (s) \frac{\pi_{r_0, k_0}^* (a | s)}{\pi_0(a | s)}.$$

By Lemma 12, we have

$$\begin{aligned} \partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] [h](r_0) \Big|_{(r_0, k_0)} &= \left(I + \gamma \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0} \right) \left(\partial_r \Pi_{r,k}^* [h] \right) \Big|_{(r_0, k_0)} (\mathcal{T}_{r_0, k_0}^*)^{-1} (r_0) \\ &= \left(I + \gamma \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0} \right) \frac{1}{\tau^*} \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0, k_0}^*)^{-1} h \right). \end{aligned}$$

Taking expectations, we have

$$E_0 \left[\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] [h](r_0) \Big|_{(r_0, k_0)} (S) \right] = E_0 \left[\left(I + \gamma \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0} \right) \frac{1}{\tau^*} \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0, k_0}^*)^{-1} h \right) (S) \right]$$

Using the identity in (33) with the state-only function

$$g(s) = \frac{1}{\tau^*} \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0, k_0}^*)^{-1} h \right) (s),$$

we obtain

$$\begin{aligned} E_0 \left[\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1}] [h](r_0) \Big|_{(r_0, k_0)} (S) \right] &= \frac{1}{\tau^*} E_0 \left[\left(I + \gamma \Pi_{r_0, k_0}^* (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0} \right) \right. \\ &\quad \left. \times \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0, k_0}^*)^{-1} h \right) (S) \right] \\ &= \frac{1}{\tau^*} E_0 \left[\rho_{r_0, k_0}^* (S) \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0, k_0}^*)^{-1} h \right) (S) \right]. \end{aligned}$$

Equivalently,

$$\begin{aligned}
& E_0 \left[\partial_r \left[\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \right] [h](r_0) \Big|_{(r_0,k_0)} (S) \right] \\
&= \frac{1}{\tau^*} E_0 \left[\rho_{r_0,k_0}^* (S) \text{Cov}_{\pi_{r_0,k_0}^*} \left((\mathcal{T}_{r_0,k_0}^*)^{-1} r_0, (\mathcal{T}_{r_0,k_0}^*)^{-1} h \right) (S) \right] \\
&= \frac{1}{\tau^*} E_0 \left[\rho_{r_0,k_0}^* (S) \Pi_{r_0,k_0}^* \left((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0 \right. \right. \\
&\quad \left. \left. \times (I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} h \right) (S) \right] \\
&= \frac{1}{\tau^*} E_0 \left[d_{r_0,k_0}^* (A, S) ((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0) (A, S) ((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} h) (A, S) \right] \\
&= \frac{1}{\tau^*} E_0 \left[d_{r_0,k_0}^* (A, S) ((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0) (A, S) ((\mathcal{T}_{r_0,k_0}^*)^{-1} h) (A, S) \right],
\end{aligned}$$

where the final equality uses that

$$\begin{aligned}
& E_0 \left[d_{r_0,k_0}^* (A, S) ((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0) (A, S) (\Pi_{r_0,k_0}^* (\mathcal{T}_{r_0,k_0}^*)^{-1} h) (S) \right] \\
&= \frac{1}{1 - \gamma} E_{d_{r_0,k_0}^*} \left[((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0) (A, S) (\Pi_{r_0,k_0}^* (\mathcal{T}_{r_0,k_0}^*)^{-1} h) (S) \right] \\
&= 0,
\end{aligned}$$

where $E_{d_{r_0,k_0}^*}$ is expectation under the probability distribution with density $(1 - \gamma)d_{r_0,k_0}^*$ relative to the law of (A, S) , and the final equality follows from the law of total expectation, since

$$E_{d_{r_0,k_0}^*} \left[((I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0) (A, S) \Big| S \right] = 0.$$

This holds because d_{r_0,k_0}^* is, up to normalization, a density ratio that reweights E_0 to the distribution induced by the policy π_{r_0,k_0}^* (with initial state distribution proportional to ρ_{r_0,k_0}^*).

Using that $q_{r_0,k_0}^* - V_{r_0,k_0}^* = (I - \Pi_{r_0,k_0}^*) (\mathcal{T}_{r_0,k_0}^*)^{-1} r_0$, we have

$$\begin{aligned}
E_0 \left[\partial_r \left[\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \right] [h](r_0) \Big|_{(r_0,k_0)} (S) \right] &= \frac{1}{\tau^*} E_0 \left[z_{r_0,k_0}^* (A, S) \{ (\mathcal{T}_{r_0,k_0}^*)^{-1} h \} (A, S) \right] \\
&= \frac{1}{\tau^*} E_0 \left[\tilde{d}_{r_0,k_0}^* (A, S) h(A, S) \right],
\end{aligned}$$

where the second equality is (34). Hence,

$$E_0 \left[\partial_r \left[\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} \right] [h](r_0) \Big|_{(r_0,k_0)} \right] = \frac{1}{\tau^*} E_0 [\tilde{d}_{r_0,k_0}^* (A, S) h(A, S)], \quad (36)$$

where

$$\alpha_{0,2}(a, s) := \frac{1}{\tau^*} \tilde{d}_{r_0,k_0}^* (a, s).$$

Combining (35) and (36), we conclude that

$$E_0 \left[\partial_r [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} (r)] \Big|_{(r_0, k_0)} [h] \right] = \langle h, \alpha_0 \rangle_{\mathcal{H}, 0},$$

where

$$\alpha_0(a, s) := d_{r_0, k_0}^*(a, s) + \frac{1}{\tau^*} \tilde{d}_{r_0, k_0}^*(a, s). \quad (37)$$

Determining component β_0 for k partial derivative. By Lemma 12, applied with additive kernel direction $k_0\beta$, we have

$$\partial_k [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} [\beta]] \Big|_{(r_0, k_0)} = (I - \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0})^{-1} \left[(\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta]) + \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta} \Pi_{r_0, k_0}^* \right] (\mathcal{T}_{r_0, k_0}^*)^{-1}.$$

Recall (33), which says that

$$E_0 \left[(I - \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0})^{-1} g(S) \right] = E_0 [g(S) \rho_{r_0, k_0}^*(S)].$$

Thus,

$$\begin{aligned} E_0 \left[\partial_k [\Pi_{r,k}^* (\mathcal{T}_{r,k}^*)^{-1} [\beta]] \Big|_{(r_0, k_0)} r_0(S) \right] &= E_0 \left[(I - \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0})^{-1} \left(\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta] + \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta} \Pi_{r_0, k_0}^* \right) (\mathcal{T}_{r_0, k_0}^*)^{-1} (r_0)(S) \right] \\ &= E_0 \left[\rho_{r_0, k_0}^*(S) \left(\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta] + \gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta} \Pi_{r_0, k_0}^* \right) (\mathcal{T}_{r_0, k_0}^*)^{-1} (r_0)(S) \right] \\ &= E_0 \left[\rho_{r_0, k_0}^*(S) (\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta]) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(S) \right] \\ &\quad + E_0 [\rho_{r_0, k_0}^*(S) (\gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta} \Pi_{r_0, k_0}^*) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(S)]. \end{aligned} \quad (38)$$

The first term on the right-hand side of (38) can be written as

$$\begin{aligned} E_0 \left[\rho_{r_0, k_0}^*(S) (\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta]) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(S) \right] &= \frac{1}{\tau^*} E_0 \left[\rho_{r_0, k_0}^*(S) \text{Cov}_{\pi_{r_0, k_0}^*} \left((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0, \gamma \tilde{\partial}_k v_{r_0, k_0}^* [k_0\beta] \right) (S) \right] \\ &= \frac{1}{\tau^*} E_0 \left[\rho_{r_0, k_0}^*(S) \left(\Pi_{r_0, k_0}^* [((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0) (\gamma \tilde{\partial}_k v_{r_0, k_0}^* [k_0\beta])] - \Pi_{r_0, k_0}^* [(\mathcal{T}_{r_0, k_0}^*)^{-1} r_0] \Pi_{r_0, k_0}^* [\gamma \tilde{\partial}_k v_{r_0, k_0}^* [k_0\beta]] \right) (S) \right] \\ &= \frac{\gamma}{\tau^*} E_0 \left[d_{r_0, k_0}^*(A, S) ((I - \Pi_{r_0, k_0}^*) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0))(A, S) (\tilde{\partial}_k v_{r_0, k_0}^* [k_0\beta])(A, S) \right], \end{aligned}$$

where $d_{r_0, k_0}^*(a, s) = \rho_{r_0, k_0}^*(s) \frac{\pi_{r_0, k_0}^*(a|s)}{\pi_0(a|s)}$.

Recall that $\tilde{\partial}_k v_{r,k}^* [k_0\beta] = (\mathcal{T}_{r,k}^*)^{-1} \mathcal{P}_{k_0\beta} \Phi_{r,k}^*$. Hence,

$$\begin{aligned} E_0 \left[\rho_{r_0, k_0}^*(S) (\tilde{\partial}_k \Pi_{r_0, k_0}^* [k_0\beta]) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(S) \right] &= \frac{\gamma}{\tau^*} E_0 \left[d_{r_0, k_0}^*(A, S) ((I - \Pi_{r_0, k_0}^*) (\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(A, S) ((\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0\beta} \Phi_{r_0, k_0}^*)(A, S) \right], \end{aligned}$$

where we recall that

$$\Phi_{r_0, k_0}^*(s) = \tau^* \log \left(\sum_{\tilde{a} \in \mathcal{A}^*} \exp((r_0(\tilde{a}, s) + \gamma v_{r_0, k_0}^*(\tilde{a}, s))/\tau^*) \right).$$

Applying (34) with $h = \mathcal{P}_{k_0\beta} \Phi_{r_0, k_0}^*$ gives

$$\begin{aligned} & \frac{\gamma}{\tau^*} E_0 \left[z_{r_0, k_0}^*(A, S) \{ (\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0\beta} \Phi_{r_0, k_0}^* \} (A, S) \right] \\ &= \frac{\gamma}{\tau^*} E_0 \left[\tilde{d}_{r_0, k_0}^*(A, S) \{ \mathcal{P}_{k_0\beta} \Phi_{r_0, k_0}^* \} (A, S) \right] \\ &= \frac{\gamma}{\tau^*} E_0 \left[\tilde{d}_{r_0, k_0}^*(A, S) \Phi_{r_0, k_0}^*(S') \beta(S' | A, S) \right]. \end{aligned}$$

The last equality follows from the definition of the additive kernel direction $k_0\beta$ and Fubini's theorem. Define

$$\begin{aligned} \beta_{0,1}(s' | a, s) &:= \frac{\gamma}{\tau^*} \tilde{d}_{r_0, k_0}^*(a, s) \\ &\quad \times \left\{ \Phi_{r_0, k_0}^*(s') - \int \Phi_{r_0, k_0}^*(\tilde{s}) k_0(\tilde{s} | a, s) \mu(d\tilde{s}) \right\}. \end{aligned}$$

Then,

$$\begin{aligned} & \frac{\gamma}{\tau^*} E_0 \left[d_{r_0, k_0}^*(A, S) ((I - \Pi_{r_0, k_0}^*)(\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(A, S) ((\mathcal{T}_{r_0, k_0}^*)^{-1} \mathcal{P}_{k_0\beta} \Phi_{r_0, k_0}^*)(A, S) \right] \\ &= \langle \beta_{0,1}, \beta \rangle_{P_0}. \end{aligned}$$

Further, by the soft Bellman equation,

$$V_{r_0, k_0}^{\star, \text{soft}}(s) = \Phi_{r_0, k_0}^*(s),$$

and

$$v_{r_0, k_0}^*(a, s) = \int V_{r_0, k_0}^*(s') k_0(s' | a, s) d\mu(s') = \int \Phi_{r_0, k_0}^*(s') k_0(s' | a, s) d\mu(s').$$

Hence,

$$\begin{aligned} \beta_{0,1}(s' | a, s) &= \frac{\gamma}{\tau^*} \tilde{d}_{r_0, k_0}^*(a, s) \left\{ \Phi_{r_0, k_0}^*(s') - \int \Phi_{r_0, k_0}^*(\tilde{s}) k_0(\tilde{s} | a, s) d\mu(\tilde{s}) \right\} \\ &= \frac{\gamma}{\tau^*} \tilde{d}_{r_0, k_0}^*(a, s) \left(V_{r_0, k_0}^{\star, \text{soft}}(s') - v_{r_0, k_0}^*(a, s) \right). \end{aligned}$$

Next, we consider the second term on the right-hand side of (38). Note that

$$\begin{aligned} E_0 \left[\rho_{r_0, k_0}^*(S) (\gamma \Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta} \Pi_{r_0, k_0}^*) ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0)(S) \right] &= \gamma E_0 \left[\rho_{r_0, k_0}^*(S) (\Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta}) \left(\Pi_{r_0, k_0}^* ((\mathcal{T}_{r_0, k_0}^*)^{-1} r_0) \right) (S) \right] \\ &= \gamma E_0 \left[\rho_{r_0, k_0}^*(S) (\Pi_{r_0, k_0}^* \mathcal{P}_{k_0\beta}) V_{r_0, k_0}^*(S) \right] \\ &= \gamma E_0 \left[\rho_{r_0, k_0}^*(S) \sum_{a \in \mathcal{A}^*} \pi_{r_0, k_0}^*(a | S) \int V_{r_0, k_0}^*(s') k_0(s' | a, S) \beta(s') \right] \\ &= \gamma E_0 \left[d_{r_0, k_0}^*(A, S) V_{r_0, k_0}^*(S') \beta(S' | A, S) \right] \\ &= \langle \beta_{0,2}, \beta \rangle_{P_0}, \end{aligned}$$

where

$$\begin{aligned}\beta_{0,2}(s' | a, s) &:= \gamma d_{r_0, k_0}^*(a, s) \\ &\times \left[V_{r_0, k_0}^*(s') - \int V_{r_0, k_0}^*(\tilde{s}) k_0(\tilde{s} | a, s) \mu(d\tilde{s}) \right].\end{aligned}$$

By Bellman's equation, we have

$$\gamma \left[V_{r_0, k_0}^*(S') - \int V_{r_0, k_0}^*(s') k_0(s' | A, S) d\mu(s') \right] = r_0(A, S) + \gamma V_{r_0, k_0}^*(S') - q_{r_0, k_0}^*(A, S).$$

Thus,

$$\beta_{0,2}(s' | a, s) := d_{r_0, k_0}^*(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^*(s') - q_{r_0, k_0}^*(a, s) \right\}.$$

Combining the identities above, we obtain from (38) that

$$E_0 \left[\partial_k \left[\Pi_{r, k}^* (\mathcal{T}_{r, k}^*)^{-1} \right] [\beta] \Big|_{(r_0, k_0)} r_0(S) \right] = \left\langle \beta_0, \beta \right\rangle_{P_0}, \quad (39)$$

where

$$\beta_0(s' | a, s) = \beta_{0,1}(s' | a, s) + \beta_{0,2}(s' | a, s).$$

Determining the nonparametric EIF. By Theorem 4, the nonparametric EIF is given by

$$\chi_0(s, a, s') = \alpha_0(a, s) - \sum_{a' \in \mathcal{A}} \pi_0(a' | s) \alpha_0(a', s) + \beta_0(s' | a, s) + m(a, s, r_0, k_0) - \Psi(P_0).$$

Moreover, using our expression for α_0 in (37), we can write

$$\alpha_0(a, s) - \sum_{a' \in \mathcal{A}} \pi_0(a' | s) \alpha_0(a', s) = d_{r_0, k_0}^*(a, s) + \frac{1}{\tau^\star} \tilde{d}_{r_0, k_0}^*(a, s) - \rho_{r_0, k_0}^*(s) - \frac{1}{\tau^\star} \tilde{\rho}_{r_0, k_0}^*(s).$$

Thus, substituting our expression for β_0 , we conclude that

$$\begin{aligned}\chi_0(s, a, s') &= d_{r_0, k_0}^*(a, s) + \frac{1}{\tau^\star} \tilde{d}_{r_0, k_0}^*(a, s) - \rho_{r_0, k_0}^*(s) - \frac{1}{\tau^\star} \tilde{\rho}_{r_0, k_0}^*(s) \\ &\quad + \frac{\gamma}{\tau^\star} \tilde{d}_{r_0, k_0}^*(a, s) \left(V_{r_0, k_0}^{\star, \text{soft}}(s') - v_{r_0, k_0}^*(a, s) \right) \\ &\quad + d_{r_0, k_0}^*(a, s) \left\{ r_0(a, s) + \gamma V_{r_0, k_0}^*(s') - q_{r_0, k_0}^*(a, s) \right\} \\ &\quad + m(a, s, r_0, k_0) - \Psi(P_0).\end{aligned}$$

□

Proof of Theorem 1. Write $\bar{r}_0 := r_{0, \nu, \gamma, g}$, and let

$$\tilde{m}(a, s, r, k) := V_{r, k}^{\star, \gamma'}(s), \quad m(a, s, r, k) := \tilde{m}(a, s, r_{r, k, \nu, \gamma, g}, k).$$

Under the stated assumptions, Theorem 10 applies with r_0 replaced by $\bar{r}_0$ and γ by γ' . Hence the

Riesz representers of the partial derivatives of $(r, k) \mapsto E_0[\tilde{m}(A, S, r, k)]$ at $(\bar{r}_0, k_0)$ are

$$\tilde{\alpha}_0(a, s) = d_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) + \frac{1}{\tau^\star} \tilde{d}_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s)$$

and

$$\begin{aligned} \tilde{\beta}_0(s' | a, s) &= \frac{\gamma'}{\tau^\star} \tilde{d}_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) \{ \Phi_{\bar{r}_0, k_0}^{\star, \gamma'}(s') - v_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) \} \\ &\quad + d_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) \{ \bar{r}_0(a, s) + \gamma' V_{\bar{r}_0, k_0}^{\star, \gamma'}(s') - q_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) \}. \end{aligned}$$

Applying Theorem 6 to the map $\tilde{m}$ gives

$$\alpha_0(a, s) = \tilde{\alpha}_0(a, s) - \frac{\nu(a | s)}{\pi_0(a | s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | s) \tilde{\alpha}_0(\tilde{a}, s),$$

and

$$\beta_0(s' | a, s) = \tilde{\beta}_0(s' | a, s) + \alpha_0(a, s) \{ r_0(a, s) - g(s) + \gamma V_{r_0-g, k_0}^{\nu, \gamma}(s') - q_{r_0-g, k_0}^{\nu, \gamma}(a, s) \}.$$

Since

$$\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | s) \tilde{\alpha}_0(\tilde{a}, s) = \left\{ \frac{1}{\tau^\star} \tilde{\rho}_{\bar{r}_0, k_0}^{\star, \gamma'}(s) + \rho_{\bar{r}_0, k_0}^{\star, \gamma'}(s) \right\},$$

we obtain

$$\alpha_0(a, s) = d_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) + \frac{1}{\tau^\star} \tilde{d}_{\bar{r}_0, k_0}^{\star, \gamma'}(a, s) - \frac{\nu(a | s)}{\pi_0(a | s)} \left\{ \rho_{\bar{r}_0, k_0}^{\star, \gamma'}(s) + \frac{1}{\tau^\star} \tilde{\rho}_{\bar{r}_0, k_0}^{\star, \gamma'}(s) \right\}.$$

It follows immediately that

$$\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | s) \alpha_0(\tilde{a}, s) = 0,$$

by the definitions of the state averages and because $\nu(\cdot | s)$ sums to one.

Finally, Theorem 4 yields

$$\chi_0(s, a, s') = \alpha_0(a, s) + \beta_0(s' | a, s) + V_{\bar{r}_0, k_0}^{\star, \gamma'}(s) - \Psi(P_0),$$

and substituting the expressions above gives the stated EIF. □

H.3 Functional von Mises expansions for examples

In the following lemmas, we define

$$V_q^\pi(s) := \sum_{a \in \mathcal{A}} \pi(a | s) q(a, s), \quad d_{\rho, r}^{\pi, \gamma}(a, s) := \rho(s) \frac{\pi(a | s)}{\pi_r(a | s)}, \quad \pi_r(a | s) := \exp\{r(a, s)\}.$$

Lemma 14 (von Mises expansion for policy value). *For any $r, q \in L^\infty(\lambda)$ and $\rho \in L^\infty(\mu)$,*

$$\begin{aligned} & E_0 \left[V_q^\pi(S) + d_{\rho,r}^{\pi,\gamma}(A, S) \{r(A, S) + \gamma V_q^\pi(S') - q(A, S)\} + \rho(S) \left\{ \frac{\pi}{\pi_r}(A | S) - 1 \right\} \right] - E_0[V_{r_0, k_0}^\pi(S)] \\ &= E_0 \left[\{d_0^{\pi,\gamma}(A, S) - d_{\rho,r}^{\pi,\gamma}(A, S)\} \{q(A, S) - q_{r_0, k_0}^{\pi,\gamma}(A, S) - \gamma(V_q^\pi(S') - V_{r_0, k_0}^{\pi,\gamma}(S'))\} \right] \\ & \quad + O(\|r - r_0\|_{\mathcal{H},0}^2), \end{aligned}$$

where the big-oh notation depends on $\|r_0\|_{L^\infty(\lambda)}$, $\|r\|_{L^\infty(\lambda)}$, and $\|\rho\|_{L^\infty(\mu)}$.

Proof. In what follows, we drop explicit dependence on π and γ in our notation, writing $q_0 := q_{r_0, k_0}^{\pi,\gamma}$ and $d_0 := d_0^{\pi,\gamma}$.

Consider the quantity

$$E_0[V_q^\pi(S)] - E_0[V_{q_0}^\pi(S)] + E_0[d_{\rho,r}(A, S) \{r(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] + E_0 \left[\rho(S) \left\{ \frac{\pi}{\pi_r}(A | S) - 1 \right\} \right]. \quad (40)$$

Adding and subtracting $r_0(A, S)$ inside the expectation, we obtain

$$\begin{aligned} E_0[d_{\rho,r}(A, S) \{r(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] &= E_0[d_{\rho,r}(A, S) \{r_0(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] \\ & \quad + E_0[d_{\rho,r}(A, S) \{r(A, S) - r_0(A, S)\}]. \end{aligned}$$

Hence, the quantity in (40) can be written

$$\begin{aligned} & E_0[V_q^\pi(S)] - E_0[V_{q_0}^\pi(S)] + E_0[d_{\rho,r}(A, S) \{r_0(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] \\ & \quad + E_0[d_{\rho,r}(A, S) \{r(A, S) - r_0(A, S)\}] + E_0 \left[\rho(S) \left\{ \frac{\pi}{\pi_r}(A | S) - 1 \right\} \right]. \end{aligned} \quad (41)$$

By Bellman's equation,

$$r_0(A, S) = q_0(A, S) - \gamma E_0[V_{q_0}^\pi(S') | A, S].$$

Thus, by the law of iterated expectations,

$$\begin{aligned} E_0[d_{\rho,r}^{\pi,\gamma}(A, S) \{r_0(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] &= E_0[d_{\rho,r}^{\pi,\gamma}(A, S) \{q_0(A, S) - \gamma V_{q_0}^\pi(S') + \gamma V_q^\pi(S') - q(A, S)\}] \\ &= E_0[d_{\rho,r}^{\pi,\gamma}(A, S) \{q_0(A, S) - q(A, S) + \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}]. \end{aligned}$$

Moreover, for any q ,

$$E_0[V_q^\pi(S)] = E_0[d_0(A, S) \{q(A, S) - \gamma V_q^\pi(S')\}].$$

Therefore,

$$E_0[V_q^\pi(S)] - E_0[V_{q_0}^\pi(S)] = E_0[d_0(A, S) \{q(A, S) - q_0(A, S) - \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}].$$

Combining these results, expression (41) can be written as

$$\begin{aligned}
& E_0[d_0(A, S)\{q(A, S) - q_0(A, S) - \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}] \\
& + E_0[d_{\rho,r}(A, S)\{q_0(A, S) - q(A, S) + \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}] \\
& + E_0[d_{\rho,r}(A, S)\{r(A, S) - r_0(A, S)\}] + E_0\left[\rho(S)\left\{\frac{\pi}{\pi_r}(A | S) - 1\right\}\right] \\
& = E_0[(d_0(A, S) - d_{\rho,r}(A, S))\{q(A, S) - q_0(A, S) - \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}] \\
& + E_0[d_{\rho,r}(A, S)\{r(A, S) - r_0(A, S)\}] + E_0\left[\rho(S)\left\{\frac{\pi}{\pi_r}(A | S) - 1\right\}\right].
\end{aligned} \tag{42}$$

We turn to the term $E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}]$. Since $\max\{\|r\|_{L^\infty(\lambda)}, \|r_0\|_{L^\infty(\lambda)}\} < M$, it follows that

$$|\exp\{r(A, S) - r_0(A, S)\} - 1 - \{r(A, S) - r_0(A, S)\}| \lesssim \{r(A, S) - r_0(A, S)\}^2,$$

with the constant depending only on M . Hence, recalling $\pi_r(a | s) = \exp\{r(a, s)\}$, we have

$$\begin{aligned}
E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}] &= E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{\exp\{r(A, S) - r_0(A, S)\} - 1\}] \\
&+ O(E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}^2]),
\end{aligned}$$

where the first term on the right-hand side satisfies

$$\begin{aligned}
E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{\exp\{r(A, S) - r_0(A, S)\} - 1\}] &= E_0\left[\rho(S)\left\{\frac{\pi}{\pi_0}(A | S) - \frac{\pi}{\pi_r}(A | S)\right\}\right] \\
&= E_0\left[\rho(S)\left\{1 - \frac{\pi}{\pi_r}(A | S)\right\}\right],
\end{aligned}$$

where we used the law of iterated expectations and the fact that $E_0\left[\frac{\pi}{\pi_0}(A | S) | S\right] = 1$. Hence,

$$\begin{aligned}
E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}] &= E_0\left[\rho(S)\left\{1 - \frac{\pi}{\pi_r}(A | S)\right\}\right] \\
&+ O(E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}^2]).
\end{aligned}$$

Equivalently,

$$\begin{aligned}
& E_0[d_{\rho,r}(A, S)\{r(A, S) - r_0(A, S)\}] + E_0\left[\rho(S)\left\{\frac{\pi}{\pi_r}(A | S) - 1\right\}\right] \\
& = O(E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}^2])
\end{aligned} \tag{43}$$

Plugging (43) into (42), we obtain

$$\begin{aligned}
& E_0[V_q^\pi(S)] - E_0[V_{q_0}^\pi(S)] + E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) + \gamma V_q^\pi(S') - q(A, S)\}] + E_0\left[\rho(S)\left\{\frac{\pi}{\pi_r}(A | S) - 1\right\}\right] \\
& = E_0[d_0(A, S)\{q(A, S) - q_0(A, S) - \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}] \\
& + E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{q_0(A, S) - q(A, S) + \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}] \\
& + O(E_0[d_{\rho,r}^{\pi,\gamma}(A, S)\{r(A, S) - r_0(A, S)\}^2]).
\end{aligned}$$

We conclude that

$$\begin{aligned}
& E_0[V_q^\pi(S)] - E_0[V_{q_0}^\pi(S)] + E_0[d_{\rho,r}^{\pi,\gamma}(A,S)\{r(A,S) + \gamma V_q^\pi(S') - q(A,S)\}] + E_0\left[\rho(S)\left\{\frac{\pi}{\pi_r}(A|S) - 1\right\}\right] \\
&= E_0\left[\{d_0(A,S) - d_{\rho,r}^{\pi,\gamma}(A,S)\}\{q(A,S) - q_0(A,S) - \gamma(V_q^\pi(S') - V_{q_0}^\pi(S'))\}\right] \\
&\quad + O\left(E_0[d_{\rho,r}^{\pi,\gamma}(A,S)\{r(A,S) - r_0(A,S)\}^2]\right).
\end{aligned}$$

Finally, since

$$d_{\rho,r}^{\pi,\gamma}(a,s) = \rho(s) \frac{\pi(a|s)}{\pi_r(a|s)} = \rho(s) \frac{\pi(a|s)}{\pi_0(a|s)} \exp\{r_0(a,s) - r(a,s)\},$$

the weight $d_{\rho,r}^{\pi,\gamma}$ is uniformly bounded by a constant depending only on $\|\rho\|_{L^\infty(\mu)}$, $\|r\|_{L^\infty(\lambda)}$, and $\|r_0\|_{L^\infty(\lambda)}$. Therefore,

$$E_0[d_{\rho,r}^{\pi,\gamma}(A,S)\{r(A,S) - r_0(A,S)\}^2] \lesssim \|r - r_0\|_{\mathcal{H},0}^2,$$

which yields the stated remainder bound. $\square$

Lemma 15 (von Mises expansion for policy value with normalization). *Let $\rho \in L^\infty(\mu)$, $r, q^{\nu,\gamma}, q_\nu^{\pi,\gamma'} \in L^\infty(\lambda)$. Denote $r_{\nu,g} := g + (I - \nu)q^{\nu,\gamma}$ and $r_{0,\nu,g} := g + (I - \nu)q_0^{\nu,\gamma}$. Then*

$$\begin{aligned}
& E_0\left[V_{q_\nu^{\pi,\gamma'}}^\pi(S) - V_{q_0^{\pi,\gamma'}}^\pi(S)\right] + E_0\left[d_{\rho,r}^{\pi,\gamma'}(A,S)\{r_{\nu,g}(A,S) + \gamma' V_{q_\nu^{\pi,\gamma'}}^\pi(S') - q_\nu^{\pi,\gamma'}(A,S)\}\right] \\
&+ E_0\left[d_{\rho,r}^{\pi,\gamma'}(A,S)\left(1 - \frac{\nu}{\pi}(A|S)\right)\{r(A,S) - g(S) + \gamma V_{q_\nu^{\pi,\gamma'}}^\pi(S') - q_\nu^{\pi,\gamma'}(A,S)\}\right] \\
&+ E_0\left[\rho(S)\left(\frac{\pi}{\pi_r}(A|S) - \frac{\nu}{\pi_r}(A|S)\right)\right] \\
&= E_0\left[(d_{\rho,r}^{\pi,\gamma'}(A,S) - d_0^{\pi,\gamma'}(A,S))\{\mathcal{T}_{k_0,\pi,\gamma'}(q_0^{\pi,\gamma'} - q_\nu^{\pi,\gamma'})(A,S)\}\right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}) \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2).
\end{aligned}$$

where the big- O constant depends on $\|\rho\|_{L^\infty(\mu)}$, $\|r\|_{L^\infty(\lambda)}$, $\|r_0\|_{L^\infty(\lambda)}$, and $\|\frac{\nu}{\pi}\|_{L^\infty(\lambda)}$.

Proof. Write

$$d_0^{\pi,\gamma'}(a,s) := \rho_0^{\pi,\gamma'}(s) \frac{\pi(a|s)}{\pi_0(a|s)}.$$

Applying the same decomposition as in the proof of Lemma 14, but with the reward term there

replaced by the surrogate normalized reward $r_{\nu,g}$, yields

$$\begin{aligned}
& E_0 \left[V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S) - V_{q_0^{\pi,\gamma'}}^{\pi}(S) \right] + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{ r_{\nu,g}(A, S) + \gamma' V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S') - q_{\nu}^{\pi,\gamma'}(A, S) \} \right] \\
& = E_0 \left[(d_{\rho,r}^{\pi,\gamma'}(A, S) - d_0^{\pi,\gamma'}(A, S)) \{ \mathcal{T}_{k_0,\pi,\gamma'}(q_0^{\pi,\gamma'} - q_{\nu}^{\pi,\gamma'})(A, S) \} \right] \\
& \quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{ r_{\nu,g}(A, S) - r_{0,\nu,g}(A, S) \} \right]. \tag{44}
\end{aligned}$$

Using that $\mathcal{T}_{k_0,\nu,\gamma}^{-1} \mathcal{T}_{k_0,\nu,\gamma} = I$, we have

$$\begin{aligned}
E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{ r_{\nu,g}(A, S) - r_{0,\nu,g}(A, S) \} \right] & = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (I - \nu)(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right]. \tag{45}
\end{aligned}$$

In the last equality, we argued as in the proof of Lemma 7 that, for any w and f ,

$$E_0 \left[w(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}^{-1} f(A, S) \right] = E_0 [w(A, S) (I - \nu) f(A, S)].$$

Continuing from (45) and adding and subtracting, we obtain

$$\begin{aligned}
& E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{ r_{\nu,g}(A, S) - r_{0,\nu,g}(A, S) \} \right] \\
& = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (1 - \frac{\nu}{\pi}(A | S)) \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& \quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left\{ \frac{\nu(A|S)}{\pi(A|S)} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) - (\nu \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma}))(S) \right\} \right]. \tag{46}
\end{aligned}$$

Note that

$$E_0 \left[d_{\rho,r_0}^{\pi,\gamma'}(A, S) \left\{ \frac{\nu(A|S)}{\pi(A|S)} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) - (\nu \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma}))(S) \right\} \right] = 0,$$

by a change of measure. Hence,

$$\begin{aligned}
& E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (I - \nu) \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (1 - \frac{\nu}{\pi}(A | S)) \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& \quad + E_0 \left[(d_{\rho,r}^{\pi,\gamma'} - d_{\rho,r_0}^{\pi,\gamma'})(A, S) \left\{ \frac{\nu(A|S)}{\pi(A|S)} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) - (\nu \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma}))(S) \right\} \right]. \tag{47}
\end{aligned}$$

Since $r_0 - g = \mathcal{T}_{k_0,\nu,\gamma}(q_0^{\nu,\gamma})$, by applying iterated expectations,

$$\begin{aligned}
& E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (1 - \frac{\nu}{\pi}(A | S)) \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) \right] \\
& = E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (1 - \frac{\nu}{\pi}(A | S)) \{ q^{\nu,\gamma}(A, S) - \gamma V_{q_{\nu}^{\pi,\gamma}}^{\pi}(S') - r_0(A, S) + g(S) \} \right]. \tag{48}
\end{aligned}$$

Furthermore, by Cauchy–Schwarz,

$$\begin{aligned}
& E_0 \left[(d_{\rho,r}^{\pi,\gamma'} - d_{\rho,r_0}^{\pi,\gamma'})(A, S) \left\{ \frac{\nu(A|S)}{\pi(A|S)} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) - (\nu \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma}))(S) \right\} \right] \\
&= O \left(\|d_{\rho,r}^{\pi,\gamma'} - d_{\rho,r_0}^{\pi,\gamma'}\|_{\mathcal{H},r_0} \left\| \frac{\nu(A|S)}{\pi(A|S)} \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})(A, S) - (\nu \mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma}))(S) \right\|_{\mathcal{H},r_0} \right) \\
&= O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}), \tag{49}
\end{aligned}$$

where the big- O constant depends on $\|\rho\|_{L^\infty(\mu)}$, $\|r\|_{L^\infty(\lambda)}$, $\|r_0\|_{L^\infty(\lambda)}$, and $\|\frac{\nu}{\pi}\|_{L^\infty(\lambda)}$. We also used that $\|\nu f\|_{\mathcal{H},r_0} \lesssim \|f\|_{\mathcal{H},r_0}$ by positivity of ν and π_0 . Combining (46)-(49), we obtain

$$\begin{aligned}
& E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{r_{\nu,g}(A, S) - r_{0,\nu,g}(A, S)\} \right] \\
&= E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) (1 - \frac{\nu}{\pi}(A | S)) \{q^{\nu,\gamma}(A, S) - \gamma V_{q^{\nu,\gamma}}^\nu(S') - r_0(A, S) + g(S)\} \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}). \tag{50}
\end{aligned}$$

Plugging (50) into (44), we obtain

$$\begin{aligned}
& E_0 \left[V_{q_\nu^{\pi,\gamma'}}^\pi(S) - V_{q_{0,\nu}^{\pi,\gamma'}}^\pi(S) \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{r_{\nu,g}(A, S) + \gamma' V_{q_\nu^{\pi,\gamma'}}^\pi(S') - q_\nu^{\pi,\gamma'}(A, S)\} \right] \\
&= E_0 \left[(d_{\rho,r}^{\pi,\gamma'}(A, S) - d_{0,\nu}^{\pi,\gamma'}(A, S)) \{q_{0,\nu}^{\pi,\gamma'}(A, S) - \gamma' V_{q_{0,\nu}^{\pi,\gamma'}}^\pi(S') + \gamma' V_{q_\nu^{\pi,\gamma'}}^\pi(S') - q_\nu^{\pi,\gamma'}(A, S)\} \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S)\right) \{q^{\nu,\gamma}(A, S) - \gamma V_{q^{\nu,\gamma}}^\nu(S') - r_0(A, S) + g(S)\} \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}) \\
&= E_0 \left[(d_{\rho,r}^{\pi,\gamma'}(A, S) - d_{0,\nu}^{\pi,\gamma'}(A, S)) \{q_{0,\nu}^{\pi,\gamma'}(A, S) - \gamma' V_{q_{0,\nu}^{\pi,\gamma'}}^\pi(S') + \gamma' V_{q_\nu^{\pi,\gamma'}}^\pi(S') - q_\nu^{\pi,\gamma'}(A, S)\} \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S)\right) \{q^{\nu,\gamma}(A, S) - \gamma V_{q^{\nu,\gamma}}^\nu(S') - r(A, S) + g(S)\} \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S)\right) \{r(A, S) - r_0(A, S)\} \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}). \tag{51}
\end{aligned}$$

where in the final equality we have added and subtracted $r(A, S) - g(S)$.

Arguing as in the derivation of (43) in the proof of Lemma 14, we obtain

$$\begin{aligned}
E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S) \right) \{r - r_0\}(A, S) \right] &= E_0 \left[\rho(S) \frac{\pi}{\pi_r}(A | S) \left(1 - \frac{\nu}{\pi}(A | S) \right) \{r - r_0\}(A, S) \right] \\
&= E_0 \left[\rho(S) \left(\frac{\pi}{\pi_r}(A | S) - \frac{\nu}{\pi_r}(A | S) \right) \left(\frac{\pi_r}{\pi_0} - 1 \right) (A, S) \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2) \\
&= E_0 \left[\rho(S) \left(\frac{\pi}{\pi_0}(A | S) - \frac{\nu}{\pi_0}(A | S) \right) \right] \\
&\quad - E_0 \left[\rho(S) \left(\frac{\pi}{\pi_r}(A | S) - \frac{\nu}{\pi_r}(A | S) \right) \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2) \\
&= -E_0 \left[\rho(S) \left(\frac{\pi}{\pi_r}(A | S) - \frac{\nu}{\pi_r}(A | S) \right) \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2). \tag{52}
\end{aligned}$$

where in the second equality we expanded $r - r_0$ around r_0 and collected higher-order terms. In the final step, we used the law of iterated expectations to show that

$$E_0 \left[\rho(S) \left(\frac{\pi}{\pi_0}(A | S) - \frac{\nu}{\pi_0}(A | S) \right) \right] = 0.$$

Plugging (52) into (51), we conclude that

$$\begin{aligned}
&E_0 \left[V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S) - V_{q_{0,\nu}^{\pi,\gamma'}}^{\pi}(S) \right] + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{r_{\nu,g}(A, S) + \gamma' V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S') - q_{\nu}^{\pi,\gamma'}(A, S)\} \right] \\
&= E_0 \left[(d_{\rho,r}^{\pi,\gamma'}(A, S) - d_0^{\pi,\gamma'}(A, S)) \{q_{0,\nu}^{\pi,\gamma'}(A, S) - \gamma' V_{q_{0,\nu}^{\pi,\gamma'}}^{\pi}(S') + \gamma' V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S') - q_{\nu}^{\pi,\gamma'}(A, S)\} \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S) \right) \{q^{\nu,\gamma}(A, S) - \gamma V_{q^{\nu,\gamma}}^{\nu}(S') - r(A, S) + g(S)\} \right] \\
&\quad - E_0 \left[\rho(S) \left(\frac{\pi}{\pi_r}(A | S) - \frac{\nu}{\pi_r}(A | S) \right) \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}) \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2).
\end{aligned}$$

Rearranging terms,

$$\begin{aligned}
&E_0 \left[V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S) - V_{q_{0,\nu}^{\pi,\gamma'}}^{\pi}(S) \right] + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \{r_{\nu,g}(A, S) + \gamma' V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S') - q_{\nu}^{\pi,\gamma'}(A, S)\} \right] \\
&\quad + E_0 \left[d_{\rho,r}^{\pi,\gamma'}(A, S) \left(1 - \frac{\nu}{\pi}(A | S) \right) \{r(A, S) - g(S) + \gamma V_{q^{\nu,\gamma}}^{\nu}(S') - q^{\nu,\gamma}(A, S)\} \right] \\
&\quad + E_0 \left[\rho(S) \left(\frac{\pi}{\pi_r}(A | S) - \frac{\nu}{\pi_r}(A | S) \right) \right] \\
&= E_0 \left[(d_{\rho,r}^{\pi,\gamma'}(A, S) - d_0^{\pi,\gamma'}(A, S)) \{q_{0,\nu}^{\pi,\gamma'}(A, S) - \gamma' V_{q_{0,\nu}^{\pi,\gamma'}}^{\pi}(S') + \gamma' V_{q_{\nu}^{\pi,\gamma'}}^{\pi}(S') - q_{\nu}^{\pi,\gamma'}(A, S)\} \right] \\
&\quad + O(\|r - r_0\|_{\mathcal{H},r_0} \|\mathcal{T}_{k_0,\nu,\gamma}(q^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H},r_0}) \\
&\quad + O(\|r - r_0\|_{\mathcal{H},0}^2).
\end{aligned}$$

□

For the next lemma, write

$$\pi_{r,v}^*(a \mid s) \propto \mathbf{1}\{a \in \mathcal{A}^*\} \exp[\tau^{*-1}\{r(a, s) + \gamma v(a, s)\}],$$

and define

$$\Phi_{r,v}^*(s) := \tau^* \log \sum_{a \in \mathcal{A}^*} \exp[\tau^{*-1}\{r(a, s) + \gamma v(a, s)\}], \quad V_{q,r,v}^*(s) := \sum_{a \in \mathcal{A}^*} \pi_{r,v}^*(a \mid s) q(a, s).$$

For arbitrary state-action functions d and $\tilde{d}$, set

$$\rho_{d,r}(s) := \sum_{a \in \mathcal{A}} \pi_r(a \mid s) d(a, s), \quad \tilde{\rho}_{\tilde{d},r}(s) := \sum_{a \in \mathcal{A}} \pi_r(a \mid s) \tilde{d}(a, s),$$

where $\pi_r = \exp(r)$. Throughout this lemma, r ranges over normalized log-policy functions, so $\sum_a \exp\{r(a, s)\} = 1$ almost surely.

Lemma 16 (von Mises expansion for soft optimal value). *Let $q, r, v, d, \tilde{d} \in L^\infty(\lambda)$, with r normalized as above, and define the uncentered score*

$$\begin{aligned} \varphi_{r,v,q,d,\tilde{d}}(s', a, s) &:= V_{q,r,v}^*(s) + d(a, s)\{r(a, s) + \gamma V_{q,r,v}^*(s') - q(a, s)\} \\ &\quad + d(a, s) - \rho_{d,r}(s) \\ &\quad + \frac{1}{\tau^*} \left[\tilde{d}(a, s) + \gamma \tilde{d}(a, s) \{\Phi_{r,v}^*(s') - v(a, s)\} - \tilde{\rho}_{\tilde{d},r}(s) \right]. \end{aligned}$$

Suppose these functions and their true counterparts lie in a common bounded $L^\infty(\lambda)$ ball. Then, in a sufficiently small neighborhood of the truth in $\|\cdot\|_{\mathcal{H},r_0}$,

$$\begin{aligned} \left| E_0\{\varphi_{r,v,q,d,\tilde{d}}(S', A, S)\} - \psi_0 \right| &\lesssim (\|d - d_{r_0,k_0}^*\|_{\mathcal{H},r_0} + \|r - r_0\|_{\mathcal{H},r_0} + \|v - v_{r_0,k_0}^*\|_{\mathcal{H},r_0}) \\ &\quad \times (\|\mathcal{T}_{r_0,k_0}^*(q - q_{r_0,k_0}^*)\|_{\mathcal{H},r_0} + \|r - r_0\|_{\mathcal{H},r_0} + \|v - v_{r_0,k_0}^*\|_{\mathcal{H},r_0}) \\ &\quad + \|\tilde{d} - \tilde{d}_{r_0,k_0}^*\|_{\mathcal{H},r_0} (\|r - r_0\|_{\mathcal{H},r_0} + \|v - v_{r_0,k_0}^*\|_{\mathcal{H},r_0}). \end{aligned} \quad (53)$$

The implicit constant depends only on the uniform bounds, γ , τ^ , and the local operator bounds used below.*

Proof. Abbreviate

$$q_0 := q_{r_0,k_0}^*, \quad v_0 := v_{r_0,k_0}^*, \quad d_0 := d_{r_0,k_0}^*, \quad \tilde{d}_0 := \tilde{d}_{r_0,k_0}^*,$$

and let $\pi^* := \pi_{r,v}^*$, $\pi_0^* := \pi_{r_0,v_0}^*$, $\Delta q := q - q_0$, and $\Delta Q := (r - r_0) + \gamma(v - v_0)$. Split the left-hand side of (53) into the terms linear in Δq , denoted R_q , and the remaining terms, denoted R_{pol} .

For the first part,

$$R_q = E_0[V_{\Delta q,r,v}^*(S)] + E_0[d(A, S)\{\gamma V_{\Delta q,r,v}^*(S') - \Delta q(A, S)\}].$$

The discounted occupancy identity gives

$$E_0[V_{\Delta q, r_0, v_0}^*(S)] + E_0[d_0(A, S)\{\gamma V_{\Delta q, r_0, v_0}^*(S') - \Delta q(A, S)\}] = 0.$$

Subtracting this identity, using Cauchy–Schwarz, and applying the local Lipschitz bound for the softmax map yields

$$|R_q| \lesssim \{\|d - d_0\|_{\mathcal{H}, r_0} + \|r - r_0\|_{\mathcal{H}, r_0} + \|v - v_0\|_{\mathcal{H}, r_0}\} \|\Delta q\|_{\mathcal{H}, r_0}.$$

By Condition D2 and Lemma 8, the inverse of $\mathcal{T}_{r_0, k_0}^*$ is locally bounded, so

$$\|\Delta q\|_{\mathcal{H}, r_0} \lesssim \|\mathcal{T}_{r_0, k_0}^* \Delta q\|_{\mathcal{H}, r_0}.$$

This proves the first product in (53) that contains the Bellman error.

It remains to control R_{pol} . The identity

$$E_0[d(A, S)\{r - r_0\}(A, S)] + E_0[d(A, S) - \rho_{d, r}(S)] = E_0[d(A, S)\{r - r_0 + 1 - \exp(r - r_0)\}(A, S)]$$

uses $\pi_r/\pi_0 = \exp(r - r_0)$. Taylor’s theorem therefore bounds this quantity by $O(\|r - r_0\|_{\mathcal{H}, r_0}^2)$, uniformly over bounded d .

The first-order change of the target softmax satisfies

$$\pi_{r, v}^* - \pi_0^* = \frac{\pi_0^*}{\tau^*} \{\Delta Q - \Pi_{r_0, k_0}^* \Delta Q\} + O(\|\Delta Q\|_{\mathcal{H}, r_0}^2).$$

Substitution into the plug-in and ordinary Bellman terms, followed by the discounted occupancy identity, gives the first-order policy contribution

$$\frac{1}{\tau^*} E_0[z_{r_0, k_0}^*(A, S) \Delta Q(A, S)].$$

Replacing d_0 by d in these terms contributes at most

$$O(\|d - d_0\|_{\mathcal{H}, r_0} \|\Delta Q\|_{\mathcal{H}, r_0}).$$

The second-order softmax remainder is $O(\|\Delta Q\|_{\mathcal{H}, r_0}^2)$; this follows from the bounded second derivative in Lemma 20.

For the signed augmentation, Taylor expansion of $\Phi_{r, v}^*$ and the same log-policy identity give

$$\begin{aligned} & \frac{1}{\tau^*} E_0 \left[\tilde{d}(A, S) + \gamma \tilde{d}(A, S) \{\Phi_{r, v}^*(S') - v(A, S)\} - \tilde{\rho}_{\tilde{d}, r}(S) \right] \\ &= -\frac{1}{\tau^*} E_0 \left[\tilde{d}(A, S) \{\mathcal{T}_{r_0, k_0}^* \Delta Q\}(A, S) \right] + O(\|\Delta Q\|_{\mathcal{H}, r_0}^2). \end{aligned}$$

By the defining adjoint identity (34),

$$E_0[\tilde{d}(A, S) \{\mathcal{T}_{r_0, k_0}^* \Delta Q\}(A, S)] = E_0[z_{r_0, k_0}^*(A, S) \Delta Q(A, S)].$$

The two first-order policy terms thus cancel. Cauchy–Schwarz bounds the effect of replacing $\tilde{d}_0$ by $\tilde{d}$ by

$$O\left(\|\tilde{d} - \tilde{d}_0\|_{\mathcal{H}, r_0} \{\|r - r_0\|_{\mathcal{H}, r_0} + \|v - v_0\|_{\mathcal{H}, r_0}\}\right).$$

Combining the bounds for R_q and R_{pol} proves (53). $\square$

H.4 Proofs for Section 6.2

Proof of Theorem 12. Note that $\psi_n = P_n \varphi_n$, where we define the estimated uncentered influence function by

$$\varphi_n(s', a, s) := V_n^{\pi, \gamma}(s) + \rho_n^{\pi, \gamma}(s) \frac{\pi(a | s)}{\pi_n(a | s)} \{r_n(a, s) + \gamma V_n^{\pi, \gamma}(s') - q_n^{\pi, \gamma}(a, s)\} + \rho_n^{\pi, \gamma}(s) \left\{ \frac{\pi(a | s)}{\pi_n(a | s)} - 1 \right\}.$$

Let φ_0 denote the true uncentered influence function φ_{r_0, k_0} . Then

$$\begin{aligned} \psi_n - \psi_0 &= P_n \varphi_n - P_0 \varphi_0 \\ &= P_n(\varphi_0 - P_0 \varphi_0) + P_n(\varphi_n - \varphi_0) \\ &= P_n \chi_0 + P_n(\varphi_n - \varphi_0) \\ &= P_n \chi_0 + (P_n - P_0)(\varphi_n - \varphi_0) + P_0(\varphi_n - \varphi_0) \\ &= P_n \chi_0 + (P_n - P_0)(\varphi_n - \varphi_0) + \{P_0 \varphi_n - \psi_0\}. \end{aligned}$$

By Assumptions E1 and E3, $\|\varphi_n - \varphi_0\|_{L^2(P_0)} = o_p(1)$. Moreover, by Assumption E2 and Markov's inequality,

$$(P_n - P_0)(\varphi_n - \varphi_0) = O_p\left(n^{-1/2} \|\varphi_n - \varphi_0\|_{L^2(P_0)}\right) = o_p(n^{-1/2}).$$

Thus,

$$\psi_n - \psi_0 = P_n \chi_0 + o_p(n^{-1/2}) + \{P_0 \varphi_n - \psi_0\}.$$

By Lemma 14, recalling that $d_{\rho_n, r_n}^{\pi, \gamma} = \rho_n^{\pi, \gamma} \frac{\pi}{\pi_n}$,

$$\begin{aligned} \{P_0 \varphi_n - \psi_0\} &= O\left(\langle d_{\rho_n, r_n}^{\pi, \gamma} - d_{\rho_0, r_0}^{\pi, \gamma}, \mathcal{T}_{k_0, \pi, \gamma}(q_n^{\pi, \gamma} - q_{r_0, k_0}^{\pi, \gamma}) \rangle_{\mathcal{H}, 0}\right) + O(\|r_n - r_0\|_{\mathcal{H}, 0}^2) \\ &= O\left(\|d_{\rho_n, r_n}^{\pi, \gamma} - d_{\rho_0, r_0}^{\pi, \gamma}\|_{\mathcal{H}, 0} \|\mathcal{T}_{k_0, \pi, \gamma}(q_n^{\pi, \gamma} - q_{r_0, k_0}^{\pi, \gamma})\|_{\mathcal{H}, 0}\right) + O(\|r_n - r_0\|_{\mathcal{H}, 0}^2). \end{aligned}$$

By Assumption E1,

$$\|d_{\rho_n, r_n}^{\pi, \gamma} - d_{\rho_0, r_0}^{\pi, \gamma}\|_{\mathcal{H}, 0} \lesssim \|\rho_n^{\pi, \gamma} - \rho_0^{\pi, \gamma}\|_{\mathcal{H}, 0} + \|r_n - r_0\|_{\mathcal{H}, 0}.$$

Hence,

$$\{P_0 \varphi_n - \psi_0\} = O\left(\{\|\rho_n^{\pi, \gamma} - \rho_0^{\pi, \gamma}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, 0}\} \|\mathcal{T}_{k_0, \pi, \gamma}(q_n^{\pi, \gamma} - q_0^{\pi, \gamma})\|_{\mathcal{H}, 0}\right) + O(\|r_n - r_0\|_{\mathcal{H}, 0}^2).$$

The first term is $o_p(n^{-1/2})$ by Assumption E1, and the second is $o_p(n^{-1/2})$ by C5(b). Therefore,

$$\psi_n - \psi_0 = P_n \chi_0 + o_p(n^{-1/2}),$$

and the result follows. $\square$

Proof of Theorem 13. Note that $\psi_n = P_n \varphi_n$, where we define the estimated uncentered influence function by

$$\begin{aligned} \varphi_n(s', a, s) &:= V_n^*(s) \\ &+ d_n^*(a, s) \{r_n(a, s) + \gamma V_n^*(s') - q_n^*(a, s)\} \\ &+ \frac{\gamma}{\tau^*} \tilde{d}_n^*(a, s) \left[\tau^* \log \left(\sum_{\tilde{a} \in \mathcal{A}^*} \exp \left(\frac{1}{\tau^*} (r_n(\tilde{a}, s') + \gamma v_n^*(\tilde{a}, s')) \right) \right) - v_n^*(a, s) \right] \\ &+ d_n^*(a, s) + \frac{1}{\tau^*} \tilde{d}_n^*(a, s) - \rho_n^*(s) - \frac{1}{\tau^*} \tilde{\rho}_n^*(s). \end{aligned}$$

Let φ_0 denote the true uncentered influence function corresponding to the nuisances $(r_0, v_{r_0, k_0}^*, q_{r_0, k_0}^*, d_{r_0, k_0}^*, \tilde{d}_{r_0, k_0}^*)$, so that $P_0 \varphi_0 = \psi_0$ and $P_0 \chi_0 = 0$, where χ_0 is the influence function in Theorem 10. Then, arguing as in the proof of Theorem 12,

$$\psi_n - \psi_0 = P_n \chi_0 + (P_n - P_0)(\varphi_n - \varphi_0) + \{P_0 \varphi_n - \psi_0\}.$$

By Assumptions E1* and E3*, $\|\varphi_n - \varphi_0\|_{L^2(P_0)} = o_p(1)$. Moreover, by Assumption E2* and Markov's inequality,

$$(P_n - P_0)(\varphi_n - \varphi_0) = O_p \left(n^{-1/2} \|\varphi_n - \varphi_0\|_{L^2(P_0)} \right) = o_p(n^{-1/2}).$$

Thus

$$\psi_n - \psi_0 = P_n \chi_0 + o_p(n^{-1/2}) + \{P_0 \varphi_n - \psi_0\}.$$

It remains to control the remainder term $P_0 \varphi_n - \psi_0$. By construction, φ_n equals $\varphi_{r_n, v_n^*, q_n^*, d_n^*, \tilde{d}_n^*}$ from Lemma 16, with $\rho_n^* = \rho_{d_n^*, r_n}$ and $\tilde{\rho}_n^* = \tilde{\rho}_{\tilde{d}_n^*, r_n}$. Consequently, (53) gives

$$\begin{aligned} |P_0 \varphi_n - \psi_0| &\lesssim (\|d_n^* - d_{r_0, k_0}^*\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^* - v_{r_0, k_0}^*\|_{\mathcal{H}, r_0}) \\ &\quad \times (\|\mathcal{T}_{r_0, k_0}^*(q_n^* - q_{r_0, k_0}^*)\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^* - v_{r_0, k_0}^*\|_{\mathcal{H}, r_0}) \\ &\quad + \|\tilde{d}_n^* - \tilde{d}_{r_0, k_0}^*\|_{\mathcal{H}, r_0} (\|r_n - r_0\|_{\mathcal{H}, r_0} + \|v_n^* - v_{r_0, k_0}^*\|_{\mathcal{H}, r_0}). \end{aligned}$$

The first product is $o_p(n^{-1/2})$ by Assumption E1*(c), and the second is $o_p(n^{-1/2})$ by Assumption E1*(b). Hence $\{P_0 \varphi_n - \psi_0\} = o_p(n^{-1/2})$, and therefore

$$\psi_n - \psi_0 = P_n \chi_0 + o_p(n^{-1/2}).$$

Theorem 8 then yields the claimed asymptotic linearity and efficiency with influence function χ_0 from Theorem 10, and the result follows. $\square$

Proof of Theorem 14. Note that $\psi_n = P_n \varphi_n$, where we define the estimated uncentered influence

function by

$$\begin{aligned}
\varphi_n(s', a, s) &:= V_{n,\nu}^{\pi,\gamma'}(s) \\
&+ \rho_n^{\pi,\gamma'}(s) \frac{\pi(a | s)}{\pi_n(a | s)} \{r_{n,\nu,\gamma,g}(a, s) + \gamma' V_{n,\nu}^{\pi,\gamma'}(s') - q_{n,\nu}^{\pi,\gamma'}(a, s)\} \\
&+ \rho_n^{\pi,\gamma'}(s) \left(\frac{\pi(a | s)}{\pi_n(a | s)} - \frac{\nu(a | s)}{\pi_n(a | s)} \right) \{r_n(a, s) - g(s) + \gamma V_n^{\nu,\gamma}(s') - q_n^{\nu,\gamma}(a, s)\} \\
&+ \rho_n^{\pi,\gamma'}(s) \left(\frac{\pi(a | s)}{\pi_n(a | s)} - \frac{\nu(a | s)}{\pi_n(a | s)} \right).
\end{aligned}$$

Let φ_0 denote the true uncentered influence function corresponding to the nuisances $(r_0, q_0^{\nu,\gamma}, q_{0,\nu}^{\pi,\gamma'}, \rho_0^{\pi,\gamma'})$, so that $P_0\varphi_0 = \psi_0$ and $P_0\chi_0 = 0$, where χ_0 is the influence function in Theorem 11. Write

$$V_{0,\nu}^{\pi,\gamma'}(s) := V_{q_{0,\nu}^{\pi,\gamma'}}^{\pi}(s).$$

Then

$$\psi_n - \psi_0 = P_n\chi_0 + (P_n - P_0)(\varphi_n - \varphi_0) + \{P_0\varphi_n - \psi_0\}.$$

By Assumptions E1** and E3**, $\|\varphi_n - \varphi_0\|_{L^2(P_0)} = o_p(1)$. Moreover, by Assumption E2** and Markov's inequality,

$$(P_n - P_0)(\varphi_n - \varphi_0) = O_p\left(n^{-1/2} \|\varphi_n - \varphi_0\|_{L^2(P_0)}\right) = o_p(n^{-1/2}).$$

Thus,

$$\psi_n - \psi_0 = P_n\chi_0 + o_p(n^{-1/2}) + \{P_0\varphi_n - \psi_0\}.$$

It remains to control the remainder term $P_0\varphi_n - \psi_0$. By construction of φ_n and ψ_n , and using the notation of Lemma 15, we may write

$$P_0\varphi_n - \psi_0 = E_0\left[V_{n,\nu}^{\pi,\gamma'}(S) - V_{0,\nu}^{\pi,\gamma'}(S)\right]$$

plus the three augmentation terms from Lemma 15, with

$$\rho = \rho_n^{\pi,\gamma'}, \quad r = r_n, \quad q^{\nu,\gamma} = q_n^{\nu,\gamma}, \quad q_\nu^{\pi,\gamma'} = q_{n,\nu}^{\pi,\gamma'}.$$

Recalling that $d_{\rho_n^{\pi,\gamma'}, r_n}^{\pi,\gamma'}(A, S)$ gathers $\rho_n^{\pi,\gamma'}(S)$ and the relevant importance ratio (cf. Example 2), Lemma 15 yields

$$\begin{aligned}
\{P_0\varphi_n - \psi_0\} &= E_0\left[\left(d_{\rho_n^{\pi,\gamma'}, r_n}^{\pi,\gamma'}(A, S) - d_{\rho_0^{\pi,\gamma'}, r_0}^{\pi,\gamma'}(A, S)\right) \{\mathcal{T}_{k_0, \pi, \gamma'}(q_{0,\nu}^{\pi,\gamma'} - q_{n,\nu}^{\pi,\gamma'})(A, S)\}\right] \\
&+ O(\|r_n - r_0\|_{\mathcal{H}, r_0} \|\mathcal{T}_{k_0, \nu, \gamma}(q_n^{\nu,\gamma} - q_0^{\nu,\gamma})\|_{\mathcal{H}, r_0}) \\
&+ O(\|r_n - r_0\|_{\mathcal{H}, r_0}^2).
\end{aligned}$$

By Cauchy–Schwarz and Assumption E1**,

$$\|d_{\rho_n^{\pi,\gamma'}, r_n}^{\pi,\gamma'} - d_{\rho_0^{\pi,\gamma'}, r_0}^{\pi,\gamma'}\|_{\mathcal{H}, r_0} \lesssim \|\rho_n^{\pi,\gamma'} - \rho_0^{\pi,\gamma'}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, r_0},$$

and therefore

$$\begin{aligned} \{P_0\varphi_n - \psi_0\} &= O\left(\left\{\|\rho_n^{\pi,\gamma'} - \rho_0^{\pi,\gamma'}\|_{L^2(\rho_0)} + \|r_n - r_0\|_{\mathcal{H}, r_0}\right\} \|\mathcal{T}_{k_0, \pi, \gamma'}(q_{n, \nu}^{\pi, \gamma'} - q_{0, \nu}^{\pi, \gamma'})\|_{\mathcal{H}, r_0}\right) \\ &\quad + O(\|r_n - r_0\|_{\mathcal{H}, r_0} \|\mathcal{T}_{k_0, \nu, \gamma'}(q_n^{\nu, \gamma} - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0}) + O(\|r_n - r_0\|_{\mathcal{H}, r_0}^2). \end{aligned}$$

By Assumption E1**(a),

$$\|\rho_n^{\pi,\gamma'} - \rho_0^{\pi,\gamma'}\|_{L^2(\rho_0)} \|\mathcal{T}_{k_0, \pi, \gamma'}(q_{n, \nu}^{\pi, \gamma'} - q_{0, \nu}^{\pi, \gamma'})\|_{\mathcal{H}, r_0} = o_p(n^{-1/2}).$$

By Assumption E1**(b),

$$\|r_n - r_0\|_{\mathcal{H}, r_0} \left\{ \|\mathcal{T}_{k_0, \nu, \gamma'}(q_n^{\nu, \gamma} - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0} + \|\mathcal{T}_{k_0, \pi, \gamma'}(q_{n, \nu}^{\pi, \gamma'} - q_{0, \nu}^{\pi, \gamma'})\|_{\mathcal{H}, r_0} \right\} = o_p(n^{-1/2}).$$

Finally, by C5(b),

$$\|r_n - r_0\|_{\mathcal{H}, r_0}^2 = o_p(n^{-1/2}).$$

Combining these bounds yields

$$\{P_0\varphi_n - \psi_0\} = o_p(n^{-1/2}).$$

Therefore,

$$\psi_n - \psi_0 = P_n\chi_0 + o_p(n^{-1/2}),$$

and Theorem 8 implies the claimed asymptotic linearity and efficiency with influence function χ_0 given in Theorem 11. This completes the proof. $\square$

H.5 Structural coefficients under reward normalization

Proof of Theorem 2. For each coordinate $j \in \{1, \dots, p\}$, define

$$\tilde{m}_j(a, s, r, k) := x_j(a, s) r(a, s), \quad b_j(P) := E_P[x_j(A, S) r_{P, \nu, \gamma, g}(A, S)].$$

Since

$$\partial_r \tilde{m}_j(a, s, r, k)[h] = x_j(a, s) h(a, s), \quad \partial_k \tilde{m}_j(a, s, r, k)[\beta] = 0,$$

the lower-level Riesz representers are

$$\tilde{\alpha}_{0, j}(a, s) = x_j(a, s), \quad \tilde{\beta}_{0, j} = 0.$$

Applying Theorem 6 coordinatewise therefore yields

$$\alpha_{0, j}^x(a, s) = x_j(a, s) - \frac{\nu(a \mid s)}{\pi_0(a \mid s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} \mid s) x_j(\tilde{a}, s)$$

and

$$\frac{\beta_{0,j}^x}{k_0}(s' | a, s) = \alpha_{0,j}^x(a, s) \{r_0(a, s) - g(s) + \gamma V_{r_0-g, k_0}^{\nu, \gamma}(s') - q_{r_0-g, k_0}^{\nu, \gamma}(a, s)\}.$$

Moreover,

$$\sum_{\tilde{a} \in \mathcal{A}} \pi_0(\tilde{a} | s) \alpha_{0,j}^x(\tilde{a}, s) = 0$$

for every j , so the generic EIF from Theorem 4 reduces to

$$\begin{aligned} \chi_{b_j, 0}(s, a, s') &= \alpha_{0,j}^x(a, s) \\ &\quad + \alpha_{0,j}^x(a, s) \{r_0(a, s) - g(s) + \gamma V_{r_0-g, k_0}^{\nu, \gamma}(s') - q_{r_0-g, k_0}^{\nu, \gamma}(a, s)\} \\ &\quad + x_j(a, s) \bar{r}_0(a, s) - b_{0,j}. \end{aligned}$$

Stacking the coordinates gives the stated vector influence function $\chi_{b, 0}$ for $b(P)$.

Since $\theta(P) = G(P)^{-1}b(P)$ with $G(P) = E_P[x(A, S)x(A, S)^\top]$, the matrix delta method gives, for perturbations $(\Delta G, \Delta b)$,

$$\partial \theta_{(G_0, b_0)}[\Delta G, \Delta b] = G_0^{-1} \Delta b - G_0^{-1} \Delta G \theta_0.$$

The influence function of $G(P)$ is $x(A, S)x(A, S)^\top - G_0$. Hence

$$\chi_{\theta, 0}(s, a, s') = G_0^{-1} \left[\chi_{b, 0}(s, a, s') - \{x(a, s)x(a, s)^\top - G_0\} \theta_0 \right],$$

which is the desired EIF. □

Lemma 17 (von Mises expansion for structural coefficient moments). *Let $r, q \in L^\infty(\lambda)$, write*

$$q_0^{\nu, \gamma} := q_{r_0-g, k_0}^{\nu, \gamma}, \quad \bar{r}_q := g + q - V_q^\nu,$$

and define

$$\alpha_r^x(a, s) := x(a, s) - \frac{\nu(a | s)}{\pi_r(a | s)} \sum_{\tilde{a} \in \mathcal{A}} \pi_r(\tilde{a} | s) x(\tilde{a}, s), \quad \pi_r(a | s) := \exp\{r(a, s)\}.$$

Then

$$\begin{aligned} &\left\| E_0[x(A, S) \bar{r}_q(A, S)] - b_0 + E_0[\alpha_r^x(A, S)] \right. \\ &\quad \left. + E_0[\alpha_r^x(A, S) \{r(A, S) - g(S) + \gamma V_q^\nu(S') - q(A, S)\}] \right\|_2 \\ &\lesssim \|r - r_0\|_{\mathcal{H}, r_0} \|\mathcal{T}_{k_0, \nu, \gamma}(q - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0} + \|r - r_0\|_{\mathcal{H}, 0}^2, \end{aligned}$$

where the implicit constant depends only on $\|x\|_{L^\infty(\lambda)}$, $\|r\|_{L^\infty(\lambda)}$, and $\|r_0\|_{L^\infty(\lambda)}$.

Proof. Since

$$\bar{r}_q - \bar{r}_0 = (I - \nu)(q - q_0^{\nu, \gamma}),$$

the same identity used in the proof of Lemma 15 gives

$$E_0[x(A, S)\{\bar{r}_q - \bar{r}_0\}(A, S)] = E_0[x(A, S)(I - \nu) \mathcal{T}_{k_0, \nu, \gamma}(q - q_0^{\nu, \gamma})(A, S)].$$

Define

$$\delta_q := \mathcal{T}_{k_0, \nu, \gamma}(q - q_0^{\nu, \gamma}) = q - \gamma V_q^\nu - r_0 + g.$$

By the definition of α_0^x ,

$$E_0[x(A, S)(I - \nu)\delta_q(A, S)] = E_0[\alpha_0^x(A, S)\delta_q(A, S)].$$

Therefore,

$$E_0[x(A, S)\{\bar{r}_q - \bar{r}_0\}(A, S)] = E_0[\alpha_r^x(A, S)\delta_q(A, S)] + E_0[(\alpha_0^x - \alpha_r^x)(A, S)\delta_q(A, S)].$$

Since

$$\delta_q = \{q - \gamma V_q^\nu - r + g\} + (r - r_0),$$

we obtain

$$\begin{aligned} E_0[x(A, S)\{\bar{r}_q - \bar{r}_0\}(A, S)] &= E_0[\alpha_r^x(A, S)\{q(A, S) - \gamma V_q^\nu(S') - r(A, S) + g(S)\}] \\ &\quad + E_0[\alpha_r^x(A, S)\{r - r_0\}(A, S)] \\ &\quad + E_0[(\alpha_0^x - \alpha_r^x)(A, S)\delta_q(A, S)]. \end{aligned}$$

By boundedness of x and the first-order expansion of π_r around π_0 ,

$$\|\alpha_r^x - \alpha_0^x\|_{\mathcal{H}, r_0} \lesssim \|r - r_0\|_{\mathcal{H}, r_0},$$

so Cauchy–Schwarz gives

$$E_0[(\alpha_0^x - \alpha_r^x)(A, S)\delta_q(A, S)] = O\left(\|r - r_0\|_{\mathcal{H}, r_0} \|\mathcal{T}_{k_0, \nu, \gamma}(q - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0}\right).$$

It remains to control the reward term. By the second-order expansion of $\exp\{r - r_0\}$ around zero,

$$\{r - r_0\}(A, S) = \frac{\pi_r}{\pi_0}(A \mid S) - 1 + O(\|r - r_0\|_{\mathcal{H}, 0}^2),$$

and therefore

$$\begin{aligned} E_0[\alpha_r^x(A, S)\{r - r_0\}(A, S)] + E_0[\alpha_r^x(A, S)] &= E_0\left[\alpha_r^x(A, S)\frac{\pi_r}{\pi_0}(A \mid S)\right] + O(\|r - r_0\|_{\mathcal{H}, 0}^2) \\ &= E_0\left[\sum_{a \in \mathcal{A}} \pi_r(a \mid S)\alpha_r^x(a, S)\right] + O(\|r - r_0\|_{\mathcal{H}, 0}^2) \\ &= O(\|r - r_0\|_{\mathcal{H}, 0}^2), \end{aligned}$$

because $\sum_{a \in \mathcal{A}} \pi_r(a \mid s)\alpha_r^x(a, s) = 0$ by construction. Combining the displays above yields the

claim. □

Proof of Theorem 3. Define the estimated uncentered influence function for b_0 by

$$\begin{aligned}\varphi_n^b(s', a, s) := & x(a, s)\bar{r}_n(a, s) + \alpha_n^x(a, s) \\ & + \alpha_n^x(a, s)\{r_n(a, s) - g(s) + \gamma V_n^{\nu, \gamma}(s') - q_n^{\nu, \gamma}(a, s)\},\end{aligned}$$

and let

$$\varphi_0^b(s', a, s) := \chi_{b,0}(s, a, s') + b_0.$$

Then $b_n = P_n \varphi_n^b$, $P_0 \varphi_0^b = b_0$, and

$$b_n - b_0 = P_n \chi_{b,0} + (P_n - P_0)(\varphi_n^b - \varphi_0^b) + \{P_0 \varphi_n^b - b_0\}.$$

By Assumptions E2*** and E4***,

$$\|\varphi_n^b - \varphi_0^b\|_{L^2(P_0)} = o_p(1).$$

Here $\|f\|_{L^2(P_0)} := \{E_0[\|f(S', A, S)\|_2^2]\}^{1/2}$ for vector-valued f . Since p is fixed, Assumption E3*** and Markov's inequality imply

$$\|(P_n - P_0)(\varphi_n^b - \varphi_0^b)\|_2 = O_p\left(n^{-1/2} \|\varphi_n^b - \varphi_0^b\|_{L^2(P_0)}\right) = o_p(n^{-1/2}).$$

Applying Lemma 17 with $r = r_n$ and $q = q_n^{\nu, \gamma}$ gives

$$\|P_0 \varphi_n^b - b_0\|_2 = O_p\left(\|r_n - r_0\|_{\mathcal{H}, r_0} \|\mathcal{T}_{k_0, \nu, \gamma}(q_n^{\nu, \gamma} - q_0^{\nu, \gamma})\|_{\mathcal{H}, r_0} + \|r_n - r_0\|_{\mathcal{H}, 0}^2\right).$$

By Assumption E1***,

$$\|P_0 \varphi_n^b - b_0\|_2 = o_p(n^{-1/2}).$$

Hence

$$b_n - b_0 = P_n \chi_{b,0} + o_p(n^{-1/2}).$$

Next, write

$$G_n - G_0 = \frac{1}{n} \sum_{i=1}^n \{x(A_i, S_i)x(A_i, S_i)^\top - G_0\}.$$

Since each coordinate of x is bounded, $G_n - G_0 = O_p(n^{-1/2})$ entrywise. Because $\lambda_{\min}(G_0) > 0$, it follows that G_n is invertible with probability tending to one and

$$G_n^{-1} = G_0^{-1} - G_0^{-1}(G_n - G_0)G_0^{-1} + o_p(n^{-1/2}).$$

Therefore,

$$\begin{aligned}
\theta_n - \theta_0 &= G_n^{-1}b_n - G_0^{-1}b_0 \\
&= G_0^{-1}(b_n - b_0) - G_0^{-1}(G_n - G_0)\theta_0 + o_p(n^{-1/2}) \\
&= \frac{1}{n} \sum_{i=1}^n G_0^{-1} \left[\chi_{b,0}(S_i, A_i, S'_i) - \{x(A_i, S_i)x(A_i, S_i)^\top - G_0\}\theta_0 \right] + o_p(n^{-1/2}) \\
&= \frac{1}{n} \sum_{i=1}^n \chi_{\theta,0}(S_i, A_i, S'_i) + o_p(n^{-1/2}).
\end{aligned}$$

The multivariate central limit theorem then gives

$$\sqrt{n}(\theta_n - \theta_0) \rightsquigarrow N(0, \Sigma_0), \quad \Sigma_0 = \text{Var}_{P_0}(\chi_{\theta,0}(S, A, S')).$$

This establishes asymptotic linearity and efficiency. $\square$

I Smoothness of the Soft Bellman Equation and Its Solutions

In this section, we establish several technical lemmas on the smoothness of the soft Bellman equation and its solutions. Some of these results are used in deriving the EIF and the von Mises expansion for Example 1b. Other results are included for completeness.

I.1 Notation

We study the functional smoothness of the soft Bellman equation and its fixed-point solution for a fixed temperature parameter $\tau \in (0, \infty)$. Let $\lambda := \# \otimes \mu$ denote the product measure on $\mathcal{A} \times \mathcal{S}$, where $\#$ is the uniform measure on the finite set $\mathcal{A}$ and μ is a finite dominating measure on $\mathcal{S}$. For any measurable $B \subseteq \mathcal{A} \times \mathcal{S}$,

$$\lambda(B) = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \int 1_B(a, s) \mu(ds).$$

For a linear operator B , define $\|B\|_{L^\infty \rightarrow L^2} := \sup_{\|f\|_{L^\infty(\lambda)} \leq 1} \|Bf\|_{L^2(\lambda)}$ and more generally $\|B\|_{L^p \rightarrow L^q} := \sup_{\|f\|_{L^p(\lambda)} \leq 1} \|Bf\|_{L^q(\lambda)}$.

For $r \in \mathcal{H}$ and $k \in \mathcal{K}$, let $v_{r,k}$ denote the unique solution in $L^\infty(\lambda)$ of the soft Bellman equation

$$v_{r,k}(a, s) = \int \tau \log \left(\sum_{a' \in \mathcal{A}} \exp \left(\frac{r(a', s') + \gamma v_{r,k}(a', s')}{\tau} \right) \right) k(s' | a, s) \mu(ds').$$

Define $F : L^\infty(\lambda) \times L^\infty(\lambda) \times \mathcal{K} \rightarrow L^\infty(\lambda)$ by

$$F(r, v, k)(a, s) := v(a, s) - \int \tau \log \left(\sum_{a' \in \mathcal{A}} \exp \left(\frac{r(a', s') + \gamma v(a', s')}{\tau} \right) \right) k(s' | a, s) \mu(ds'),$$

and for each $k \in \mathcal{K}$, let $F_k(r, v) := F(r, v, k)$.

We write

$$\begin{aligned} \text{logsumexp}_\tau(q)(s) &:= \tau \log \left(\sum_{a' \in \mathcal{A}} \exp \left\{ \frac{q(a', s)}{\tau} \right\} \right), \\ \pi_{r,k}(a' | s) &:= \frac{\exp \left(\frac{r(a', s) + \gamma v_{r,k}(a', s)}{\tau} \right)}{\sum_{b \in \mathcal{A}} \exp \left(\frac{r(b, s) + \gamma v_{r,k}(b, s)}{\tau} \right)}, \quad q_{r,k}(a, s) := r(a, s) + \gamma v_{r,k}(a, s), \end{aligned}$$

and for general q let

$$\tilde{\pi}_{q,\tau}(a' | s) := \frac{\exp \left(\frac{q(a', s)}{\tau} \right)}{\sum_{b \in \mathcal{A}} \exp \left(\frac{q(b, s)}{\tau} \right)}.$$

For fixed s' , the conditional covariance under π is

$$\begin{aligned} \text{Cov}_{\pi(\cdot | s')}(g(\cdot, s'), h(\cdot, s')) &:= \sum_{a \in \mathcal{A}} \pi(a | s') g(a, s') h(a, s') \\ &\quad - \left(\sum_{a \in \mathcal{A}} \pi(a | s') g(a, s') \right) \left(\sum_{a \in \mathcal{A}} \pi(a | s') h(a, s') \right). \end{aligned}$$

Throughout this subsection, the temperature τ is fixed and all implicit constants may depend on τ .

I.2 Technical lemmas

In this subsection, additive perturbations of the kernel are indexed by elements of $\tilde{T}_{\mathcal{K}}$, equipped with the norm $\|\cdot\|_{\mathcal{K}}$, and their additive Fréchet derivative is denoted by $\tilde{\partial}_k$. The score derivative ∂_k is reserved for the semiparametric statements above.

We begin with the following lemma, which ensures that the map $(r, k) \mapsto v_{r,k}$ is well defined.

Lemma 18 (Existence and uniqueness of solution to soft Bellman equation). *For each $r \in L^\infty(\lambda)$ and $k \in \mathcal{K}$, there exists a unique solution $v_{r,k} \in L^\infty(\lambda)$ to the soft Bellman equation $F_k(r, v) = 0$ over $v \in L^\infty(\lambda)$.*

Proof. For every $r \in L^\infty(\lambda)$, the map $v \mapsto v - F_k(r, v)$ is a contraction on $L^\infty(\lambda)$. By Banach's fixed-point theorem, there exists a unique $v_{r,k} \in L^\infty(\lambda)$ such that $v_{r,k} = v_{r,k} - F_k(r, v_{r,k})$, and hence $F_k(r, v_{r,k}) = 0$. $\square$

The following lemma establishes Fréchet differentiability of F_k for a fixed k .

Lemma 19 (Fréchet differentiability of F_k). *Let $k \in \mathcal{K}$. The map F_k is Fréchet differentiable in supremum norm, with partial derivatives at $(r, v) \in L^\infty(\lambda) \times L^\infty(\lambda)$, for directions $u, h \in L^\infty(\lambda)$, given by*

$$\begin{aligned} \partial_v F_k(r, v)[h](a, s) &= h(a, s) - \gamma \int \sum_{a' \in \mathcal{A}} \tilde{\pi}_{r+\gamma v, \tau}(a' | s') h(a', s') k(s' | a, s) \mu(ds') \\ \partial_r F_k(r, v)[u](a, s) &= - \int \sum_{a' \in \mathcal{A}} \tilde{\pi}_{r+\gamma v, \tau}(a' | s') u(a', s') k(s' | a, s) \mu(ds'). \end{aligned}$$

In particular, there exists a fixed constant $C_\tau < \infty$, depending only on the fixed temperature τ , such that, uniformly over r, v, h, u ,

$$\|F_k(r+u, v+h) - F_k(r, v) - \partial_r F_k(r, v)[u] - \partial_v F_k(r, v)[h]\|_{L^\infty(\lambda)} \leq C_\tau \|u + \gamma h\|_{L^2(\lambda)}^2.$$

Proof. It is straightforward to show that F_k is Gâteaux differentiable with partial derivatives given in the theorem statement. We now show that the map is Fréchet differentiable.

By Taylor's theorem, we have, for each direction $f \in L^\infty(\lambda)$, that

$$\begin{aligned} |\log \text{sumexp}_\tau(q+f)(s') - \log \text{sumexp}_\tau(q)(s') - \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q,\tau}(a' | s') f(a', s')| &\leq \frac{1}{2\tau} \int_0^1 \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q+tf,\tau}(a' | s') |f(a', s')|^2 dt \\ &\leq \frac{1}{2\tau} \sum_{a' \in \mathcal{A}} |f(a', s')|^2. \end{aligned}$$

Substituting $q := r + \gamma v$ and $f := u + \gamma h$ in the above, we find that

$$\begin{aligned} &|F_k(r+u, v+h)(a, s) - F_k(r, v)(a, s) - \partial_v F_k(r, v)[h](a, s) - \partial_r F_k(r, v)[u](a, s)| \\ &\leq \frac{1}{2\tau} \int_0^1 \int \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q+tf,\tau}(a' | s') |f(a', s')|^2 k(s' | a, s) \mu(ds') dt \\ &\leq \frac{1}{2\tau} \int \sum_{a' \in \mathcal{A}} |u(a', s') + \gamma h(a', s')|^2 k(s' | a, s) \mu(ds'). \end{aligned}$$

By definition of $k \in \mathcal{K}$, we have that $k(s' | a, s) \mu(ds') \lesssim \mu(ds')$. Hence,

$$\begin{aligned} &|F_k(r+u, v+h)(a, s) - F_k(r, v)(a, s) - \partial_v F_k(r, v)[h](a, s) - \partial_r F_k(r, v)[u](a, s)| \\ &\lesssim \int \sum_{a' \in \mathcal{A}} |u(a', s') + \gamma h(a', s')|^2 \mu(ds') \\ &\lesssim \|u + \gamma h\|_{L^2(\lambda)}^2. \end{aligned}$$

Taking the L^∞ -norm of both sides, we find

$$\|F_k(r+u, v+h) - F_k(r, v) - \partial_v F_k(r, v)[h] - \partial_r F_k(r, v)[u]\|_{L^\infty(\lambda)} \lesssim \|u + \gamma h\|_{L^2(\lambda)}^2.$$

□

Lemma 20 (Fréchet differentiability of F). *The map $F : L^\infty(\lambda) \times L^\infty(\lambda) \times \mathcal{K} \rightarrow L^\infty(\lambda)$ is Fréchet differentiable in supremum norm, with partial derivatives at (r, v, k) given by*

$$\begin{aligned} \tilde{\partial}_k F(r, v, k)[k' - k](a, s) &= - \int \log \text{sumexp}_\tau(r + \gamma v)(s') (k' - k)(s' | a, s) \mu(ds'), \\ \partial_v F(r, v, k)[h](a, s) &= \partial_v F_k(r, v)[h](a, s), \\ \partial_r F(r, v, k)[u](a, s) &= \partial_r F_k(r, v)[u](a, s), \end{aligned}$$

for $k' \in \mathcal{K}$ and $h, u \in L^\infty(\lambda)$. In particular, we have

$$F(r+u, v+h, k') - F(r, v, k) - \partial_r F(r, v, k)[u] - \partial_v F(r, v, k)[h] - \tilde{\partial}_k F(r, v, k)[k' - k] = R_2(u) + R_2(h) + R_\infty,$$

where

$$\begin{aligned} \|R_2(u)\|_{L^2(\lambda)} &\lesssim \|u\|_{L^2(\lambda)} \|k - k'\|_{L^2(\mu \otimes \lambda)}, & \|R_2(u)\|_{L^\infty(\lambda)} &\lesssim \|u\|_{L^2(\lambda)} \|k - k'\|_{L^\infty(\mu \otimes \lambda)}, \\ \|R_2(h)\|_{L^2(\lambda)} &\lesssim \|h\|_{L^2(\lambda)} \|k - k'\|_{L^2(\mu \otimes \lambda)}, & \|R_2(h)\|_{L^\infty(\lambda)} &\lesssim \|h\|_{L^2(\lambda)} \|k - k'\|_{L^\infty(\mu \otimes \lambda)}, \\ \|R_\infty\|_{L^\infty(\lambda)} &\lesssim \|u + \gamma h\|_{L^2(\lambda)}^2. \end{aligned}$$

Proof. Adding and subtracting the term $F(r, v, k')$, we have

$$F(r + u, v + h, k') - F(r, v, k) = F(r + u, v + h, k') - F(r, v, k') + F(r, v, k') - F(r, v, k).$$

By Lemma 19,

$$F(r + u, v + h, k') - F(r, v, k') = \partial_v F_{k'}(r, v)[h] + \partial_r F_{k'}(r, v)[u] + R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2),$$

where $\|R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2)\|_{L^\infty(\lambda)} \lesssim \|u + \gamma h\|_{L^2(\lambda)}^2$. By linearity of $k \mapsto F(r, v, k)$, we have the equality:

$$F(r, v, k') - F(r, v, k) = \tilde{\partial}_k F(r, v, k)[k' - k].$$

Hence,

$$\begin{aligned} F(r + u, v + h, k') - F(r, v, k) &= \partial_v F_{k'}(r, v)[h] + \partial_r F_{k'}(r, v)[u] \\ &\quad + \tilde{\partial}_k F(r, v, k)[k' - k] + R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2) \\ &= \partial_v F_k(r, v)[h] + \partial_r F_k(r, v)[u] + \tilde{\partial}_k F(r, v, k)[k' - k] \\ &\quad + \partial_v F_{k'}(r, v)[h] - \partial_v F_k(r, v)[h] \\ &\quad + \partial_r F_{k'}(r, v)[u] - \partial_r F_k(r, v)[u] \\ &\quad + R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2). \end{aligned} \tag{54}$$

We now consider the differences $\partial_v F_{k'}(r, v)[h] - \partial_v F_k(r, v)[h]$ and $\partial_r F_{k'}(r, v)[u] - \partial_r F_k(r, v)[u]$. By the expressions of the partial derivatives in Lemma 19, we have that

$$|\partial_r F_{k'}(r, v)[u] - \partial_r F_k(r, v)[u]|(a, s) \leq \left| \int \sum_{a' \in \mathcal{A}} u(a', s')(k' - k)(s' \mid a, s) \mu(ds') \right|.$$

For each (a, s) ,

$$\begin{aligned}
\left| \int \sum_{a' \in \mathcal{A}} u(a', s') (k' - k)(s' | a, s) \mu(ds') \right| &= \left| \int \sum_{a' \in \mathcal{A}} u(a', s') (\sqrt{k'} - \sqrt{k})(s' | a, s) (\sqrt{k'} + \sqrt{k})(s' | a, s) \mu(ds') \right| \\
&\leq \left(\int \left(\sum_{a' \in \mathcal{A}} u(a', s') \right)^2 (\sqrt{k'} + \sqrt{k})^2 \mu(ds') \right)^{1/2} \\
&\quad \times \left(\int (\sqrt{k'} - \sqrt{k})^2 (s' | a, s) \mu(ds') \right)^{1/2} \\
&\lesssim \|u\|_{L^2(\lambda)} \times \left(\int (\sqrt{k'} - \sqrt{k})^2 (s' | a, s) \mu(ds') \right)^{1/2},
\end{aligned}$$

where we used that $\sqrt{k'} + \sqrt{k}$ is uniformly bounded by the definition of $\mathcal{K}$. Taking the $L^2(\lambda)$ norm over (a, s) , we have

$$\begin{aligned}
\|\partial_r F_{k'}(r, v)[u] - \partial_r F_k(r, v)[u]\|_{L^2(\lambda)} &\lesssim \|u\|_{L^2(\lambda)} \left\{ \int (\sqrt{k'} - \sqrt{k})^2 (s' | a, s) \mu(ds') \lambda(da, ds) \right\}^{1/2} \\
&\lesssim \|u\|_{L^2(\lambda)} \|k' - k\|_{L^2(\lambda \otimes \mu)},
\end{aligned}$$

where we used that $(\sqrt{k'} - \sqrt{k})^2 (s' | a, s) \lesssim (k' - k)^2 (s' | a, s)$, due to the lower boundedness (positivity) of the elements of $\mathcal{K}$. Taking the supremum over (a, s) instead, the same pointwise bound yields

$$\|\partial_r F_{k'}(r, v)[u] - \partial_r F_k(r, v)[u]\|_{L^\infty(\lambda)} \lesssim \|u\|_{L^2(\lambda)} \|k' - k\|_{L^\infty(\lambda \otimes \mu)}.$$

For the other partial derivative, an identical argument gives

$$\|\partial_v F_{k'}(r, v)[h] - \partial_v F_k(r, v)[h]\|_{L^2(\lambda)} \lesssim \|h\|_{L^2(\lambda)} \|k' - k\|_{L^2(\lambda \otimes \mu)}.$$

and

$$\|\partial_v F_{k'}(r, v)[h] - \partial_v F_k(r, v)[h]\|_{L^\infty(\lambda)} \lesssim \|h\|_{L^2(\lambda)} \|k' - k\|_{L^\infty(\lambda \otimes \mu)}.$$

Returning to (54), we have that

$$\begin{aligned}
F(r + u, v + h, k') - F(r, v, k) &= \partial_v F_k(r, v)[h] + \partial_r F_k(r, v)[u] + \tilde{\partial}_k F(r, v, k)[k' - k] \\
&\quad + R_2(u) + R_2(h) + R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2),
\end{aligned}$$

where $\|R_2(u)\|_{L^2(\lambda)} \lesssim \|u\|_{L^2(\lambda)} \|k - k'\|_{L^2(\mu \otimes \lambda)}$, $\|R_2(u)\|_{L^\infty(\lambda)} \lesssim \|u\|_{L^2(\lambda)} \|k - k'\|_{L^\infty(\mu \otimes \lambda)}$, $\|R_2(h)\|_{L^2(\lambda)} \lesssim \|h\|_{L^2(\lambda)} \|k - k'\|_{L^2(\mu \otimes \lambda)}$, $\|R_2(h)\|_{L^\infty(\lambda)} \lesssim \|h\|_{L^2(\lambda)} \|k - k'\|_{L^\infty(\mu \otimes \lambda)}$, and $\|R_\infty(\|u + \gamma h\|_{L^2(\lambda)}^2)\|_{L^\infty(\lambda)} \lesssim \|u + \gamma h\|_{L^2(\lambda)}^2$. The result follows. $\square$

Lemma 21 (Sup-norm Fréchet differentiability of the soft value function). *Let $(r, k) \in \mathcal{H} \times \mathcal{K}$.*

Suppose $\gamma < 1$, so that $\mathcal{T}_{r,k} : L^\infty(\lambda) \rightarrow L^\infty(\lambda)$ is invertible. Then the map $(r, k) \mapsto v_{r,k}$ is Fréchet differentiable, with partial derivatives

$$\partial_r v_{r,k} = \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}, \quad \tilde{\partial}_k v_{r,k}[w] = \mathcal{T}_{r,k}^{-1} \mathcal{P}_w \Phi(r + \gamma v),$$

where $\Phi(r + \gamma v)(s') := \text{logsumexp}_\tau(r + \gamma v)(s')$ and

$$\mathcal{P}_w \Phi(r + \gamma v)(a, s) := \int \text{logsumexp}_\tau(r + \gamma v)(s') w(s' \mid a, s) \mu(ds').$$

Proof. Denote

$$\mathcal{Q}_{r,k}(w) := \int \text{logsumexp}_\tau(r + \gamma v_{r,k})(s') w(s' \mid a, s) \mu(ds').$$

We apply the implicit function theorem to obtain joint differentiability of $v_{r,k}$ in (r, k) . Moreover, since $\gamma < 1$, the map F is Fréchet differentiable as a map $L^\infty(\lambda) \times L^\infty(\lambda) \times \mathcal{K} \rightarrow L^\infty(\lambda)$, and its derivative depends jointly and continuously on (r, v, k) . By the invertibility assumption in the statement, the partial derivative

$$\partial_v F_k(r, v_{r,k}) : L^\infty(\lambda) \rightarrow L^\infty(\lambda)$$

is a bounded linear isomorphism. The implicit function theorem for Banach spaces therefore implies that the solution map $(r, k) \mapsto v_{r,k}$ is Fréchet differentiable at (r, k) as a map $L^\infty(\lambda) \times \mathcal{K} \rightarrow L^\infty(\lambda)$, with derivatives

$$\begin{aligned} \partial_r v_{r,k}[u] &= -(\partial_v F_k(r, v_{r,k}))^{-1}(\partial_r F_k(r, v_{r,k})[u]), \\ \tilde{\partial}_k v_{r,k}[w] &= -(\partial_v F_k(r, v_{r,k}))^{-1}(\tilde{\partial}_k F(r, v_{r,k}, k)[w]). \end{aligned}$$

Since $\partial_v F_k(r, v) = \mathcal{T}_{r,k}$, $\partial_r F_k(r, v) = -\mathcal{P}_{r,k}$, and $\tilde{\partial}_k F(r, v, k) = -\mathcal{Q}_{r,k}$, we find that

$$\partial_r v_{r,k} = \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}, \quad \tilde{\partial}_k v_{r,k} = \mathcal{T}_{r,k}^{-1} \mathcal{Q}_{r,k}.$$

□

Lemma 22. *Under the conditions of Lemma 21, it holds that*

$$\partial_r q_{r,k}[u] = \mathcal{T}_{r,k}^{-1} u$$

for every $u \in L^\infty(\lambda)$.

Proof. By Lemma 21,

$$\begin{aligned}
\partial_r q_{r,k}[u] &= u + \gamma \partial_r v_{r,k}(u) \\
&= \left(I + \gamma \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} \right) (u) \\
&= \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k} + \gamma \mathcal{P}_{r,k}) (u) \\
&= \mathcal{T}_{r,k}^{-1} (I - \gamma \mathcal{P}_{r,k} + \gamma \mathcal{P}_{r,k}) (u) \\
&= \mathcal{T}_{r,k}^{-1} u.
\end{aligned}$$

□

In the following lemma, we define the hard Bellman operator $\tilde{\mathcal{T}}_k : L^\infty(\lambda) \rightarrow L^\infty(\lambda)$ by

$$\tilde{\mathcal{T}}_k(f)(a, s) := f(a, s) - \gamma \int \max_{a' \in \mathcal{A}} f(a', s') k(s' | a, s) \mu(ds').$$

Lemma 23 (Lipschitz continuity of value function map). *Suppose that $\gamma < 1$. Then, for all $r, r' \in L^\infty(\lambda)$ and $k, k' \in \mathcal{K}$, we have*

$$\begin{aligned}
\|v_{r',k} - v_{r,k}\|_{L^\infty(\lambda)} &\lesssim (1 - \gamma)^{-1} \|r' - r\|_{L^2(\lambda)} \\
\|\tilde{\mathcal{T}}_k(v_{r,k'} - v_{r,k})\|_{L^2(\lambda)} &\lesssim (1 + \|r\|_{L^\infty(\lambda)}) \cdot \|k' - k\|_{L^2(\mu \otimes \lambda)}.
\end{aligned}$$

Proof. By properties of the log-sum-exp map, we have

$$\begin{aligned}
v_{r',k}(a, s) - v_{r,k}(a, s) &= \int \left[\text{logsumexp}_\tau(q_{r',k}(\cdot, s')) - \text{logsumexp}_\tau(q_{r,k}(\cdot, s')) \right] k(s' | a, s) \mu(ds') \\
&\leq \int \max_{a' \in \mathcal{A}} \{r(a', s') - r'(a', s') + \gamma \{v_{r',k}(a', s') - v_{r,k}(a', s')\}\} k(s' | a, s) \mu(ds') \\
&\leq \int \max_{a' \in \mathcal{A}} \{r(a', s') - r'(a', s')\} k(s' | a, s) \mu(ds') + \gamma \int \max_{a' \in \mathcal{A}} \{v_{r',k}(a', s') - v_{r,k}(a', s')\} \\
&\leq \int \max_{a' \in \mathcal{A}} \{r(a', s') - r'(a', s')\} k(s' | a, s) \mu(ds') + \gamma \|v_{r',k} - v_{r,k}\|_{L^\infty(\lambda)}.
\end{aligned}$$

where $q_{r,k}(a', s') = r(a', s') + \gamma v_{r,k}(a', s')$. Hence, rearranging terms,

$$\begin{aligned}
v_{r',k}(a, s) - v_{r,k}(a, s) - \gamma \|v_{r',k} - v_{r,k}\|_{L^\infty(\lambda)} &\leq \int \max_{a' \in \mathcal{A}} \{r(a', s') - r'(a', s')\} k(s' | a, s) \mu(ds') \\
&\lesssim \|r' - r\|_{L^2(\lambda)},
\end{aligned}$$

where we used the boundedness of k and the definition of λ . Hence, taking the supremum norm of both sides, we find

$$\|v_{r',k} - v_{r,k}\|_{L^\infty(\lambda)} \lesssim (1 - \gamma)^{-1} \|r' - r\|_{L^2(\lambda)}.$$

Next, we turn to $v_{r,k'} - v_{r,k}$. By definition of the fixed points and arguing as in the proof of

Lemma 20, we have that

$$\begin{aligned}
v_{r,k'}(a, s) - v_{r,k}(a, s) &= \int \log \text{sumexp}_\tau(r + \gamma v_{r,k'})(s') (k' - k)(s' | a, s) \mu(ds') \\
&\quad + \int \left(\log \text{sumexp}_\tau(r + \gamma v_{r,k'})(s') - \log \text{sumexp}_\tau(r + \gamma v_{r,k})(s') \right) k(s' | a, s) \mu(ds') \\
&\lesssim \|r + \gamma v_{r,k'}\|_{L^\infty(\lambda)} \cdot \|(\sqrt{k'} - \sqrt{k})(\cdot | a, s)\|_{L^2(\mu)} \\
&\quad + \gamma \int \max_{a' \in \mathcal{A}} \{v_{r,k'}(a', s') - v_{r,k}(a', s')\} k(s' | a, s) \mu(ds').
\end{aligned}$$

Hence, using that $\|r + \gamma v_{r,k'}\|_{L^\infty(\lambda)} \lesssim 1 + \|r\|_{L^\infty(\lambda)}$,

$$\tilde{\mathcal{T}}_k(v_{r,k'} - v_{r,k}) \lesssim (1 + \|r\|_{L^\infty(\lambda)}) \cdot \|(\sqrt{k'} - \sqrt{k})(\cdot | a, s)\|_{L^2(\mu)}.$$

In fact, we can also show that $\tilde{\mathcal{T}}_k|v_{r,k'} - v_{r,k}| \lesssim (1 + \|r\|_{L^\infty(\lambda)}) \cdot \|(\sqrt{k'} - \sqrt{k})(\cdot | a, s)\|_{L^2(\mu)}$. Thus,

$$\|\tilde{\mathcal{T}}_k(v_{r,k'} - v_{r,k})\|_{L^2(\lambda)} \lesssim (1 + \|r\|_{L^\infty(\lambda)}) \cdot \|k' - k\|_{L^2(\mu \otimes \lambda)},$$

where, as shown in the proof of Lemma 20, the $L^2(\lambda)$ -norm of $(a, s) \mapsto \|(\sqrt{k'} - \sqrt{k})(\cdot | a, s)\|_{L^2(\mu)}$ is bounded above by $\|k' - k\|_{L^2(\mu \otimes \lambda)}$. □

Lemma 24 (Quadratic remainder for partial Fréchet derivative of $r \mapsto v_{r,k}$). *Suppose $\gamma < 1$. For $r, r' \in \mathcal{H}$ and $k \in \mathcal{K}$, we have*

$$\|v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r]\|_{L^\infty(\lambda)} \lesssim \|r' - r\|_{L^2(\lambda)}^2.$$

Proof. Let $q_{r,k} := r + \gamma v_{r,k}$. We apply the implicit function theorem to obtain differentiability of $v_{r,k}$ in r . By Lemma 20, the map F is Fréchet differentiable, satisfying the expansion:

$$\begin{aligned}
0 &= F(r', v_{r',k}, k) - F(r, v_{r,k}, k) \\
&= \partial_{r,v} F(r, v_{r,k}, k)[r' - r, v_{r',k} - v_{r,k}] + R(r' - r) \\
&= \partial_r F(r, v_{r,k}, k)[r' - r] + \partial_v F(r, v_{r,k}, k)[v_{r',k} - v_{r,k}] + R(r' - r),
\end{aligned}$$

where $\|R(r' - r)\|_{L^\infty(\lambda)} = O(\|q_{r',k} - q_{r,k}\|_{L^2(\lambda)}^2)$ by Lemma 19. Moreover, differentiating both sides of $F(r, v_{r,k}, k) = 0$ with respect to r , we find, by the chain rule, that

$$\partial_{r,v} F(r, v_{r,k}, k)[r' - r, \partial_r v_{r,k}[r' - r]] = \partial_r F(r, v_{r,k}, k)[r' - r] + \partial_v F(r, v_{r,k}, k)[\partial_r v_{r,k}[r' - r]] = 0.$$

Hence,

$$0 = \partial_v F(r, v_{r,k}, k)[v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r]] + R(r' - r).$$

Equivalently,

$$\mathcal{T}_{r,k}(v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r]) = -R(r' - r).$$

Since $\gamma < 1$, the map $\mathcal{T}_{r,k} : L^\infty(\lambda) \rightarrow L^\infty(\lambda)$ is invertible by Banach's fixed point theorem. Hence,

$$v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r] = -\mathcal{T}_{r,k}^{-1} R(r' - r),$$

so that

$$\|v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r]\|_{L^\infty(\lambda)} \leq \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty(\lambda)} \|R(r' - r)\|_{L^\infty(\lambda)} \lesssim \|q_{r',k} - q_{r,k}\|_{L^2(\lambda)}^2.$$

Finally, by Lemma 23, $\|q_{r',k} - q_{r,k}\|_{L^2(\lambda)}^2 \lesssim \|r' - r\|_{L^2(\lambda)}^2$. Hence,

$$\|v_{r',k} - v_{r,k} - \partial_r v_{r,k}[r' - r]\|_{L^\infty(\lambda)} \lesssim \|r' - r\|_{L^2(\lambda)}^2.$$

□

Lemma 25 (Quadratic remainder for partial Fréchet derivative of $k \mapsto v_{r,k}$). *Suppose $\gamma < 1$. For $r \in \mathcal{H}$ and $k, k' \in \mathcal{K}$, we have*

$$\|(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k])\|_{L^2(\lambda)} \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}.$$

Proof. We apply the implicit function theorem to obtain differentiability of $v_{r,k}$ in k . By Lemma 20, the map F is Fréchet differentiable, satisfying the expansion:

$$\begin{aligned} 0 &= F(r, v_{r,k'}, k') - F(r, v_{r,k}, k) \\ &= \partial_v F(r, v_{r,k}, k)[v_{r,k'} - v_{r,k}] + \tilde{\partial}_k F(r, v_{r,k}, k)[k' - k] + R_2, \end{aligned}$$

where $\|R_2\|_{L^2(\lambda)} = O(\|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)})$. Moreover, differentiating both sides of $F(r, v_{r,k}, k) = 0$ with respect to k , we find, by the chain rule, that

$$\partial_v F(r, v_{r,k}, k)[\tilde{\partial}_k v_{r,k}[k' - k]] + \tilde{\partial}_k F(r, v_{r,k}, k)[k' - k] = 0.$$

Hence,

$$0 = \partial_v F(r, v_{r,k}, k)[v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k]] + R_2.$$

Equivalently,

$$\mathcal{T}_{r,k}(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k]) = -R_2.$$

Hence,

$$\|\mathcal{T}_{r,k}(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k])\|_{L^2(\lambda)} \leq \|R_2\|_{L^2(\lambda)} \lesssim \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)},$$

and, therefore,

$$\|(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k])\|_{L^2(\lambda)} \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}.$$

□

Lemma 26 (Lipschitz properties of softmax operator). *For all $r, r' \in L^\infty(\lambda)$ and $k, k' \in \mathcal{K}$, we have that*

$$\|\Pi_{r',k'} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \lesssim \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.$$

Proof. By the τ^{-1} -Lipschitz property of the temperature- τ softmax and by Cauchy–Schwarz, we have that

$$\begin{aligned} |\Pi_{r',k'}(f) - \Pi_{r,k}(f)|(s) &= \left| \sum_{a \in \mathcal{A}} f(a, s) (\pi_{r',k'}(a | s) - \pi_{r,k}(a | s)) \right| \\ &= |\langle f(\cdot, s), \pi_{r',k'}(\cdot | s) - \pi_{r,k}(\cdot | s) \rangle_{\ell_2(\mathcal{A})}| \\ &\leq \|f(\cdot, s)\|_{\ell_2(\mathcal{A})} \|\pi_{r',k'}(\cdot | s) - \pi_{r,k}(\cdot | s)\|_{\ell_2(\mathcal{A})} \\ &\lesssim \tau^{-1} \|f(\cdot, s)\|_{\ell_2(\mathcal{A})} \left(\sum_{a \in \mathcal{A}} (|r'(a, s) - r(a, s)| + |v_{r',k'}(a, s) - v_{r,k}(a, s)|)^2 \right)^{1/2}, \end{aligned}$$

where $\langle g, h \rangle_{\ell_2(\mathcal{A})} := \sum_{a \in \mathcal{A}} g(a)h(a)$.

Let $L^2(\mu; \ell_2(\mathcal{A}))$ denotes the space of measurable functions $g : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ with finite mixed norm $\|g\|_{L^2(\mu; \ell_2(\mathcal{A}))}^2 := \int_{\mathcal{S}} \sum_{a \in \mathcal{A}} |g(a, s)|^2 \mu(ds)$. Then, by Cauchy–Schwarz,

$$\begin{aligned} \|\Pi_{r',k'}(f) - \Pi_{r,k}(f)\|_{L^1(\mu)} &= \int |\Pi_{r',k'}(f) - \Pi_{r,k}(f)|(s) \mu(ds) \\ &\leq \int \|f(\cdot, s)\|_{\ell_2(\mathcal{A})} \|\pi_{r',k'}(\cdot | s) - \pi_{r,k}(\cdot | s)\|_{\ell_2(\mathcal{A})} \mu(ds) \\ &\leq \|f\|_{L^2(\mu; \ell_2(\mathcal{A}))} \|\pi_{r',k'} - \pi_{r,k}\|_{L^2(\mu; \ell_2(\mathcal{A}))} \\ &\lesssim \|f\|_{L^2(\mu; \ell_2(\mathcal{A}))} \| (r' - r) + (v_{r',k'} - v_{r,k}) \|_{L^2(\mu; \ell_2(\mathcal{A}))} \\ &\lesssim \|f\|_{L^2(\mu; \ell_2(\mathcal{A}))} \left(\|r' - r\|_{L^2(\mu; \ell_2(\mathcal{A}))} + \|v_{r',k'} - v_{r,k}\|_{L^2(\mu; \ell_2(\mathcal{A}))} \right). \end{aligned}$$

Finally, since $\mathcal{A}$ is finite, the $L^2(\mu; \ell_2(\mathcal{A}))$ and $L^2(\lambda)$ norms are equal up to a multiplicative factor depending on $\#\mathcal{A}$. Hence,

$$\|\Pi_{r',k'}(f) - \Pi_{r,k}(f)\|_{L^1(\mu)} \lesssim \|f\|_{L^2(\lambda)} \left(\|r' - r\|_{L^2(\lambda)} + \|v_{r',k'} - v_{r,k}\|_{L^2(\lambda)} \right).$$

Moreover, by Lemma 23, we have

$$\begin{aligned} \|v_{r',k'} - v_{r,k}\|_{L^2(\lambda)} &\leq \|v_{r',k'} - v_{r,k'}\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \\ &\lesssim \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}. \end{aligned}$$

Putting everything together, we conclude that

$$\|\Pi_{r',k'}(f) - \Pi_{r,k}(f)\|_{L^1(\mu)} \lesssim \|f\|_{L^2(\lambda)} \left(\|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \right),$$

which implies, by definition, that $\|\Pi_{r',k'} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \lesssim \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}$. $\square$

Lemma 27. *There exists a constant $C < \infty$ (uniform on the parameter neighborhood) such that*

$$\|\mathcal{P}_{r',k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq C \left(\|k' - k\|_{\mathcal{K}} + \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \right).$$

Proof. By definition, $\mathcal{P}_{r,k} = \mathcal{P}_k \Pi_{r,k}$. Hence

$$\begin{aligned} \|\mathcal{P}_{r',k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &= \|\mathcal{P}_{k'} \Pi_{r',k'} - \mathcal{P}_k \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\leq \|\mathcal{P}_{k'} - \mathcal{P}_k\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Pi_{r',k'}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\quad + \|\mathcal{P}_k\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Pi_{r',k'} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}, \end{aligned}$$

by the triangle inequality and submultiplicativity. By Lemma 10,

$$\|\mathcal{P}_{k'} - \mathcal{P}_k\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|k' - k\|_{\mathcal{K}}.$$

By Lemma 26,

$$\|\Pi_{r',k'} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.$$

Using the uniform boundedness of $\|\Pi_{r',k'}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ and $\|\mathcal{P}_k\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ on the neighborhood yields

$$\|\mathcal{P}_{r',k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|k' - k\|_{\mathcal{K}} + \|r' - r\|_{L^2(\lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.$$

□

Lemma 28 (Fréchet differentiability of linear operators). *Let $r, r' \in L^\infty(\lambda)$, $k, k' \in \mathcal{K}$, and $f \in L^\infty(\lambda)$. Suppose $\|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} < \infty$. Then the partial Gâteaux derivatives of the maps $(r, k) \mapsto \Pi_{r,k} f$ and $(r, k) \mapsto \mathcal{P}_{r,k} f$ in the direction $r' - r$ and $k' - k$ are*

$$\begin{aligned} \partial_r \Pi_{r,k} [r' - r](f)(s') &= \frac{1}{\tau} \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \partial_r q_{r,k} [r' - r](\cdot, s')), \\ \partial_r \mathcal{P}_{r,k} [r' - r](f)(a, s) &= \frac{1}{\tau} \int \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \partial_r q_{r,k} [r' - r](\cdot, s')) k(s' | a, s) \mu(ds'), \\ \tilde{\partial}_k \Pi_{r,k} [k' - k](f)(s') &= \frac{1}{\tau} \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \tilde{\partial}_k q_{r,k} [k' - k](\cdot, s')), \\ \tilde{\partial}_k \mathcal{P}_{r,k} [k' - k](f)(a, s) &= \frac{1}{\tau} \int \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \tilde{\partial}_k q_{r,k} [k' - k](\cdot, s')) k(s' | a, s) \mu(ds') \\ &\quad + \int \Pi_{r,k}(f)(s') (k' - k)(s' | a, s) \mu(ds'). \end{aligned}$$

Moreover, the following bounds hold:

$$\begin{aligned}
\|\Pi_{r',k}f - \Pi_{r,k}f - \partial_r \Pi_{r,k}[r' - r](f)\|_{L^1(\mu)} &\lesssim \|f\|_{L^\infty(\lambda)} \|r' - r\|_{L^2(\lambda)}^2, \\
\|\mathcal{P}_{r',k}f - \mathcal{P}_{r,k}f - \partial_r \mathcal{P}_{r,k}[r' - r](f)\|_{L^\infty(\lambda)} &\lesssim \|f\|_{L^\infty(\lambda)} \|r' - r\|_{L^2(\lambda)}^2, \\
\|\Pi_{r,k'}f - \Pi_{r,k}f - \tilde{\partial}_k \Pi_{r,k}[k' - k](f)\|_{L^1(\mu)} &\lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 \right. \\
&\quad \left. + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)} \right\}, \\
\|\mathcal{P}_{r,k'}(f) - \mathcal{P}_{r,k}(f) &- \tilde{\partial}_k \mathcal{P}_{r,k}[k' - k](f)\|_{L^2(\lambda)} \lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 \right. \\
&\quad \left. + (1 + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}) \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)} \right\}.
\end{aligned}$$

Proof. Differentiability of softmax operator. Define the operator $\tilde{\Pi}_{q,\tau} : L^2(\lambda) \rightarrow L^2(\mu)$ by

$$(\tilde{\Pi}_{q,\tau}f)(s) := \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q,\tau}(a' | s) f(a', s), \quad \tilde{\pi}_{q,\tau}(a' | s) := \frac{\exp\left(\frac{q(a', s)}{\tau}\right)}{\sum_{b \in \mathcal{A}} \exp\left(\frac{q(b, s)}{\tau}\right)}.$$

For fixed $s' \in \mathcal{S}$, let

$$\Phi(t) := (\tilde{\Pi}_{q+tf_1,\tau}f)(s') = \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q+tf_1,\tau}(a' | s') f(a', s').$$

The derivative of $\tilde{\pi}_{q+tf_1,\tau}(a' | s')$ with respect to t is

$$\frac{d}{dt} \tilde{\pi}_{q+tf_1,\tau}(a' | s') = \frac{1}{\tau} \tilde{\pi}_{q+tf_1,\tau}(a' | s') [f_1(a', s') - E_{\tilde{\pi}_{q+tf_1,\tau}}[f_1(\cdot, s')]].$$

Hence, $\Phi'(0) = \frac{1}{\tau} \text{Cov}_{\tilde{\pi}_{q,\tau}(\cdot | s')}(f(\cdot, s'), f_1(\cdot, s'))$. The second derivative $\Phi''(t)$ is uniformly bounded in absolute value by $2\tau^{-2} \|f(\cdot, s')\|_\infty \|f_1(\cdot, s')\|_{L^2(\tilde{\pi}_{q+tf_1,\tau}(\cdot | s'))}^2$. By Taylor's theorem with integral remainder,

$$\begin{aligned}
|\Phi(t) - \Phi(0) - t\Phi'(0)| &\leq t^2 \tau^{-2} \|f(\cdot, s')\|_\infty \sup_{u \in [0,1]} \|f_1(\cdot, s')\|_{L^2(\tilde{\pi}_{q+uf_1,\tau}(\cdot | s'))}^2 \\
&\leq t^2 \tau^{-2} \|f(\cdot, s')\|_\infty \sum_{a' \in \mathcal{A}} \{f_1(a', s')\}^2.
\end{aligned}$$

Setting $t = 1$ and applying the previous bound pointwise in s' , we obtain

$$\begin{aligned}
\left| \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q+f_1,\tau}(a' | s') f(a', s') - \sum_{a' \in \mathcal{A}} \tilde{\pi}_{q,\tau}(a' | s') f(a', s') - \frac{1}{\tau} \text{Cov}_{\tilde{\pi}_{q,\tau}(\cdot | s')}(f(\cdot, s'), f_1(\cdot, s')) \right| \\
\leq \tau^{-2} \max_{a' \in \mathcal{A}} |f(a', s')| \sum_{a' \in \mathcal{A}} \{f_1(a', s')\}^2.
\end{aligned}$$

Thus, letting

$$\partial_q \tilde{\Pi}_{q,\tau}[f_1](f) := (s' \mapsto \frac{1}{\tau} \text{Cov}_{\tilde{\pi}_{q,\tau}(\cdot|s')} (f(\cdot, s'), f_1(\cdot, s'))),$$

we obtain the pointwise bound

$$\left| (\tilde{\Pi}_{q+f_1,\tau} f)(s') - (\tilde{\Pi}_{q,\tau} f)(s') - \partial_q \tilde{\Pi}_{q,\tau}[f_1](f)(s') \right| \lesssim \max_{a' \in \mathcal{A}} |f(a', s')| \sum_{a' \in \mathcal{A}} \{f_1(a', s')\}^2.$$

Partial differentiability of $\Pi_{r,k}$ in the reward. We now turn to establishing partial differentiability of the map $(r, v) \mapsto \Pi_{r,k}$. We begin with the dependence on r , i.e., the map $r \mapsto \Pi_{r,k}$. By the chain rule, we have that $\partial_r \Pi_{r,k}[u] = \partial_q \tilde{\Pi}_{q_r,k,\tau}[\partial_r q_{r,k}[u]]$ and, hence,

$$\partial_r \Pi_{r,k}[u](f)(s') = \frac{1}{\tau} \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \partial_r q_{r,k}[u](\cdot, s')).$$

Next, we determine the remainder in the first-order Taylor expansion. Adding and subtracting terms and applying the triangle inequality, we have

$$\begin{aligned} & |\Pi_{r+u,k} f - \Pi_{r,k} f - \partial_r \Pi_{r,k}[u](f)| \\ &= \left| \Pi_{r+u,k} f - \Pi_{r,k} f - \partial_q \tilde{\Pi}_{q_r,k,\tau}[\partial_r q_{r,k}[u]](f) \right| \\ &\leq \left| \tilde{\Pi}_{q_{r+u,k},\tau} f - \tilde{\Pi}_{q_r,k,\tau} f - \partial_q \tilde{\Pi}_{q_r,k,\tau}[q_{r+u,k} - q_{r,k}](f) \right| \\ &\quad + \left| \partial_q \tilde{\Pi}_{q_r,k,\tau}[q_{r+u,k} - q_{r,k} - \partial_r q_{r,k}[u]](f) \right|, \end{aligned}$$

where $q_{r+u,k} - q_{r,k} - \partial_r q_{r,k}[u] = \gamma(v_{r+u,k} - v_{r,k} - \partial_r v_{r,k}[u])$. Hence, applying our earlier bounds, we find that

$$\begin{aligned} & |\Pi_{r+u,k} f - \Pi_{r,k} f - \partial_r \Pi_{r,k}[u](f)| \\ &\lesssim \|f\|_{L^\infty(\lambda)} \sum_{a' \in \mathcal{A}} (q_{r+u,k} - q_{r,k})^2(a', \cdot) + \left| \partial_q \tilde{\Pi}_{q_r,k,\tau}[q_{r+u,k} - q_{r,k} - \partial_r q_{r,k}[u]](f) \right| \\ &\lesssim \|f\|_{L^\infty(\lambda)} \sum_{a' \in \mathcal{A}} (q_{r+u,k} - q_{r,k})^2(a', \cdot) + \gamma \|f\|_{L^\infty(\lambda)} \cdot \Pi_{r,k} |v_{r+u,k} - v_{r,k} - \partial_r v_{r,k}[u]|, \end{aligned}$$

where the final bound follows from the covariance properties in the definition of $\partial_q \tilde{\Pi}_{q_r,k,\tau}$. We have, by Lemma 23,

$$\|q_{r',k} - q_{r,k}\|_{L^2(\lambda)} \leq \|r' - r\|_{L^2(\lambda)} + \gamma \|v_{r',k} - v_{r,k}\|_{L^\infty(\lambda)} \lesssim \|r' - r\|_{L^2(\lambda)}.$$

Moreover, by Lemma 24, $\|v_{r+u,k} - v_{r,k} - \partial_r v_{r,k}[u]\|_{L^\infty(\lambda)} \lesssim \|u\|_{L^2(\lambda)}^2$. Hence, we conclude that

$$\|\Pi_{r+u,k} f - \Pi_{r,k} f - \partial_r \Pi_{r,k}[u](f)\|_{L^1(\mu)} \lesssim \|f\|_{L^\infty(\lambda)} \|u\|_{L^2(\lambda)}^2 \quad (55)$$

Partial differentiability of $\Pi_{r,k}$ in the kernel. Next we establish partial differentiability of the

map $k \mapsto \Pi_{r,k}$. By the chain rule, we have that $\tilde{\partial}_k \Pi_{r,k}[k' - k] = \partial_q \tilde{\Pi}_{q_{r,k},\tau}[\tilde{\partial}_k q_{r,k}[k' - k]]$ and, hence,

$$\tilde{\partial}_k \Pi_{r,k}[k' - k](f)(s') = \frac{1}{\tau} \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \tilde{\partial}_k q_{r,k}[k' - k](\cdot, s')).$$

Next, we determine the remainder in the first-order Taylor expansion. Adding and subtracting terms and applying the triangle inequality, we have

$$\begin{aligned} & \left| \Pi_{r,k'} f - \Pi_{r,k} f - \tilde{\partial}_k \Pi_{r,k}[k' - k](f) \right| \\ &= \left| \Pi_{r,k'} f - \Pi_{r,k} f - \partial_q \tilde{\Pi}_{q_{r,k},\tau}[\tilde{\partial}_k q_{r,k}[k' - k]](f) \right| \\ &\leq \left| \tilde{\Pi}_{q_{r,k'},\tau} f - \tilde{\Pi}_{q_{r,k},\tau} f - \partial_q \tilde{\Pi}_{q_{r,k},\tau}[q_{r,k'} - q_{r,k}](f) \right| \\ &\quad + \left| \partial_q \tilde{\Pi}_{q_{r,k},\tau}[q_{r,k'} - q_{r,k} - \tilde{\partial}_k q_{r,k}[k' - k]](f) \right|. \end{aligned}$$

Since $q_{r,k} = r + \gamma v_{r,k}$ and r is fixed here, we have

$$q_{r,k'} - q_{r,k} - \tilde{\partial}_k q_{r,k}[k' - k] = \gamma(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k]).$$

Hence, we find that

$$\begin{aligned} & \left| \Pi_{r,k'} f - \Pi_{r,k} f - \tilde{\partial}_k \Pi_{r,k}[k' - k](f) \right| \\ &\lesssim \|f\|_{L^\infty(\lambda)} \sum_{a' \in \mathcal{A}} (q_{r,k'} - q_{r,k})^2(a', \cdot) + \left| \partial_q \tilde{\Pi}_{q_{r,k},\tau}[q_{r,k'} - q_{r,k} - \tilde{\partial}_k q_{r,k}[k' - k]](f) \right| \\ &\lesssim \|f\|_{L^\infty(\lambda)} \sum_{a' \in \mathcal{A}} (v_{r,k'} - v_{r,k})^2(a', \cdot) + \gamma \|f\|_{L^\infty(\lambda)} \Pi_{r,k} \left| v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k] \right|. \end{aligned}$$

Lemma 25 shows that

$$\|(v_{r,k'} - v_{r,k} - \tilde{\partial}_k v_{r,k}[k' - k])\|_{L^2(\lambda)} \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}.$$

Thus, applying Cauchy-Schwarz,

$$\|\Pi_{r,k'} f - \Pi_{r,k} f - \tilde{\partial}_k \Pi_{r,k}[k' - k](f)\|_{L^1(\mu)} \lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 \right. \quad (56)$$

$$\left. + \|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)} \right\} \quad (57)$$

Partial differentiability of $\mathcal{P}_{r,k}$ in the reward. Building on the above, we turn to establishing partial differentiability of $(r, k) \mapsto \mathcal{P}_{r,k}$. We begin with the r component. By definition, $(\mathcal{P}_{r,k} f)(a, s) = \int \Pi_{r,k}(f)(s') k(s' | a, s) \mu(ds')$. By linearity and the chain rule,

$$\begin{aligned} \partial_r \mathcal{P}_{r,k}[u](f)(a, s) &= \int \partial_r \Pi_{r,k}[u](f)(s') k(s' | a, s) \mu(ds') \\ &= \frac{1}{\tau} \int \text{Cov}_{\pi_{r,k}(\cdot|s')} (f(\cdot, s'), \partial_r q_{r,k}[u](\cdot, s')) k(s' | a, s) \mu(ds'). \end{aligned}$$

Therefore,

$$\begin{aligned}
& |\mathcal{P}_{r+u,k}f - \mathcal{P}_{r,k}f - \partial_r \mathcal{P}_{r,k}[u](f)|(a, s) \\
&= \left| \int \left(\Pi_{r+u,k}f - \Pi_{r,k}f - \partial_r \Pi_{r,k}[u](f) \right)(s') k(s' | a, s) \mu(ds') \right| \\
&\leq \|k(\cdot | a, s)\|_{L^\infty(\mu)} \|\Pi_{r+u,k}f - \Pi_{r,k}f - \partial_r \Pi_{r,k}[u](f)\|_{L^1(\mu)} \\
&\lesssim \|\Pi_{r+u,k}f - \Pi_{r,k}f - \partial_r \Pi_{r,k}[u](f)\|_{L^1(\mu)} \lesssim \|f\|_{L^\infty(\lambda)} \|u\|_{L^2(\lambda)}^2,
\end{aligned}$$

where the last step uses (55) and the uniform bound $\sup_{k \in \mathcal{K}} \sup_{(a,s)} \|k(\cdot | a, s)\|_{L^\infty(\mu)} < \infty$. In particular, taking the supremum over (a, s) yields

$$\|\mathcal{P}_{r+u,k}f - \mathcal{P}_{r,k}f - \partial_r \mathcal{P}_{r,k}[u](f)\|_{L^\infty(\lambda)} \lesssim \|f\|_{L^\infty(\lambda)} \|u\|_{L^2(\lambda)}^2.$$

Partial differentiability of $\mathcal{P}_{r,k}$ in the kernel. By the chain rule,

$$\begin{aligned}
\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k](f)(a, s) &= \int \tilde{\partial}_k \Pi_{r,k}[k' - k](f)(s') k(s' | a, s) \mu(ds') \\
&\quad + \int \Pi_{r,k}(f)(s') (k' - k)(s' | a, s) \mu(ds') \\
&= \frac{1}{\tau} \int \text{Cov}_{\pi_{r,k}(\cdot | s')} \left(f(\cdot, s'), \tilde{\partial}_k q_{r,k}[k' - k](\cdot, s') \right) k(s' | a, s) \mu(ds') \\
&\quad + \int \Pi_{r,k}(f)(s') (k' - k)(s' | a, s) \mu(ds').
\end{aligned}$$

Hence,

$$\begin{aligned}
& \left| \mathcal{P}_{r,k'}(f)(a, s) - \mathcal{P}_{r,k}(f)(a, s) - \tilde{\partial}_k \mathcal{P}_{r,k}[k' - k](f)(a, s) \right| \\
&= \left| \int \left(\Pi_{r,k'}(f) - \Pi_{r,k}(f) - \tilde{\partial}_k \Pi_{r,k}[k' - k](f) \right)(s') k(s' | a, s) \mu(ds') \right. \\
&\quad \left. + \int (\Pi_{r,k'}(f) - \Pi_{r,k}(f))(s') (k' - k)(s' | a, s) \mu(ds') \right|.
\end{aligned}$$

By (56), we have

$$\begin{aligned}
& \left| \int \left(\Pi_{r,k'}(f) - \Pi_{r,k}(f) - \tilde{\partial}_k \Pi_{r,k}[k' - k](f) \right)(s') k(s' | a, s) \mu(ds') \right| \\
&\lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)} \right\}.
\end{aligned}$$

For the second integral, using that the temperature- τ softmax map is τ^{-1} -Lipschitz from (ℓ_∞, ℓ_1) ,

$$|\Pi_{r,k'}(f)(s') - \Pi_{r,k}(f)(s')| \leq \|f\|_{L^\infty(\lambda)} \|\pi_{r,k'}(\cdot | s') - \pi_{r,k}(\cdot | s')\|_1 \leq \frac{\gamma}{\tau} \|f\|_{L^\infty(\lambda)} \sum_{a' \in \mathcal{A}} |v_{r,k'}(a', s') - v_{r,k}(a', s')|.$$

Arguing as in the proof of Lemma 20, we find that

$$\begin{aligned} \left\| (a, s) \mapsto \int (\Pi_{r,k'}(f) - \Pi_{r,k}(f))(s') (k' - k)(s' \mid a, s) \mu(ds') \right\|_{L^2(\lambda)} \\ \lesssim \|f\|_{L^\infty(\lambda)} \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}. \end{aligned}$$

Combining the two bounds,

$$\begin{aligned} \left\| \mathcal{P}_{r,k'}(f) - \mathcal{P}_{r,k}(f) - \tilde{\partial}_k \mathcal{P}_{r,k}[k' - k](f) \right\|_{L^2(\lambda)} &\lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 \right. \\ &\quad \left. + (1 + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}) \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)} \right\}. \end{aligned}$$

□

Lemma 29. *Let $r \in L^2(\lambda)$ and $h \in L^2(\lambda)$, and fix k . Suppose the linear operators $\mathcal{T}_{r,k} : L^2(\lambda) \rightarrow L^2(\lambda)$ and $\mathcal{T}_{r+h,k} : L^2(\lambda) \rightarrow L^2(\lambda)$ are invertible. Then*

$$\left\| \Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] \right\|_{L^1(\lambda)} = O(\|h\|_{L^2(\lambda)}^2),$$

with the big- O constant depending on $\|r\|_{L^\infty(\lambda)}$ and $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$.

Proof. By the chain rule,

$$\partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] = (\partial_r \Pi_{r,k}[h]) \mathcal{T}_{r,k}^{-1}(r) + \Pi_{r,k} (\partial_r \mathcal{T}_{r,k}^{-1}[h])(r) + \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(h),$$

where $\partial_r \mathcal{T}_{r,k}^{-1}[h] = -\mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h]) \mathcal{T}_{r,k}^{-1}$. By adding and subtracting, we obtain

$$\begin{aligned} &\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] \\ &= (\Pi_{r+h,k} - \Pi_{r,k} - \partial_r \Pi_{r,k}[h]) \mathcal{T}_{r,k}^{-1}(r) \\ &\quad + \Pi_{r,k} (\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} - \partial_r \mathcal{T}_{r,k}^{-1}[h])(r) \\ &\quad + (\Pi_{r+h,k} - \Pi_{r,k}) (\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(r) \\ &\quad + (\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1} - \Pi_{r,k} \mathcal{T}_{r,k}^{-1})(h). \end{aligned}$$

Taking norms and applying Lemma 28, we have

$$\begin{aligned} &\left\| \Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] \right\|_{L^1(\lambda)} \\ &\lesssim \left\| \Pi_{r+h,k} - \Pi_{r,k} - \partial_r \Pi_{r,k}[h] \right\|_{L^\infty(\lambda) \rightarrow L^1(\lambda)} \left\| \mathcal{T}_{r,k}^{-1}(r) \right\|_{L^\infty(\lambda)} \\ &\quad + \left\| \mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} - \partial_r \mathcal{T}_{r,k}^{-1}[h] \right\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \|r\|_{L^\infty(\lambda)} \\ &\quad + \left\| \Pi_{r+h,k} - \Pi_{r,k} \right\|_{L^2(\lambda) \rightarrow L^1(\mu)} \left\| (\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(r) \right\|_{L^2(\lambda)} \\ &\quad + \left\| (\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1} - \Pi_{r,k} \mathcal{T}_{r,k}^{-1})(h) \right\|_{L^1(\lambda)}. \end{aligned}$$

We will show that each term is bounded above by $\|h\|_{L^2(\lambda)}^2$, up to constants depending on γ, r ,

$\|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$, and $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. If so, the claim follows since

$$\|\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h]\|_{L^1(\lambda)} \lesssim \|h\|_{L^2(\lambda)}^2.$$

We study each term in turn.

Term 1. By Lemma 28,

$$\|\Pi_{r+h,k} - \Pi_{r,k} - \partial_r \Pi_{r,k}[h]\|_{L^\infty(\lambda) \rightarrow L^1(\lambda)} \lesssim \|h\|_{L^2(\lambda)}^2.$$

Hence,

$$\|\Pi_{r+h,k} - \Pi_{r,k} - \partial_r \Pi_{r,k}[h]\|_{L^\infty(\lambda) \rightarrow L^1(\lambda)} \|\mathcal{T}_{r,k}^{-1}(r)\|_{L^\infty(\lambda)} \lesssim \|\mathcal{T}_{r,k}^{-1}(r)\|_{L^\infty(\lambda)} \|h\|_{L^2(\lambda)}^2.$$

Term 2. Using the resolvent identity, the inverse-operator remainder admits the second-order expansion

$$\begin{aligned} & \mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1} \\ &= -\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1} \\ &+ \mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k})\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k})\mathcal{T}_{r+h,k}^{-1}. \end{aligned}$$

Hence,

$$\begin{aligned} & \|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \\ & \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} + \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty}^2 \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}\|_{L^\infty \rightarrow L^\infty} \|h\|_{L^2(\lambda)}^2. \end{aligned}$$

The remainder bound in Lemma 28 for $\mathcal{P}_{r,k}$ implies

$$\|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}^2.$$

Consequently,

$$\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}^2,$$

and thus

$$\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} - \partial_r \mathcal{T}_{r,k}^{-1}[h]\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \|r\|_{L^\infty(\lambda)} \lesssim \|h\|_{L^2(\lambda)}^2 \|r\|_{L^\infty(\lambda)}.$$

Term 3. We have

$$\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \leq \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k}^{-1}\|_{L^\infty \rightarrow L^\infty}.$$

Furthermore, by Lemma 28,

$$\|\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^\infty} \lesssim \|\partial_r \mathcal{P}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} + \|h\|_{L^2(\lambda)}^2,$$

and $\|\partial_r \mathcal{P}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}$. Hence,

$$\|\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}, \quad \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}.$$

Therefore,

$$\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)},$$

and

$$\|(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(r)\|_{L^2(\lambda)} \lesssim \|h\|_{L^2(\lambda)} \|r\|_{L^\infty(\lambda)}.$$

Finally, by Lemma 26,

$$\|\Pi_{r+h,k} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \lesssim \|h\|_{L^2(\lambda)}.$$

Putting everything together,

$$\|\Pi_{r+h,k} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(r)\|_{L^2(\lambda)} \lesssim \|h\|_{L^2(\lambda)}^2 \|r\|_{L^\infty(\lambda)}.$$

Term 4.

$$\begin{aligned} \|(\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1} - \Pi_{r,k} \mathcal{T}_{r,k}^{-1})(h)\|_{L^1(\lambda)} &= \|(\Pi_{r+h,k} - \Pi_{r,k}) \mathcal{T}_{r+h,k}^{-1}(h) + \Pi_{r,k} (\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(h)\|_{L^1(\lambda)} \\ &\leq \|\Pi_{r+h,k} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|h\|_{L^2(\lambda)} \\ &\quad + \|\Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|h\|_{L^2(\lambda)}. \end{aligned}$$

By Lemma 26, $\|\Pi_{r+h,k} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \lesssim \|h\|_{L^2}$. Moreover, arguing as in Term 3 and applying Lemma 27, we find $\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|h\|_{L^2}$, using that $\|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ and $\|\Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)}$ are uniformly bounded under the standing assumptions. Therefore,

$$\|(\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1} - \Pi_{r,k} \mathcal{T}_{r,k}^{-1})(h)\|_{L^1(\lambda)} \lesssim \|h\|_{L^2(\lambda)}^2.$$

The implicit constant depends on $\|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. However, we showed that

$$\begin{aligned} \|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} + \|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} + O(\|h\|_{L^2(\lambda)}^2). \end{aligned}$$

Thus,

$$\|\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h]\|_{L^1(\lambda)} = O(\|h\|_{L^2(\lambda)}^2),$$

where the big- O constant depends on $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ but not directly on $\|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. $\square$

Lemma 30. *Let $r \in L^2(\lambda)$, $k \in \mathcal{K}$, and $g \in \tilde{T}_{\mathcal{K}}$. Suppose the linear operators $\mathcal{T}_{r,k} : L^2(\lambda) \rightarrow L^2(\lambda)$*

and $\mathcal{T}_{r,k+g} : L^2(\lambda) \rightarrow L^2(\lambda)$ are invertible. Then

$$\|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} = O\left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right)$$

with the big- O constant depending on γ , $\|r\|_{L^\infty(\lambda)}$, and $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$.

Proof. By the chain rule,

$$\tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g] = (\tilde{\partial}_k \Pi_{r,k}[g]) \mathcal{T}_{r,k}^{-1}(r) + \Pi_{r,k} (\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g])(r),$$

where $\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g] = -\mathcal{T}_{r,k}^{-1}(\tilde{\partial}_k \mathcal{T}_{r,k}[g])\mathcal{T}_{r,k}^{-1}$. By adding and subtracting, we obtain

$$\begin{aligned} & \Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g] \\ &= (\Pi_{r,k+g} - \Pi_{r,k} - \tilde{\partial}_k \Pi_{r,k}[g]) \mathcal{T}_{r,k}^{-1}(r) \\ &+ \Pi_{r,k} (\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} - \tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g])(r) \\ &+ (\Pi_{r,k+g} - \Pi_{r,k})(\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1})(r). \end{aligned}$$

Taking norms and applying the k -direction analogues of Lemma 28, we have

$$\begin{aligned} & \|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \\ & \lesssim \|\Pi_{r,k+g} - \Pi_{r,k} - \tilde{\partial}_k \Pi_{r,k}[g]\|_{L^\infty(\lambda) \rightarrow L^1(\lambda)} \|\mathcal{T}_{r,k}^{-1}(r)\|_{L^\infty(\lambda)} \\ &+ \|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} - \tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g]\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \|r\|_{L^\infty(\lambda)} \\ &+ \|\Pi_{r,k+g} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|(\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1})(r)\|_{L^2(\lambda)}. \end{aligned}$$

Term 1. By Lemma 28, we have

$$\begin{aligned} \|\Pi_{r,k+g} f - \Pi_{r,k} f - \tilde{\partial}_k \Pi_{r,k}[g](f)\|_{L^1(\mu)} & \lesssim \|f\|_{L^\infty(\lambda)} \left\{ \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 \right. \\ & \left. + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} \|g\|_{L^2(\mu \otimes \lambda)} \right\}. \end{aligned}$$

Hence,

$$\begin{aligned} \|\Pi_{r,k+g} - \Pi_{r,k} - \tilde{\partial}_k \Pi_{r,k}[g]\|_{L^\infty(\lambda) \rightarrow L^1(\lambda)} \|\mathcal{T}_{r,k}^{-1}(r)\|_{L^\infty(\lambda)} & \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 \\ & + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} \|g\|_{\mathcal{K}}, \end{aligned}$$

where the implicit constant depends on $\|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2}$ and $\|r\|_\infty$.

Term 2. Using the resolvent identity, the inverse-operator remainder admits the second-order expansion

$$\begin{aligned} & \mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\tilde{\partial}_k \mathcal{T}_{r,k}[g])\mathcal{T}_{r,k}^{-1} \\ &= -\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k}[g])\mathcal{T}_{r,k}^{-1} \\ &+ \mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k})\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k})\mathcal{T}_{r,k+g}^{-1}. \end{aligned}$$

Hence,

$$\begin{aligned} & \|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\tilde{\partial}_k \mathcal{T}_{r,k}[g])\mathcal{T}_{r,k}^{-1}\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \\ & \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k}[g]\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ & \quad + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k}\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)}^2. \end{aligned}$$

The remainder bound in Lemma 28 for $\mathcal{P}_{r,k}$ implies

$$\|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k}[g]\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} \|g\|_{\mathcal{K}},$$

and similarly, by Lemma 27,

$$\|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k}\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}.$$

Consequently,

$$\begin{aligned} & \|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\tilde{\partial}_k \mathcal{T}_{r,k}[g])\mathcal{T}_{r,k}^{-1}\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \\ & \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \\ & \quad \times \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} \|g\|_{\mathcal{K}} \right) \\ & \quad + \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ & \quad \times \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}} \right)^2. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1} - \tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g]\|_{L^\infty(\lambda) \rightarrow L^2(\lambda)} \\ & \quad \times \|r\|_{L^\infty(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 \\ & \quad + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} \|g\|_{\mathcal{K}}. \end{aligned}$$

Term 3. We have

$$\|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \leq \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}.$$

Furthermore, by Lemma 28,

$$\|\mathcal{P}_{r,k+g} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}},$$

which implies

$$\|\mathcal{T}_{r,k+g} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}.$$

Therefore,

$$\|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim (\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}).$$

Consequently,

$$\|(\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1})(r)\|_{L^2(\lambda)} \lesssim (\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}) \|r\|_{L^\infty(\lambda)}.$$

Finally, by Lemma 26,

$$\|\Pi_{r,k+g} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}.$$

Putting everything together,

$$\begin{aligned} \|\Pi_{r,k+g} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} &\times \|(\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1})(r)\|_{L^2(\lambda)} \\ &\lesssim (\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}})^2 \|r\|_{L^\infty(\lambda)}. \end{aligned}$$

Final bound. Combining the above estimates, we obtain

$$\|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \lesssim (\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}})^2 \|r\|_{L^\infty(\lambda)}.$$

Equivalently (absorbing $\|r\|_{L^\infty(\lambda)}$ into the implicit constant),

$$\|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \lesssim \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2.$$

The implicit constant depends on $\|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. However, we showed that

$$\begin{aligned} \|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} + \|\mathcal{T}_{r,k+g}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} + O(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2). \end{aligned}$$

Thus,

$$\|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} = O\left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right),$$

where the big- O constant depends on $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ but not directly on $\|\mathcal{T}_{r,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. $\square$

Lemma 31. *Let $r \in L^2(\lambda)$, $k \in \mathcal{K}$, $h \in L^2(\lambda)$, and $g \in \tilde{T}_{\mathcal{K}}$. Suppose the linear operator $\mathcal{T}_{r,k} : L^2(\lambda) \rightarrow L^2(\lambda)$ is invertible and $\|h\|_{L^2(\lambda)} + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}} \rightarrow 0$. Then*

$$\begin{aligned} &\|\Pi_{r+h,k+g} \mathcal{T}_{r+h,k+g}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \\ &= O\left(\|h\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right), \end{aligned}$$

with the big- O constant depending on γ , $\|r\|_{L^\infty(\lambda)}$, and $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$; moreover, the inverses appearing above exist for all sufficiently small perturbations.

Proof. Invertibility of the perturbed operators. We first show that $\mathcal{T}_{r+h,k} : L^2(\lambda) \rightarrow L^2(\lambda)$ and $\mathcal{T}_{r+h,k+g} : L^2(\lambda) \rightarrow L^2(\lambda)$ are invertible whenever $O\left(\|h\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right)$ is sufficiently small.

Since $\mathcal{T}_{r,k}$ is invertible, it suffices to control the perturbations

$$\Delta_1 := \mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}, \quad \Delta_2 := \mathcal{T}_{r+h,k+g} - \mathcal{T}_{r,k}.$$

By Lemma 27,

$$\begin{aligned}\|\Delta_1\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &= O(\|h\|_{L^2(\lambda)}), \\ \|\Delta_2\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &= O(\|h\|_{L^2(\lambda)} + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}).\end{aligned}$$

If $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Delta_i\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < 1$ for $i = 1, 2$, then by the Neumann-series inversion lemma, both $\mathcal{T}_{r+h,k}$ and $\mathcal{T}_{r+h,k+g}$ are invertible, with

$$\begin{aligned}\|\mathcal{T}_{r+h,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\leq \frac{\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}}{1 - \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Delta_1\|_{L^2(\lambda) \rightarrow L^2(\lambda)}}, \\ \|\mathcal{T}_{r+h,k+g}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\leq \frac{\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}}{1 - \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Delta_2\|_{L^2(\lambda) \rightarrow L^2(\lambda)}}.\end{aligned}$$

As $\|h\|_{L^2(\lambda)} + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}} \rightarrow 0$, it holds that $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\Delta_i\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < 1$ for $i = 1, 2$. Hence, $\mathcal{T}_{r+h,k} : L^2(\lambda) \rightarrow L^2(\lambda)$ and $\mathcal{T}_{r+h,k+g} : L^2(\lambda) \rightarrow L^2(\lambda)$ become invertible. Without loss of generality, we proceed assuming these inverses exist.

Proof of bound. Lemma 29 establishes that

$$\|\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h]\|_{L^1(\lambda)} = O(\|h\|_{L^2(\lambda)}^2).$$

Lemma 30 establishes that

$$\|\Pi_{r,k+g} \mathcal{T}_{r,k+g}^{-1}(r) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} = O(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2).$$

Therefore, adding and subtracting,

$$\begin{aligned}&\|\Pi_{r+h,k+g} \mathcal{T}_{r+h,k+g}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \\ &\leq \|\Pi_{r+h,k+g} \mathcal{T}_{r+h,k+g}^{-1}(r+h) - \Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)][g]\|_{L^1(\lambda)} \\ &\quad + \|\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h]\|_{L^1(\lambda)} \\ &\quad + \|(\tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)])(g)\|_{L^1(\lambda)}.\end{aligned}$$

Applying the lemmas, we obtain

$$\begin{aligned}&\|\Pi_{r+h,k+g} \mathcal{T}_{r+h,k+g}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g]\|_{L^1(\lambda)} \\ &\leq O(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2) + O(\|h\|_{L^2(\lambda)}^2) \\ &\quad + \|(\tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)])(g)\|_{L^1(\lambda)}.\end{aligned}$$

We claim that

$$\|(\tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)])(g)\|_{L^1(\lambda)} = O(\|h\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2).$$

Assuming this claim, it follows that

$$\begin{aligned} & \left\| \Pi_{r+h,k+g} \mathcal{T}_{r+h,k+g}^{-1}(r+h) - \Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r) - \partial_r[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][h] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g] \right\|_{L^1(\lambda)} \\ &= O\left(\|h\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right), \end{aligned}$$

as desired.

Proof of claim. We bound

$$\left\| \left(\tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)] \right) [g] \right\|_{L^1(\lambda)}.$$

By the product rule,

$$\tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)][g] = (\tilde{\partial}_k \Pi_{r,k}[g]) \mathcal{T}_{r,k}^{-1}(r) + \Pi_{r,k} (\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g])(r),$$

and similarly at $(r+h, k)$. Hence, adding and subtracting intermediate terms,

$$\begin{aligned} & (\tilde{\partial}_k[\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k[\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)])(g) \\ &= \underbrace{(\tilde{\partial}_k \Pi_{r+h,k}[g] - \tilde{\partial}_k \Pi_{r,k}[g]) \mathcal{T}_{r+h,k}^{-1}(r+h)}_{A_1} + \underbrace{\tilde{\partial}_k \Pi_{r,k}[g] (\mathcal{T}_{r+h,k}^{-1}(r+h) - \mathcal{T}_{r,k}^{-1}(r))}_{A_2} \\ &+ \underbrace{(\Pi_{r+h,k} - \Pi_{r,k}) (\tilde{\partial}_k \mathcal{T}_{r+h,k}^{-1}[g])(r+h)}_{A_3} + \underbrace{\Pi_{r,k} ((\tilde{\partial}_k \mathcal{T}_{r+h,k}^{-1}[g]) - (\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g]))(r+h)}_{A_4} \\ &+ \underbrace{\Pi_{r,k} (\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g])(h)}_{A_5}. \end{aligned}$$

Bounds for A_1 – A_5 . Using Lemma 28, Lemma 27, and the bounds derived in the proofs of

Lemmas 29 and 30, we obtain:

$$\begin{aligned}
\|A_1\|_{L^1(\lambda)} &\lesssim \|\tilde{\partial}_k \Pi_{r+h,k}[g] - \tilde{\partial}_k \Pi_{r,k}[g]\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\mathcal{T}_{r+h,k}^{-1}(r+h)\|_{L^2(\lambda)} \\
&= O\left(\|h\|_{L^2(\lambda)} \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}\right)\right), \\
\|A_2\|_{L^1(\lambda)} &\lesssim \|\tilde{\partial}_k \Pi_{r,k}[g]\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\mathcal{T}_{r+h,k}^{-1}(r+h) - \mathcal{T}_{r,k}^{-1}(r)\|_{L^2(\lambda)} \\
&= O\left(\|h\|_{L^2(\lambda)} \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}\right)\right), \\
\|A_3\|_{L^1(\lambda)} &\lesssim \|\Pi_{r+h,k} - \Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\tilde{\partial}_k \mathcal{T}_{r+h,k}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|r+h\|_{L^\infty(\lambda)} \\
&= O\left(\|h\|_{L^2(\lambda)} \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}\right)\right), \\
\|A_4\|_{L^1(\lambda)} &\lesssim \|\Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\tilde{\partial}_k \mathcal{T}_{r+h,k}^{-1}[g] - \tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|r+h\|_{L^\infty(\lambda)} \\
&= O\left(\|h\|_{L^2(\lambda)} \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}\right)\right), \\
\|A_5\|_{L^1(\lambda)} &\lesssim \|\Pi_{r,k}\|_{L^2(\lambda) \rightarrow L^1(\mu)} \|\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[g]\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|h\|_{L^2(\lambda)} \\
&= O\left(\|h\|_{L^2(\lambda)} \left(\|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)} + \|g\|_{\mathcal{K}}\right)\right).
\end{aligned}$$

Combining the bounds and using that all operator norms involved are uniformly bounded on the neighborhood, we conclude

$$\left\| \left(\tilde{\partial}_k [\Pi_{r+h,k} \mathcal{T}_{r+h,k}^{-1}(r+h)] - \tilde{\partial}_k [\Pi_{r,k} \mathcal{T}_{r,k}^{-1}(r)] \right) [g] \right\|_{L^1(\lambda)} = O\left(\|h\|_{L^2(\lambda)}^2 + \|v_{r,k+g} - v_{r,k}\|_{L^2(\lambda)}^2 + \|g\|_{\mathcal{K}}^2\right).$$

□

Lemma 32 (Second-order partial Fréchet differentiability of $r \mapsto v_{r,k}$). *Suppose $\gamma < 1$. Then the map $r \mapsto \partial_r v_{r,k} = \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}$ is Fréchet differentiable as a map from $L^\infty(\lambda)$ to $L^\infty(\lambda)$, with derivative*

$$\partial_r^2 v_{r,k}[h_1, h_2] := \partial_r[\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][h_1](h_2) = \frac{1}{\gamma} \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} \left[\text{Cov}_{\pi_{r,k}} \left(\mathcal{T}_{r,k}^{-1} h_1, \mathcal{T}_{r,k}^{-1} h_2 \right) \right],$$

where $\partial_r q_{r,k}[h](\cdot, s) := h(\cdot, s) + \gamma \partial_r v_{r,k}[h](\cdot, s)$. In particular, we have the quadratic expansion

$$\|\partial_r v_{r+h,k} - \partial_r v_{r,k} - \partial_r^2 v_{r,k}[\cdot, h]\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}^2.$$

Proof. By the chain rule, we have that $r \mapsto \mathcal{T}_{r,k}$ is Gateaux differentiable with derivative operator

$$\partial_r \mathcal{T}_{r,k}[h] = -\gamma \partial_r \mathcal{P}_{r,k}[h], \quad \text{i.e.} \quad (\partial_r \mathcal{T}_{r,k}[h])(f) = -\gamma (\partial_r \mathcal{P}_{r,k}[h])(f).$$

Applying the inverse-map differentiation formula, we define the derivative of the inverse $\mathcal{T}_{r,k}^{-1}$ as

$$\partial_r \mathcal{T}_{r,k}^{-1}[h] = -\mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{T}_{r,k}[h]) \mathcal{T}_{r,k}^{-1}.$$

Applying the product rule and differentiability of $r \mapsto \mathcal{T}_{r,k}$ and $r \mapsto \mathcal{P}_{r,k}$, we obtain that

$$\partial_r[\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][h] = -\mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{T}_{r,k}[h]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} + \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h]).$$

Later, we will show that $r \mapsto \partial_r v_{r,k}$ is Fréchet differentiable, in an appropriate sense, with the second derivative given above. Before doing so, we simplify the expression for this second derivative as follows:

$$\begin{aligned} \partial_r^2 v_{r,k}[h_1, h_2] &= -\mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{T}_{r,k}[h_2]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}[h_1] + \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h_2])[h_1] \\ &= \gamma \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h_2]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}[h_1] + \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h_2])[h_1] \\ &= \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h_2]) (I + \gamma \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k})[h_1] \\ &= \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{P}_{r,k}[h_2]) (I + \gamma \partial_r v_{r,k})[h_1] \\ &= \frac{1}{\gamma} \mathcal{T}_{r,k}^{-1} \left[(a, s) \mapsto \int \text{Cov}_{\pi_{r,k}(\cdot|s')} \left(\partial_r q_{r,k}[h_1](\cdot, s'), \partial_r q_{r,k}[h_2](\cdot, s') \right) k(s' | a, s) \mu(ds') \right], \end{aligned}$$

$\partial_r q_{r,k}[h](\cdot, s) := h(\cdot, s) + \gamma \partial_r v_{r,k}[h](\cdot, s)$. We can write:

$$\partial_r^2 v_{r,k}[h_1, h_2] = \frac{1}{\gamma} \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} \left[(a', s') \mapsto \text{Cov}_{\pi_{r,k}(\cdot|s')} \left(\partial_r q_{r,k}[h_1](\cdot, s'), \partial_r q_{r,k}[h_2](\cdot, s') \right) \right].$$

Finally, by Lemma 22, $\partial_r q_{r,k}[h] = \mathcal{T}_{r,k}^{-1} h$. Hence,

$$\partial_r^2 v_{r,k}[h_1, h_2] = \frac{1}{\gamma} \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} \left[\text{Cov}_{\pi_{r,k}} \left(\mathcal{T}_{r,k}^{-1} h_1, \mathcal{T}_{r,k}^{-1} h_2 \right) \right].$$

Now, we establish a quadratic bound for the remainder of the first-order Taylor expansion of $r \mapsto \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}$; namely, $\mathcal{T}_{r+h,k}^{-1} \mathcal{P}_{r+h,k} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \partial_r[\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][h]$. Adding and subtracting $\mathcal{T}_{r+h,k}^{-1} \mathcal{P}_{r,k}$

and $\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r+h,k}$, we obtain

$$\begin{aligned}
& \mathcal{T}_{r+h,k}^{-1} \mathcal{P}_{r+h,k} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \partial_r [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}] [h] \\
&= \left(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} \right) \mathcal{P}_{r,k} + \mathcal{T}_{r,k}^{-1} \left(\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \right) \\
&+ \left(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} \right) \left(\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \right) \\
&+ \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \partial_r [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}] [h].
\end{aligned}$$

Thus,

$$\begin{aligned}
& \mathcal{T}_{r+h,k}^{-1} \mathcal{P}_{r+h,k} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \partial_r [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}] [h] \\
&= \underbrace{\left[\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1} (\partial_r \mathcal{T}_{r,k} [h]) \mathcal{T}_{r,k}^{-1} \right] \mathcal{P}_{r,k}}_{\text{Inverse-operator remainder}} \\
&+ \underbrace{\mathcal{T}_{r,k}^{-1} \left[\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} - \partial_r \mathcal{P}_{r,k} [h] \right]}_{\text{Direct } \mathcal{P} \text{ remainder}} \\
&+ \underbrace{\left(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} \right) \left(\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \right)}_{\text{Mixed interaction term}}.
\end{aligned}$$

We analyze each term, beginning with the direct $\mathcal{P}$ remainder and the mixed interaction term, and then returning to the inverse-operator remainder. We will use that the Bellman operators $\mathcal{T}_{r+h,k}$ and $\mathcal{T}_{r,k}$ admit bounded inverses on $L^\infty(\lambda)$ since $\gamma < 1$.

For the direct $\mathcal{P}$ remainder, by Lemma 28 and the definition of operator norms, we obtain

$$\begin{aligned}
\| \mathcal{T}_{r,k}^{-1} [\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} - \partial_r \mathcal{P}_{r,k} [h]] \|_{L^\infty \rightarrow L^\infty} &\leq \| \mathcal{T}_{r,k}^{-1} \|_{L^\infty \rightarrow L^\infty} \| \mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} - \partial_r \mathcal{P}_{r,k} [h] \|_{L^\infty \rightarrow L^\infty} \\
&\lesssim \| h \|_{L^2(\lambda)}^2,
\end{aligned}$$

where we used that $\| \mathcal{T}_{r,k}^{-1} \|_{L^\infty \rightarrow L^\infty} < \infty$ since $\gamma < 1$.

For the mixed interaction term, we note that

$$\| (\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1}) (\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k}) \|_{L^\infty \rightarrow L^\infty} \leq \| \mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} \|_{L^\infty \rightarrow L^\infty} \| \mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \|_{L^\infty \rightarrow L^\infty}.$$

Moreover,

$$\| \mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} \|_{L^\infty \rightarrow L^\infty} \leq \| \mathcal{T}_{r,k}^{-1} \|_{L^\infty \rightarrow L^\infty} \| \mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} \|_{L^\infty \rightarrow L^\infty} \| \mathcal{T}_{r+h,k}^{-1} \|_{L^\infty \rightarrow L^\infty}.$$

Furthermore, by Lemma 28, we have

$$\| \mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \|_{L^\infty \rightarrow L^\infty} \lesssim \| \partial_r \mathcal{P}_{r,k} [h] \|_{L^\infty \rightarrow L^\infty} + \| h \|_{L^2(\lambda)}^2.$$

Moreover, $\| \partial_r \mathcal{P}_{r,k} [h] \|_{L^\infty \rightarrow L^\infty} \lesssim \| h \|_{L^2(\lambda)}$. Applying this bound, we obtain

$$\| \mathcal{P}_{r+h,k} - \mathcal{P}_{r,k} \|_{L^\infty \rightarrow L^\infty} \lesssim \| h \|_{L^2(\lambda)}, \quad \| \mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} \|_{L^\infty \rightarrow L^\infty} \lesssim \| h \|_{L^2(\lambda)}.$$

Hence,

$$\begin{aligned} \|(\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1})(\mathcal{P}_{r+h,k} - \mathcal{P}_{r,k})\|_{L^\infty \rightarrow L^\infty} &\lesssim (1 + \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty}) \|h\|_{L^2(\lambda)}^2 \\ &\lesssim \|h\|_{L^2(\lambda)}^2. \end{aligned}$$

We now turn to the inverse-operator remainder. Using the resolvent identity, the inverse-operator remainder can itself be expressed using the second-order expansion:

$$\begin{aligned} &\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1} \\ &= -\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1} \\ &\quad + \mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k})\mathcal{T}_{r,k}^{-1}(\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k})\mathcal{T}_{r+h,k}^{-1}. \end{aligned}$$

Hence, taking operator norms, we have that

$$\begin{aligned} &\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \\ &\leq \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} \\ &\quad + \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \\ &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} + \|\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty}^2 \|\mathcal{T}_{r+h,k}\|_{L^\infty \rightarrow L^\infty} \|h\|_{L^2(\lambda)}^2. \end{aligned}$$

The remainder bound in Lemma 28 for $\mathcal{P}_{r,k}$ implies that $\|\mathcal{T}_{r+h,k} - \mathcal{T}_{r,k} - \partial_r \mathcal{T}_{r,k}[h]\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}^2$. We conclude that

$$\|\mathcal{T}_{r+h,k}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1}(\partial_r \mathcal{T}_{r,k}[h])\mathcal{T}_{r,k}^{-1}\|_{L^\infty \rightarrow L^\infty} \lesssim \|h\|_{L^2(\lambda)}^2.$$

Combining the bounds for all three terms and using that $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} < \infty$, we conclude that

$$\begin{aligned} \|\partial_r v_{r+h,k} - \partial_r v_{r,k} - \partial_r^2 v_{r,k}[\cdot, h]\|_{L^\infty \rightarrow L^\infty} &= \|\mathcal{T}_{r+h,k}^{-1} \mathcal{P}_{r+h,k} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \partial_r[\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][h]\|_{L^\infty \rightarrow L^\infty} \\ &\lesssim \|h\|_{L^2(\lambda)}^2. \end{aligned}$$

□

Lemma 33 (Cross Fréchet differentiability of $(r, k) \mapsto v_{r,k}$). *Let $k, k' \in \mathcal{K}$. Suppose $\mathcal{T}_{r,k}$ and $\mathcal{T}_{r,k'}$ are both invertible as maps from $L^2(\lambda) \rightarrow L^2(\lambda)$ with $\|\mathcal{T}_{r,k}^{-1}\|_{L^2 \rightarrow L^2} + \|\mathcal{T}_{r,k'}^{-1}\|_{L^2 \rightarrow L^2} < \infty$. Then, the map $k \mapsto \partial_r v_{r,k}[h]$ is Fréchet differentiable with derivative*

$$\begin{aligned} \tilde{\partial}_k \partial_r v_{r,k}[h, k' - k] &= \mathcal{T}_{r,k}^{-1} \left[\frac{1}{\tau} \mathcal{P}_{r,k} \left[(a', s') \mapsto \text{Cov}_{\pi_{r,k}(\cdot|s')} \left(\partial_r q_{r,k}[h](\cdot, s'), \tilde{\partial}_k q_{r,k}[k' - k](\cdot, s') \right) \right] \right. \\ &\quad \left. + \mathcal{P}_{k'-k} \Pi_{r,k} \partial_r q_{r,k}[h] \right]. \end{aligned}$$

Here $\partial_r q_{r,k}[h](\cdot, s') := \mathcal{T}_{r,k}^{-1} h(\cdot, s')$ and $\tilde{\partial}_k q_{r,k}[k' - k](\cdot, s') := \gamma \tilde{\partial}_k v_{r,k}[k' - k](\cdot, s')$. In particular, we

have the expansion:

$$\|\partial_r v_{r,k'} - \partial_r v_{r,k} - \tilde{\partial}_k \partial_r v_{r,k}[\cdot, k' - k]\|_{L^\infty \rightarrow L^2} \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2.$$

Proof. By the chain rule, $k \mapsto \mathcal{T}_{r,k}$ is Gâteaux differentiable with

$$\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k] = -\gamma \tilde{\partial}_k \mathcal{P}_{r,k}[k' - k], \quad \text{i.e.} \quad (\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k])(f) = -\gamma (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k])(f).$$

Applying the inverse-map differentiation formula,

$$\tilde{\partial}_k \mathcal{T}_{r,k}^{-1}[k' - k] = -\mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k]) \mathcal{T}_{r,k}^{-1}.$$

By the product rule and differentiability of $k \mapsto \mathcal{T}_{r,k}$ and $k \mapsto \mathcal{P}_{r,k}$,

$$\tilde{\partial}_k [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][k' - k] = -\mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} + \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k]).$$

Hence, for directions $h \in L^\infty(\lambda)$ and $k' - k$, the mixed partial satisfies

$$\begin{aligned} \tilde{\partial}_k \partial_r v_{r,k}[h, k' - k] &= -\mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}[h] + \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k])[h] \\ &= \gamma \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k]) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}[h] + \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k])[h] \\ &= \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k]) (I + \gamma \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k})[h] \\ &= \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{P}_{r,k}[k' - k]) (I + \gamma \partial_r v_{r,k})[h]. \end{aligned}$$

Here we used the identity $\tilde{\partial}_k \mathcal{T}_{r,k}[k' - k] = -\gamma \tilde{\partial}_k \mathcal{P}_{r,k}[k' - k]$ and $(I + \gamma \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}) = (I + \gamma \partial_r v_{r,k})$. Plugging in the definitions of the operators from Lemma 28, we conclude that

$$\begin{aligned} \tilde{\partial}_k \partial_r v_{r,k}[h, k' - k] &= \mathcal{T}_{r,k}^{-1} \left[\frac{1}{\tau} \mathcal{P}_{r,k} \left[(a', s') \mapsto \text{Cov}_{\pi_{r,k}(\cdot | s')} \left(\partial_r q_{r,k}[h](\cdot, s'), \tilde{\partial}_k q_{r,k}[k' - k](\cdot, s') \right) \right] \right. \\ &\quad \left. + \mathcal{P}_{k'-k} \Pi_{r,k} \partial_r q_{r,k}[h] \right]. \end{aligned}$$

where $\tilde{\partial}_k q_{r,k}[k' - k](\cdot, s') := \gamma \tilde{\partial}_k v_{r,k}[k' - k](\cdot, s')$ and $\pi_{r,k}(\cdot | s')$ denotes the softmax distribution induced by $q_{r,k}(\cdot, s')$.

Establish second-order bound for Taylor series remainder. Arguing exactly as in the proof

of Lemma 32, but differentiating in k , we have

$$\begin{aligned}
& \mathcal{T}_{r,k'}^{-1} \mathcal{P}_{r,k'} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \tilde{\partial}_k [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}] [k' - k] \\
&= \underbrace{\left[\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k} [k' - k]) \mathcal{T}_{r,k}^{-1} \right]}_{\text{Inverse-operator remainder}} \mathcal{P}_{r,k} \\
&+ \underbrace{\mathcal{T}_{r,k}^{-1} [\mathcal{P}_{r,k'} - \mathcal{P}_{r,k} - \tilde{\partial}_k \mathcal{P}_{r,k} [k' - k]]}_{\text{Direct } \mathcal{P} \text{ remainder}} \\
&+ \underbrace{\left(\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1} \right) (\mathcal{P}_{r,k'} - \mathcal{P}_{r,k})}_{\text{Mixed interaction term}}.
\end{aligned}$$

We can write the first term as

$$\begin{aligned}
& \mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1} + \mathcal{T}_{r,k}^{-1} (\tilde{\partial}_k \mathcal{T}_{r,k} [k' - k]) \mathcal{T}_{r,k}^{-1} \\
&= -\mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k]) \mathcal{T}_{r,k}^{-1} \\
&+ \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1}.
\end{aligned}$$

Combining the previous two displays, we find that

$$\begin{aligned}
& \mathcal{T}_{r,k'}^{-1} \mathcal{P}_{r,k'} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \tilde{\partial}_k [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}] [k' - k] \\
&= \underbrace{-\mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k]) \mathcal{T}_{r,k}^{-1}}_{\text{Inverse linear remainder}} \\
&+ \underbrace{\mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}}_{\text{Inverse quadratic product}} \\
&+ \underbrace{\mathcal{T}_{r,k}^{-1} [\mathcal{P}_{r,k'} - \mathcal{P}_{r,k} - \tilde{\partial}_k \mathcal{P}_{r,k} [k' - k]]}_{\text{Direct } \mathcal{P} \text{ remainder}} \\
&+ \underbrace{\left(\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1} \right) (\mathcal{P}_{r,k'} - \mathcal{P}_{r,k})}_{\text{Mixed interaction term}}.
\end{aligned}$$

For the first term, we have

$$\begin{aligned}
& \left\| \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k]) \mathcal{T}_{r,k}^{-1} \right\|_{L^\infty \rightarrow L^2} \\
&\leq \left\| \mathcal{T}_{r,k}^{-1} \right\|_{L^2 \rightarrow L^2} \left\| \mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k] \right\|_{L^\infty \rightarrow L^2} \left\| \mathcal{T}_{r,k}^{-1} \right\|_{L^\infty \rightarrow L^\infty} \\
&\lesssim \left\| \mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k] \right\|_{L^\infty \rightarrow L^2}.
\end{aligned}$$

Here we used that $\left\| \mathcal{T}_{r,k}^{-1} \right\|_{L^2 \rightarrow L^2} < \infty$ and absorbed its dependence into the notation $\lesssim$. By

Lemma 28, we thus obtain

$$\begin{aligned} & \left\| \mathcal{T}_{r,k}^{-1} \left(\mathcal{T}_{r,k'} - \mathcal{T}_{r,k} - \tilde{\partial}_k \mathcal{T}_{r,k} [k' - k] \right) \mathcal{T}_{r,k}^{-1} \right\|_{L^\infty \rightarrow L^2} \\ & \lesssim \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}. \end{aligned}$$

For the second term. At the end of this proof, we establish the claim that $\|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^2} \lesssim \|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}$ and, hence, also $\|\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}$. Using submultiplicativity of operator norms,

$$\begin{aligned} & \left\| \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}) \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} \right\|_{L^2 \rightarrow L^2} \\ & \leq \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \|\mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}^2 \\ & \lesssim \left(\|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \right)^2 \\ & \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 \end{aligned}$$

where we absorbed $\|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$, $\|\mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$, and $\|\mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$ into the $\lesssim$ notation.

For the third term, by an identical argument to the first term, we have that

$$\begin{aligned} & \left\| \mathcal{T}_{r,k}^{-1} \left(\mathcal{P}_{r,k'} - \mathcal{P}_{r,k} - \tilde{\partial}_k \mathcal{P}_{r,k} [k' - k] \right) \right\|_{L^\infty \rightarrow L^2} \\ & \lesssim \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \|k' - k\|_{L^2(\mu \otimes \lambda)}. \end{aligned}$$

For the fourth term, we note that

$$\left\| (\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1}) (\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}) \right\|_{L^\infty \rightarrow L^2} \lesssim \|\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^2}.$$

Moreover, by the resolvent identity, we have

$$\begin{aligned} \|\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &= \|\mathcal{T}_{r,k}^{-1} (\mathcal{T}_{r,k} - \mathcal{T}_{r,k'}) \mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\leq \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}. \end{aligned}$$

Hence,

$$\begin{aligned} \left\| (\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1}) (\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}) \right\|_{L^\infty \rightarrow L^2} &\lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\quad \times \|\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^2}. \end{aligned}$$

At the end of this proof, we establish the claim that

$$\begin{aligned} \|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^\infty \rightarrow L^2} &\lesssim \|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\ &\lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}, \end{aligned}$$

and, hence, also

$$\|\mathcal{T}_{r,k'} - \mathcal{T}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.$$

Assuming this claim holds, we conclude that

$$\begin{aligned}
& \|(\mathcal{T}_{r,k'}^{-1} - \mathcal{T}_{r,k}^{-1}) \times (\mathcal{P}_{r,k'} - \mathcal{P}_{r,k})\|_{L^\infty \rightarrow L^2} \\
& \lesssim \|\mathcal{T}_{r,k}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \|\mathcal{T}_{r,k'}^{-1}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} \\
& \quad \times \left\{ \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)} \right\}^2 \\
& \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2.
\end{aligned}$$

Combining bounds for all four terms, we conclude that

$$\begin{aligned}
& \|\partial_r v_{r,k'} - \partial_r v_{r,k} - \tilde{\partial}_k \partial_r v_{r,k}[\cdot, k' - k]\|_{L^\infty \rightarrow L^2} \\
& = \left\| \mathcal{T}_{r,k'}^{-1} \mathcal{P}_{r,k'} - \mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k} - \tilde{\partial}_k [\mathcal{T}_{r,k}^{-1} \mathcal{P}_{r,k}][k' - k] \right\|_{L^\infty \rightarrow L^2} \\
& \lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)}^2 + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}^2,
\end{aligned}$$

as desired.

Establishing claimed bound for $\|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)}$. We can write

$$\begin{aligned}
((\mathcal{P}_{r,k'} - \mathcal{P}_{r,k})f)(a, s) &= \int \left\langle \pi_{r,k'}(\cdot | s'), f(\cdot, s') \right\rangle \underbrace{(k' - k)(s' | a, s)}_{:= \Delta k} \mu(ds') \\
&+ \int \underbrace{\left\langle \pi_{r,k'}(\cdot | s') - \pi_{r,k}(\cdot | s'), f(\cdot, s') \right\rangle}_{:= \langle \Delta \pi, f \rangle(s')} k(s' | a, s) \mu(ds').
\end{aligned}$$

By Cauchy–Schwarz and Jensen’s inequality, the first term satisfies

$$\begin{aligned}
& \left\| \int \left\langle \pi_{r,k'}(\cdot | s'), f(\cdot, s') \right\rangle \Delta k(s' | a, s) \mu(ds') \right\|_{L^2(\lambda)} \leq \|\Delta k\|_{L^2(\mu \otimes \lambda)} \left\| \left(\sum_{a' \in \mathcal{A}} \pi_{r,k'}(a' | s') |f(a', s')|^2 \right)^{1/2} \right\|_{L^2(\mu)} \\
& \leq \sqrt{|\mathcal{A}|} \|\Delta k\|_{L^2(\mu \otimes \lambda)} \|f\|_{L^2(\lambda)}.
\end{aligned}$$

Similarly, viewing the second term as a Hilbert–Schmidt operator with kernel $K^*((a, s), (a', s')) = k(s' | a, s) \Delta \pi(a' | s')$, we obtain

$$\left\| \int \langle \Delta \pi(\cdot | s'), f(\cdot, s') \rangle k(s' | a, s) \mu(ds') \right\|_{L^2(\lambda)} \leq \|k\|_{L^2(\mu \otimes \lambda)} \|\Delta \pi(\cdot | \cdot)\|_{L^2(\lambda)} \|f\|_{L^2(\lambda)}.$$

Hence, using the Lipschitz property of the softmax map,

$$\|\pi_{r,k'} - \pi_{r,k}\|_{L^2(\lambda)} \lesssim \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.$$

we conclude that

$$\begin{aligned}
\|\mathcal{P}_{r,k'} - \mathcal{P}_{r,k}\|_{L^2(\lambda) \rightarrow L^2(\lambda)} &\leq \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|k\|_{L^2(\mu \otimes \lambda)} \|\pi_{r,k'} - \pi_{r,k}\|_{L^2(\lambda)} \\
&\lesssim \|k' - k\|_{L^2(\mu \otimes \lambda)} + \|v_{r,k'} - v_{r,k}\|_{L^2(\lambda)}.
\end{aligned}$$

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